Competition Creates Competition: Stock Prices and the Discovery of Takeover Bidders

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Abstract

A stronger incumbent bidder can attract a challenger when it makes the target's stock price more informative. I study a takeover auction in which a prospective challenger sees the price before it pays to prepare a bid. A stronger incumbent lowers the challenger's acquisition profit at every belief about its quality, and it also makes target proceeds more sensitive to that quality, so an investor who knows the challenger's quality class gains more from trade. The investor shares one order opportunity with liquidity traders, so it keeps a residual advantage after every order.

On an open set of parameters, a stronger incumbent turns an uninformative price into an informative one, with unique trading and preparation outcomes. Challenger entry rises, and so does the probability that a challenger from the high-value class acquires the target. With the information in the price held fixed, a stronger incumbent can only lower entry. Between the two economies, informative and uninformative equilibria coexist, and along the informative branch entry rises and then stays constant. When the investor pays for research, research and entry rise together over a range of strengths. In the strong-incumbent economy, access to prices raises expected target proceeds and acquisition surplus net of preparation costs.

1 Introduction

At fixed information, a stronger incumbent reduces a prospective challenger's expected return from entry into a takeover contest. The challenger must pay preparation costs before it can submit an executable offer, and a stronger rival lowers what that preparation earns (Fishman, 1988; Hirshleifer and Png, 1989). I study a force in the opposite direction. Competition can make the target's stock more informative about the challenger, and when a potential buyer observes that price before it commits to acquisition preparation, the information effect can outweigh the loss of acquisition profits. A stronger incumbent then attracts another buyer and raises the probability that a challenger from the high-value class acquires the target.

The setting is a listed target with a public sale opportunity. One buyer is already prepared, and another has not committed to the diligence that an executable proposal requires; Section 2.1 describes the decision interval in which that buyer decides while the target's stock trades. Incumbent strength is the public distribution of the prepared bidder's value; a stronger incumbent shifts it upward, and the challenger decides before any offer is announced.

The challenger and target shareholders hold two claims on the same acquisition surplus: the challenger keeps what remains if it prepares, bids, and wins, and target shareholders hold the traded claim, what the cash second-price auction pays them. The challenger's value comes from a high-value or a low-value quality class, and the classes fix the expected value gap between them. A stronger incumbent outbids the challenger more often and then pays the challenger's bid, so more of the gap reaches target proceeds; a challenger that wins pays more, so less of the gap remains as acquisition profit at every belief. Target shares therefore become more sensitive to information about the challenger at the same time as the challenger's own return to preparation falls.

An outside specialist investor with no net position in the target knows the challenger's class and trades on that sensitivity. The trading stage is one order slot, with the alternative-arrival structure of Dow and Gorton (1997): with some probability the investor fills the slot, and otherwise a liquidity trader fills it with a sale, a purchase, or no order. No one observes who placed the order, and a liquidity order can match the investor's direction, so no order reveals the class. Under every investor strategy the posterior after every order stays inside fixed bounds, and the investor keeps a residual advantage after every order; a larger gap in target proceeds can make that advantage worth the cost of trade. Target shareholders gain when the price recruits entry, because a holder through the sale receives sale proceeds, not the interim price.

The main result compares three economies on a nonempty open set of parameters: one with a weak incumbent, one with a strong incumbent, and one with a very strong incumbent against which recruitment stops. In each economy the equilibrium outcome is unique. With the weak incumbent, the investor's residual advantage cannot cover the trading cost, the price reveals nothing, and only a

zero-cost challenger prepares. With the strong incumbent, the investor buys after good news and sells after bad news, and a purchase brings in a high-cost challenger, so entry rises even though acquisition profit falls at every fixed belief, and under a condition on the high class's chance to win, which the benchmark meets, the probability that a challenger from the high-value class owns the target rises with it. With the very strong incumbent, the price stays informative, but even the most favorable price fails to justify preparation at the high cost, so entry returns to its floor.

The reversal comes from the change in information that competition induces. Market makers anticipate the preparation that each price induces, so the reversal holds under rational pricing; when the investor's orders stay fixed at their informative level while the incumbent strengthens, a stronger incumbent can only lower entry (Section 4.2).

In the benchmark of Section 4.2, entry rises from 0.250000 at the weak incumbent to 0.522757 at the strong incumbent, and high-class ownership rises from 0.125000 to 0.324401. Over the same change, gross acquisition profit at the prior falls from 4.804167 to 4.291667, while the target-payoff spread rises from 0.016667 to 0.666667. In the benchmark, high-cost preparation stops above the entry ceiling, the strength $r_C \approx 3.592659$. At the benchmark auction parameters, the hypotheses of the main result hold exactly when $82/59 < u < 2686/1559$, where u is the informed presence mass relative to the liquidity mass in one direction. Equivalently, the top posterior at the prior $1/2$ lies between the preparation thresholds of the strong and the very strong incumbent, $141/200$ and $4245/5804$. Section 4.2 reports this region and the range of priors in a table.

Informative and uninformative equilibria coexist on an interval of strengths (Section 4.3), and Appendix A.4 lists the complete equilibrium set along the benchmark path. In the benchmark that interval runs from $r_A \approx 1.393958$ to $r_N \approx 1.747878$. Along the informative branch, when the class is high the investor first mixes between a purchase and abstention, and entry rises with strength; from $r_F \approx 1.467127$, the investor buys for sure in that class, and entry stays constant on the rest of the interval. The same start of coexistence holds for a window of priors around the benchmark.

When the investor pays for research that can make it informed, research and entry rise together at intermediate strengths (Section 5). The investor chooses the probability of success up to a cap, at a quadratic cost. The unique outcome has no research with the weak incumbent and research at the cap with the strong incumbent. At the strengths 1.55, 1.60 and 1.65, research rises from 0.237901 through 0.279037 to 0.316683, and entry from 0.507630 through 0.513965 to 0.519763. Along this branch the comovement holds for every liquidity probability of the model, and the branch coexists with an equilibrium without research.

The rise in entry survives other laws of preparation costs, with or without atoms (Section 6.1). For every prior and every pair of classes that satisfy the class ranking and differ above the reserve, the reversal holds at some incumbent strengths and costs (Section 6.2); with more than two classes, it holds under a separation condition. It also survives private signals that the buyer and the investor

hold about different facts (Section 6.3); in an example with its own costs and strengths, the buyer's signal is more accurate than the investor's, and the rise in that example needs both signals.

In the strong-incumbent economy, access to prices raises expected target proceeds and acquisition surplus net of preparation costs (Section 7.1). Sale terms shape both what a buyer pays when it wins and what the stock price can reveal before anyone prepares. An alternative bargaining institution shows how the payment rule matters: the effect of competition on the target-payoff spread depends on the weight the rule places on the runner-up's value (Section 7.2). A reserve that excludes the low-value class can make the price informative against both incumbents, and if informed trade is then full and only a purchase recruits at both strengths, entry is flat in incumbent strength (Section 7.3).

A purchase raises the price by more than a sale lowers it, both measured from the price after no order, exactly when the purchase recruits the buyer (Section 8). The gap comes from the entry component of the spread between purchase and sale prices; in the benchmark, that component is 90.2% of the spread. Above the entry ceiling, orders no longer predict entry, but they still predict which bidder wins; given the entry floor and the target-payoff spread, the sale impact and the volume of orders identify the informed share.

The paper belongs to the literature on the real effects of financial markets (Bond et al., 2012). The closest antecedent is Dow et al. (2017), where a firm's investment decision feeds back into the incentive to produce information about its stock; here the sale rule divides acquisition surplus between a traded claim and a buyer that decides whether to enter. Edmans et al. (2015) show that corrective real decisions can discourage trade on bad news under rational pricing; here the real decision is the buyer's entry itself.

Firms learn from peers' stock prices (Foucault and Frésard, 2014), and target prices affect takeover activity (Edmans et al., 2012); in this model, a prospective challenger learns about its own acquisition value from the target's price before it commits.[1]

The auction-entry literature makes the bidder pool endogenous (Samuelson, 1985; McAfee and McMillan, 1987; Levin and Smith, 1994; Gentry and Stroup, 2019). Under selective entry, a bidder's pre-entry information indexes the distribution of its value, and the payment of the entry cost reveals the value (Roberts and Sweeting, 2013).[2] Here the pre-entry information is the target's stock price, a public price that moves with the orders of an investor who observes the challenger's quality class. I keep the direct deterrence force of the auction-entry literature and add a market that operates before preparation, with the informed trader outside the auction and trading a payoff that

[1] Bidders also learn from announcement returns when they decide whether to complete a deal (Luo, 2005). That decision comes after the entry margin of this paper.

[2] One limit of this framework is a bidder that knows only the common value distribution before entry (Levin and Smith, 1994). The other limit is a bidder that already knows its value (Samuelson, 1985). Selective entry also affects the choice of sale procedure (Roberts and Sweeting, 2013), and auction formats affect bidders' incentives to acquire information (Persico, 2000).

differs from the entrant's profit.[3]

The investor in this paper is an outside trader and earns only the trading residual. An informed trader can also hold a stake before a control decision: an acquirer can use liquidity trade to conceal share purchases before a bid (Kyle and Vila, 1991), large holders trade on their information (Edmans, 2009), and arbitrageurs' positions can affect tendering (Cornelli and Li, 2002).[4]

Section 2 sets out the model, and Section 3 derives the returns to information. Section 4 states the main results. Sections 5 to 8 cover research, robustness, welfare and sale terms, and empirical implications, and Section 9 concludes. Appendix A states the full form of Proposition 3 and the results of Sections 5 to 8 in full, and proves every result.

2 The model

2.1 The decision interval

The model applies to a listed target whose sale opportunity is publicly understood while its stock still trades and while a further bidder can still come in. A disclosed approach, an announced strategic review, or an open contest can create such an interval if the opportunity becomes public before the buyer set is fixed; only then can a prospective buyer watch the price during the period that matters. Section 8 states how to date this interval in a given transaction.

Within the interval, one buyer is already prepared. It may be the party that approached the target or a bidder that has completed diligence, and its strength is what the public can infer about its acquisition value from the disclosed record. Another buyer, the challenger, has not decided whether to prepare. Preparation is required for an executable proposal: before a buyer can submit a binding offer, it must verify the target's contracts, liabilities, and technology, arrange financing and approvals, and commit evaluation capacity that has other uses. Those costs are sunk once incurred, and they differ across buyers, since prior familiarity, capacity already in place, or a small deal relative to the buyer's size make preparation cheap for some buyers and expensive for others. A challenger that does not prepare stays out of the sale, so its cost is an entry cost: payment reveals its value and permits a bid.

The investor and the challenger know different things. The challenger knows its own integration

[3] Recent work studies how bidders learn about their own values and about competitors (Pernoud and Gleyze, 2026), post-auction feedback in security-payment design (Liu and Bernhardt, 2022), and bidder-pool choice with correlated values (Carlin et al., 2026). In takeovers, Betton et al. (2014) study negotiations with stock-market feedback, and Lin et al. (2025) model payment choice and trade within an initiated deal; I hold cash consideration fixed and study entry.

[4] Large holders also monitor or intervene as prices feed back to firm decisions (Edmans et al., 2015, Section II.E; Maug, 1998; Back et al., 2018). Goldstein and Guembel (2008) study manipulation through the real decision that responds to the price. The model abstracts from toeholds and dispersed-shareholder free riding (Bulow et al., 1999; Grossman and Hart, 1980).

plans, while a specialist investor may know what the target's customers do, how its technology compares with alternatives, or how its product demand will change. These are different facts about the same acquisition match, so a more accurate signal about one can leave the other unknown. The model gives the investor exact knowledge of the challenger's quality class, though not of its value, and suppresses the challenger's initial information; Section 6.3 removes both extremes, and Section 5 makes the investor's information the result of costly research. The identity of the challenger is common knowledge, and the market is uncertain only about how well the challenger fits the target.

Public incumbent strength is a distribution, not a bid. The record can reveal that a strategic acquirer with clear synergies is at the table, or that the prepared party is a financial buyer with a leveraged capital structure, and in each case it need not reveal what either buyer would pay; strategic and financial bidders differ systematically in their valuations (Gorbenko and Malenko, 2014). The model varies that distribution and keeps the incumbent's realized value private until it bids. Disclosure records also document a costly preparation stage: Imprivata's definitive proxy separates an unsolicited approach, outreach to potential buyers, and indications of interest conditional on further diligence (Imprivata, Inc., 2016), and takeover competition develops before public bids (Boone and Mulherin, 2007; Gentry and Stroup, 2019). Section 8 states how to test whether prices draw buyers into that preparation stage.

2.2 Values, preparation, and the sale

The target's known standalone value is normalized to zero, and acquisition values and preparation costs are measured per target share. A common standalone component added to every ownership outcome shifts prices and payoffs by a constant and leaves incentives unchanged.

All strategic agents are risk neutral. The target share is the only traded claim with information about the acquisition match, and neither bidder trades it. The sale mechanism binds all target shares, so shareholder tendering and holdout stay outside the model. The incumbent's value distribution is public when the sale opportunity becomes visible.

Two potential acquirers face the target. The incumbent has already prepared, incurs no further entry cost, and knows its value before it bids. Its acquisition value R has a continuous distribution F on $[0, \bar{r}]$, conditional on public information when the sale process begins, and F is extended by unity above $\bar{r}$. A first-order stochastic shift of F strengthens the incumbent. In the main case R is uniform on $[0, r]$, and an increase in r strengthens the incumbent.

The challenger belongs to one of two quality classes, $j \in \{H, L\}$, with prior $\pi = \Pr(j = H) \in (0, 1)$. Given its class, its acquisition value θ has a distribution G_j on $[0, \infty)$ with a finite mean; atoms are allowed. The probability that a class-j value falls below the reserve p is $G_j(p^-)$, and $D(y) = G_L(y) - G_H(y)$ is the difference between the two class distributions at y.

The high class, or high-value class, dominates the low class in the sense of first-order stochastic dominance. Both class supports lie above the reserve, so that every value meets it:

$$G_H(y) \le G_L(y) \quad \text{for every } y, \qquad G_H(p^-) = G_L(p^-) = 0. \tag{1}$$

I call the two restrictions in (1) the class ranking.

Under the class ranking (1), competition widens the class difference in target proceeds under every incumbent distribution (Proposition 1). Appendix A.1 characterizes the weakest restriction of this kind: when neither class has a value atom at the reserve, the spread rises under every strengthening exactly when the high class dominates above the reserve and the two classes are equally likely to fall below it, which I call equal admissibility. Section 7.3 studies a reserve that excludes the low class, where equal admissibility fails. The structure is the selective-entry structure of the auction literature: the class indexes the distribution of the challenger's value, preparation reveals the value, and the stock price is the pre-entry signal.

The benchmark is the point-mass case: G_H puts all its mass on h, G_L puts all its mass on ℓ, and $\pi = 1/2$. In the benchmark the class and the value coincide, and values, the reserve, and the uniform incumbent's strength satisfy

$$0 < p < \ell < r < h. \tag{2}$$

The general statements need no normalization of money: a common positive rescale of every value, the reserve, and every preparation, trading and research cost rescales prices and payoffs by the same factor, while orders, beliefs, entry, and ownership do not change. The benchmark measures money in units of the low value, so $\ell = 1$ there.

The challenger initially knows neither its class nor its value. After it observes the stock price, it privately learns its preparation cost $C \in \{0, c_H\}$, with $\Pr(C = 0) = \rho$, $0 < \rho < 1$, and $c_H > 0$. The low cost is zero: a zero-cost challenger is a ready buyer whose diligence costs nothing at the margin. A payment of C reveals θ and permits a bid, and a challenger that does not pay stays outside the sale.

The pair (j, θ), the incumbent's value, the preparation cost, and the draws of the order slot in Section 2.3 are mutually independent. The cost therefore carries no information about the class or the value, and conditional on the class, the challenger's value is independent of the order.

Before trade, the seller publicly commits to a cash second-price auction with reserve $p > 0$. The highest admissible bidder acquires the target and pays the larger of the reserve and the highest competing admissible bid. A bid equal to the reserve is admissible, and there is no sale without an admissible bid. Each prepared bidder whose value meets the reserve bids its value; this bid is weakly dominant (Appendix A.1).

A prepared challenger whose value is below the reserve does not bid. Without a challenger,

the incumbent buys at the reserve when its value meets it, so expected no-entry proceeds are $t_0 = p[1 - F(p)]$: the board collects its announced walk-away price from a lone prepared bidder whose value meets it. Because F is continuous, acquisition-value ties have probability zero.

2.3 Trading: the order slot

An investor observes the challenger's class j. It is an outside sector specialist, such as a sector-focused long-short fund. It holds no net position in the target before trade, has no control rights, and cannot acquire the target. Its information concerns the challenger's fit with the target, and it forms that view from its own research on the target's customers, technology, and demand.

Trade takes place in one order slot, a single public order opportunity that comes before the challenger's decision. With probability $\lambda \in (0, 1)$ the informed investor fills the slot: it sees the class j and chooses an order $q \in \{-1, 0, 1\}$, and a nonzero order costs $k > 0$. With probability $1 - \lambda$ a liquidity trader fills the slot instead and, for exogenous reasons apart from the sale, sells, abstains, or buys one unit with probabilities β, $1 - 2\beta$, and β, where $0 < \beta \leq 1/3$.[5] A liquidity sale comes from a current shareholder who sells for cash, a liquidity purchase from an outside investor, and an abstention means that the opportunity passes with no order. No one observes who filled the slot, and the identity draw is independent of the class, the costs, and the incumbent's value.

Competitive market makers see the order $X \in \{-1, 0, 1\}$, set the price $P = \mathbb{E}[V_T \mid X]$, where V_T is the terminal payoff of a target share, and absorb the order at that price; each share held at the sale, market-maker inventory included, receives V_T. If the investor fills the slot with order q, its order is the whole order that market makers see, so it trades at the price $P(q)$ and earns

$$q\,[V_T - P(q)] - k|q|. \tag{3}$$

The unit is a normalized small trading unit. The cost k is a separate friction, such as an execution cost or a cost to carry the position, and it is distinct from adverse-selection price impact, which the competitive price already generates. The investor's order after bad news is a short sale, and k includes the cost to borrow shares.

A liquidity order in a given direction arrives with probability $b = (1 - \lambda)\beta$, and the ratio $u = \lambda/b$ is the informed presence mass relative to the liquidity mass in one direction; the number $1 + u$ is the largest likelihood ratio that one order can carry. Section 3.2 shows that, under every investor strategy, the posterior after every order lies between a bottom posterior $m(\pi)$ and a top posterior $M(\pi)$:

$$M(\pi) = \frac{\pi(1 + u)}{\pi(1 + u) + 1 - \pi}, \qquad m(\pi) = \frac{\pi}{\pi + (1 - \pi)(1 + u)}. \tag{4}$$

[5] Only the posterior bound at the zero order uses $\beta \leq 1/3$, and through it the ceiling of Section 3.2; Propositions 2 and 3 hold for every $\beta < 1/2$ (Appendix A.2).

The bounds follow from one property of the slot: each class reaches each order through a liquidity trader, so the investor keeps a residual advantage after every order.

The slot uses the alternative-arrival structure of Dow and Gorton (1997), in which an informed order and a liquidity order of the same size are alternative arrivals in one opportunity; in Admati and Pfleiderer (2009), a liquidity shock forces the informed holder's own sale, so a sale occurs on path for both motives. The unit order menu, the presence probability, and the symmetric unit cost follow Edmans et al. (2015).[6] Risk-neutral market makers with zero expected profit price each order, as in Glosten and Milgrom (1985) and Kyle (1985). Here the terminal payoff depends on the price through preparation.

Market makers observe the order X and nothing else. The challenger observes the price P and its own cost C, but not X, its class, its value, or the incumbent's value; the investor observes the class j but not the value θ, and the incumbent knows its own value. The sequence is as follows. The seller announces the sale rule. The investor learns the class, the slot draw selects the investor with probability λ and a liquidity trader otherwise, and the selected trader places its order. Market makers set the price from the order, and the challenger observes the price and its cost and decides whether to prepare. Prepared bidders then learn their values and submit their bids, and ownership and financial payoffs are realized. Section 5 adds a research stage before the investor learns the class.

2.4 Equilibrium

An equilibrium specifies, for each class, a distribution over the three orders; the market makers' posterior $\mu(x) = \Pr(j = H \mid X = x)$ and price $P(x)$ after each order x; the challenger's belief at each price; its preparation choice at each price and cost; and truthful bids. The investor can mix, and wrong-signed orders are allowed. Every order has positive probability in both classes, because a liquidity trader can place it, so Bayes' rule gives a posterior after every order. Prices are competitive. A high-cost challenger prepares when gross profit at its belief exceeds c_H and stays out when it falls short, and a zero-cost challenger, whose gross profit is an expected auction rent and never negative, prepares after every price. The investor's order maximizes

$$U_j(q) = q\left[t_0 + e(q)(t_j - t_0) - P(q)\right] - k|q|$$

[6]The unit menu $\{-1, 0, 1\}$ keeps every order on path, so no off-path belief enters the analysis. A menu of continuous order sizes needs a convention for the sizes that no class uses in equilibrium, as in Edmans et al. (2015). Another route keeps every size on path through additive liquidity flow: with Laplace noise of scale s and full unit orders, the largest likelihood ratio of a favorable flow is $\mathrm{e}^{2/s}$, where e is Euler's number, printed upright to separate it from entry $e(x)$, so the top posterior equals the slot's $M(\pi)$ when $1 + u = \mathrm{e}^{2/s}$. The two set-ups then share the posterior bounds, and they differ in the probability with which each class reaches the top posterior.

at the candidate's prices and entry, where $e(q)$ is entry after the order q, the probability that the challenger prepares after the price $P(q)$; t_j is expected target proceeds after entry in class j, and t_0 is expected target proceeds without entry. Definition A.1 in Appendix A states the six conditions in full.

For a known challenger value θ, expected target proceeds after the challenger prepares are $t(\theta)$, and the challenger's expected gross acquisition profit is $g(\theta)$; as before, t_0 is expected target proceeds without entry. The class objects are expectations over the class distribution, $t_j = \mathbb{E}_j[t(\theta)]$ and $g_j = \mathbb{E}_j[g(\theta)]$ for $j \in \{H, L\}$, and the target-payoff spread is $\Delta_T = t_H - t_L$. Bids are truthful, so these objects depend on the incumbent's distribution and the reserve but not on any belief. At a belief $\mu = \Pr(j = H)$, the challenger's expected gross profit is $B(\mu) = g_L + \mu\,(g_H - g_L)$, because its value depends on the price only through its class. When the incumbent's distribution matters, I write $g_j(F)$, $\Delta_T(F)$, and B_F; for the uniform incumbent on $[0, r]$, I write $g_j(r)$, $\Delta_T(r)$, and B_r.

The investor's objective takes the expected terminal payoff of a target share given class j and order q as $t_0 + e(q)(t_j - t_0)$, because entry depends only on the price and the cost, and the cost is independent of the class and the order. The same independence, with Bayes' rule, turns the competitive price into

$$P(x) = t_0 + e(x)\left[t_L - t_0 + \Delta_T\,\mu(x)\right]. \tag{5}$$

The ex ante payoff of the investor multiplies U_j by λ, and this factor leaves the best order unchanged.

When ties arise, an indifferent investor may mix over its best orders, and an indifferent high-cost challenger prepares; Table A.3 in Appendix A.4 records each conclusion that another tie rule changes.

A unilateral deviation by the investor changes only its own order and faces the candidate equilibrium's price function and entry rule, so the deviation payoff is $U_j(q)$ read at the candidate's $P(q)$ and $e(q)$. A comparison across economies, such as across incumbent strengths, solves for the equilibrium again in each economy.

Entry E is the probability that the challenger prepares, and high-class ownership O_H is the probability that a challenger of class H acquires the target. The probability that a prepared challenger of class H wins against the incumbent distribution F is $W_H(F) = \mathbb{E}_H\left[F(\theta)\mathbf{1}\{\theta \geq p\}\right]$.

3 Competition and information

3.1 Acquisition profits and target value

The auction determines both the return to preparation and the claim that informed investors trade. Consider a prepared challenger with value $\theta \geq p$. If it wins, it pays $\max\{p, R\}$; if it loses, the incumbent pays θ. The target therefore receives $\max\{p, \min(R, \theta)\}$, and the challenger keeps

$(\theta - \max\{p, R\})_+$. A prepared challenger with $\theta < p$ cannot bid, and the target then receives the reserve if the incumbent's value meets it, exactly as without entry. For every value at or above the reserve, the two expected amounts add up to the value:

$$t(\theta) + g(\theta) = \theta, \qquad \theta \geq p.$$

The second-price rule passes the whole value of an admissible challenger to the target and the challenger together. An admissible value also raises proceeds to at least the reserve, so $t_j \geq t_0$ for both classes.

In the point-mass benchmark with the uniform incumbent, integration over R gives

$$\begin{aligned} t_0 &= p\Big(1 - \frac{p}{r}\Big), & t_H &= \frac{r}{2} + \frac{p^2}{2r}, \\ t_L &= \ell - \frac{\ell^2 - p^2}{2r}, & g_H &= h - \frac{r}{2} - \frac{p^2}{2r}, \\ g_L &= \frac{\ell^2 - p^2}{2r}, & \Delta_T(r) &= \frac{(r-\ell)^2}{2r}. \end{aligned} \tag{6}$$

The relevant derivatives are

$$\Delta_T'(r) = \frac{1}{2} - \frac{\ell^2}{2r^2} > 0, \qquad g_H'(r) = -\frac{1}{2} + \frac{p^2}{2r^2} < 0, \qquad g_L'(r) = -\frac{\ell^2 - p^2}{2r^2} < 0. \tag{7}$$

Thus a stronger incumbent lowers acquisition profit at every fixed belief while it makes target proceeds more sensitive to challenger quality. Proposition 1 extends this opposition to every continuous incumbent distribution and every pair of classes that satisfies the class ranking (1).

Proposition 1 (Competition and the two returns to information)**.** *Let the incumbent's value have a continuous distribution F on $[0, \bar{r}]$, extended by unity above $\bar{r}$. For any classes G_H, G_L with finite means,*

$$\begin{aligned} \Delta_T(F) &= pF(p)D(p^-) + \int_p^\infty [1 - F(y)]\, D(y)\, dy, \\ g_j(F) &= \int_p^\infty [1 - G_j(y)]\, F(y)\, dy, \qquad j \in \{H, L\}. \end{aligned} \tag{8}$$

Let $\tilde{F} \leq F$ be a continuous first-order stochastic strengthening of F.

(a) For every pair of classes, the strengthening weakly decreases g_H and g_L, and hence gross acquisition profit at every fixed belief.

(b) Under the class ranking (1), the strengthening weakly increases the target-payoff spread.

The decrease in g_j is strict when $\int_p^\infty [1 - G_j(y)][F(y) - \tilde{F}(y)]\, dy > 0$. The increase in (b) is strict

when $\int_p^\infty [F(y) - \tilde{F}(y)]\, D(y)\, dy > 0$.

At one incumbent value R that meets the reserve, the target receives $\min\{R, \max\{\theta, p\}\}$, and the expected class difference in proceeds is $\int_p^R D(y)\, dy$. Each slice $D(y)\, dy$ of the class value gap at a level $y > p$ reaches the target when the incumbent outbids the challenger at that level and pays its bid, and it stays with the challenger otherwise.

A stronger incumbent moves slices to the target, and the spread rises while acquisition profit falls. The reserve adds one term: below the reserve, only an admissible challenger buys, at p, and the class difference is $pD(p^-)$. A stronger incumbent fails the reserve less often, so that term shrinks, and it can outweigh the moved slices when the low class is more often inadmissible. Under the class ranking (1), every value meets the reserve, and the term is zero. Appendix A.1 gives the proof.

The sum of the two lines of (8) under the class ranking (1) is

$$\Delta_T(F) + g_H(F) - g_L(F) = \mathbb{E}_H[\max\{\theta, p\}] - \mathbb{E}_L[\max\{\theta, p\}].$$

The identity holds for every incumbent distribution, and in the benchmark the right side is $h - \ell$ at every r. A stronger incumbent leaves the value gap between the classes unchanged; it moves the gap from the challenger's profit to target proceeds.

The spread approaches zero as r approaches ℓ, since the incumbent then almost never outbids a low-value challenger. As r increases, the spread rises while gross acquisition profit falls at each belief. Whether entry rises depends on how this change in payoffs affects the information in the price.

3.2 What the challenger learns from the price

The challenger sees one price. A liquidity order in the same direction is always possible, and this bounds what any order can reveal. Each class reaches a given nonzero order through a liquidity trader with probability b, and the informed investor adds at most λ, so the likelihood ratio of the order lies between $1/(1+u)$ and $1+u$. By Bayes' rule, the posterior after every order lies in the interval $[m(\pi), M(\pi)]$ of (4), under every investor strategy, mixed or pure; when the high class buys and the low class sells, the posteriors after a sale, no order, and a purchase are $m(\pi)$, π, and $M(\pi)$. The challenger's belief at a price averages the posteriors of the orders with that price, so it obeys the same bounds. No price can therefore support a preparation decision that needs a belief above the top posterior. This gives the ceiling of the model: when $g_H \geq g_L$, gross profit does not fall with the belief, so once gross profit at the top posterior falls below c_H, no order recruits a high-cost challenger, in or out of equilibrium.

When $g_H > g_L$, gross profit covers the high cost exactly when the belief reaches the preparation

threshold $\tau = (c_H - g_L)/(g_H - g_L)$: $B(\mu) \geq c_H$ if and only if $\mu \geq \tau$. For the uniform incumbent I write $\tau(r)$. A zero-cost challenger prepares after every price, so entry after every order is at least ρ, and competitive pricing gives (5). When the classes differ in target proceeds, the price rises strictly with the posterior. Suppose two orders have one price. Entry after them is the same, because entry depends on the order only through the price, and at least ρ; the price formula then gives the two orders the same posterior. The challenger therefore reads the market makers' posterior from the price, and its gross profit at the price $P(x)$ is $B(\mu(x))$, so high-cost preparation follows exactly the orders whose posterior reaches τ. Appendix A.2 gives the formal statements and their proofs.

The zero-cost challenger gives each step its floor: entry stays at least ρ, so the price reveals the posterior, and the gross return of a correctly signed order stays at least a fixed share of the spread, the trading bound of Section 3.3. Its preparation does not respond to the price, so information can only add entry.

3.3 Informed trading

Rational pricing removes the anticipated increase in proceeds from the investor's advantage. What remains is the gross return of a purchase in class H and of a sale in class L,

$$K_H = e(1)\,\Delta_T\,[1 - \mu(1)], \qquad K_L = e(-1)\,\Delta_T\,\mu(-1), \qquad \rho\,\kappa(\pi)\,\Delta_T \leq K_H,\, K_L \leq \Delta_T,$$

with $\kappa(\pi) = \min\{m(\pi), 1 - M(\pi)\}$. A wrong-signed order earns at most $-k$, so it never pays. Class H buys when K_H exceeds k and abstains when it falls short, and class L compares K_L with k in the same way.

Market makers price the entry response, and the posterior bounds of Section 3.2 leave some uncertainty about the class after every order. The investor's gross return is the entry probability times the spread times that residual uncertainty. The lower bound uses the entry floor ρ and the posterior bounds, and the upper bound is the spread. Both bounds hold under every candidate profile, mixed or pure, and their comparison with k decides when information survives in equilibrium.

Since $1 - M(\pi) = m(1 - \pi)$, the bound takes the form $\kappa(\pi) = m(\min\{\pi, 1 - \pi\})$. It is largest at the prior $1/2$, where it equals $m(1/2) = 1/(2 + u)$, and smaller at every other prior. The class that the prior favors binds, because the price already leans toward it.

The no-entry proceeds t_0 cancel from each return, because they move the price and the terminal payoff together; the terms that matter are the entry gains $t_H - t_0$ and $t_L - t_0$, which are nonnegative.

Each return depends only on the posterior and entry after its own order, so entry after the zero order enters no payoff, and the investor's problem splits by class. Appendix A.2 gives the formal statement and proof of the trading bound, and Appendix A.4 uses this split to list every equilibrium along the benchmark path.

4 Equilibrium results

4.1 Competition creates competition

The main result compares three economies that differ only in incumbent strength. Against the weak incumbent the price carries no information and only a zero-cost challenger prepares; against the strong incumbent the price is informative and a purchase recruits a high-cost challenger; against the very strong incumbent the price stays informative, while entry returns to its floor. I write F_0, F_1 and F_2 for the three incumbent distributions.

Proposition 2 (Competition creates competition)**.** *Let the classes satisfy the class ranking* (1)*, let F_1 be a continuous first-order stochastic strengthening of the continuous incumbent distribution F_0, and let F_2 strengthen F_1 in the same sense. Let $0 < \pi < 1$, $0 < \rho < 1$, $k > 0$ and a slot (λ, β) be given, and suppose that the high cost lies between gross profit at the prior against F_0 and gross profit at the top posterior against F_1, that the trading cost lies between the target-payoff spread at F_0 and the fraction $\rho\kappa(\pi)$ of the spread at F_1, and that gross profit at the top posterior against F_2 falls short of the high cost:*

$$\begin{aligned} &\textit{(H1)} && B_{F_0}(\pi) < c_H < B_{F_1}(M(\pi)), \\ &\textit{(H2)} && \Delta_T(F_0) < k < \rho\,\kappa(\pi)\,\Delta_T(F_1), \\ &\textit{(H3)} && B_{F_2}(M(\pi)) < c_H. \end{aligned}$$

Each economy then has a unique equilibrium outcome.

(i) The weak incumbent F_0. *The investor does not trade, and both classes abstain with probability one. Every posterior equals π and entry equals ρ after every order, and every order has the price $t_0 + \rho\,[t_L - t_0 + \Delta_T(F_0)\pi]$, so the price carries no information.*

(ii) The strong incumbent F_1. *The investor trades in both directions: class H buys and class L sells, each with probability one. The posteriors after a sale, no order and a purchase are $m(\pi)$, π and $M(\pi)$, and only a purchase recruits the high-cost challenger, so entry after the three orders is ρ, ρ and* 1*, and $P(-1) < P(0) < P(1)$. Entry is*

$$\rho + (1-\rho)(b + \pi\lambda) > \rho.$$

Entry given class H is $\rho + (1-\rho)(b+\lambda)$. High-class ownership rises from $\pi\rho W_H(F_0)$ to $\pi[\rho + (1-\rho)(b+\lambda)]W_H(F_1)$ if and only if

$$[\rho + (1-\rho)(b+\lambda)]\,W_H(F_1) > \rho\,W_H(F_0).$$

(iii) The very strong incumbent F_2. *The investor again trades in both directions, with posteriors $m(\pi)$, π and $M(\pi)$ after a sale, no order and a purchase. Entry equals ρ after every order, and the three prices differ, so the price stays informative while entry returns to ρ.*

The high-cost window (H1) sets entry through acquisition profit: gross profit at the prior falls short of the high cost against F_0, so an uninformative price recruits no high-cost challenger, while gross profit at the top posterior exceeds the high cost against F_1, so a purchase can recruit one. The trading-cost window (H2) fixes trading before any order profile is known: by the trading bound of Section 3.3, every correct order earns at most $\Delta_T(F_0) - k < 0$ against the weak incumbent and at least $\rho\kappa(\pi)\Delta_T(F_1) - k > 0$ against the strong incumbent, so a correct order cannot cover the trading cost in the first economy and covers it in the second. The ceiling (H3) stops recruitment against F_2, where gross profit at the top posterior falls short of the high cost (the ceiling of Section 3.2).

The trading-cost window is nonempty exactly when $\Delta_T(F_0)/\Delta_T(F_1) < \rho\kappa(\pi)$, that is, when competition widens the target-payoff spread enough between the two incumbent distributions. The condition involves the two distributions, the classes, ρ, the order slot and π, and no unit of money, and a prior away from one half narrows the window, since $\kappa(\pi)$ falls when the price already leans toward the favored class. For uniform incumbents, (H1)–(H3) are strict inequalities between continuous functions of $(r_0, r_1, r_2, c_H, k, \rho, \pi, \lambda, \beta)$, so they define an open set; Section 4.2 gives its slice through the benchmark.

The additional entry favors the high class, which places a purchase with probability $b + \lambda$, while the low class places one with probability b. Ownership needs its own condition when the high class can lose to the incumbent, because a stronger incumbent then lowers W_H. The condition in part (ii) holds whenever the stronger incumbent leaves the high class a good chance to win, $(1 + \rho)W_H(F_1) \geq 2\rho W_H(F_0) > 0$, and the high class places a purchase more often than not, $b + \lambda > 1/2$; in the point-mass benchmark $W_H = 1$, so ownership rises.

Part (iii) completes the pattern, in which entry first rises and later falls as the incumbent strengthens. Against the very strong incumbent even the top posterior fails to justify high-cost preparation, while trading stays informative, and Section 4.3 describes the strengths between the three economies. Appendix A.3 states the three equilibrium tuples and gives the proof. Every comparison behind the result is strict, so the unique outcome in each economy is the same however ties are broken.

4.2 The benchmark and the fixed-information control

The benchmark has point-mass classes $h = 10$ and $\ell = 1$ at the prior $\pi = 1/2$, with reserve $p = 1/2$, zero-cost probability $\rho = 1/4$, high cost $c_H = 6$ and trading cost $k = 1/50$. The incumbent is uniform on $[0, r]$, at the strengths $r_0 = 6/5$, $r_1 = 3$ and $r_2 = 18/5$, and money is measured in units

Table 1. Acquisition payoffs in the benchmark.

	$r_0 = 6/5$	$r_1 = 3$	$r_2 = 18/5$
Proceeds without entry, t_0	0.291667	0.416667	0.430556
Proceeds with a high-class challenger, t_H	0.704167	1.541667	1.834722
Proceeds with a low-class challenger, t_L	0.687500	0.875000	0.895833
Target-payoff spread, Δ_T	0.016667	0.666667	0.938889
Gross profit of class H, g_H	9.295833	8.458333	8.165278
Gross profit of class L, g_L	0.312500	0.125000	0.104167
Gross profit at the prior, $B_r(1/2)$	4.804167	4.291667	4.134722
Preparation threshold, $\tau(r)$	0.633117	0.705000	0.731392

Notes. Point-mass classes h and ℓ, reserve p, incumbent uniform on $[0, r]$. The entries follow from equation (6). They do not depend on the slot.

of the low value.

In the benchmark slot the largest likelihood ratio of an order is $1 + u = \mathrm{e}$, and the liquidity probability is $\beta = 0.294655$. The presence probability is therefore $\lambda = u\beta/(1 + u\beta) = 0.336122$, with $b = 0.195615$, and the posterior bounds at the prior are $M(1/2) = \mathrm{e}/(1 + \mathrm{e}) = 0.731059$ and $m(1/2) = 0.268941$; Appendix A.9 lists these declarations and the arithmetic behind each value.

Table 1 reports the acquisition payoffs, and the two forces move in opposite directions from r_0 to r_1: gross acquisition profit at the prior falls from 4.804167 to 4.291667, while the target-payoff spread rises from 0.016667 to 0.666667. The benchmark satisfies (H1)–(H3), so Proposition 2 applies, and entry rises from 0.250000 to 0.522757 while high-class ownership rises from 0.125000 to 0.324401. At r_2 entry returns to 0.250000 and the three prices still differ; Table 3 lists the three equilibrium tuples.

Proposition A.1 in Appendix A.3 gives the admissible region at the benchmark auction. At the prior $1/2$, (H1)–(H3) hold if and only if $82/59 < u < 2686/1559$, or equivalently the top posterior lies between the preparation thresholds $\tau(r_1) = 141/200$ and $\tau(r_2) = 4245/5804$; with $1 + u = \mathrm{e}$ they hold exactly for priors between ≈ 0.467850 and ≈ 0.500424. Below the band a purchase at r_1 does not recruit a high-cost challenger, and above it a conclusion of Proposition 2 fails at r_2 or at r_1. Table 2 gives the region in decimals, with the benchmark point and its distance to each boundary.

Both ends of the band are preparation thresholds: the lower end $\tau(r_1)$ is the belief that a purchase needs to recruit a high-cost challenger against the strong incumbent, and the upper end $\tau(r_2)$ is the belief at which a purchase would still recruit one against the very strong incumbent. The threshold rises slowly with strength, and this sets the width of the band: a high-class challenger outbids every incumbent on the path, and its expected payment $r/2 + p^2/(2r)$ rises by less than one half per unit of strength, a change that is small next to the gap $g_H - g_L$ in Table 1.

Gross profit at every belief falls strictly with strength above ℓ, so the ceiling (H3) holds at every

Table 2. The admissible region at the benchmark auction.

Parameter	Region	Benchmark	Margin to the boundary	
			lower	upper
Mass ratio u	$(1.389831, 1.722899)$	1.718282	0.328451	0.004617
Top posterior $M(1/2)$	$(0.705000, 0.731392)$	0.731059	0.026059	0.000334
Presence λ at $\beta = 0.294655$	$(0.290539, 0.336721)$	0.336122	0.045583	0.000599
Presence λ at $\beta = 1/3$	$(0.316602, 0.364797)$			
Prior π, all three parts	$(0.467850, 0.500424)$	1/2	0.032150	0.000424
Prior π, parts (i) and (ii)	$(0.467850, 0.633117)$	1/2	0.032150	
Liquidity probability β	$(0, 1/3]$	0.294655		0.038678

Notes. Benchmark auction and costs, strengths r_0, r_1, r_2. Rows 1 to 4 fix the prior at $1/2$; rows 5 and 6 fix $1 + u = \mathrm{e}$. Each region is the exact open set on which (H1)–(H3) hold, or (H1)–(H2) in row 6. Decimal endpoints are rounded to six places; Proposition A.1 gives the exact endpoints.

strength in (r_C, h), where the entry ceiling $r_C = 3.592659$ is the strength at which $\tau(r) = M(1/2)$; parts (i) and (ii) of Proposition 2 hold without the ceiling. Table A.6 in Appendix A.9 records every hypothesis and its positive margin at the benchmark.

Figure 1 shows the preparation threshold along the path: the prior lies below it at every strength, so an uninformative price never recruits a high-cost challenger, and the threshold rises with strength until it meets the top posterior at the entry ceiling r_C, just below r_2. Panel (b) shows the band of Table 2 for the top posterior (Proposition A.1).

Holding the information in the price fixed while the incumbent strengthens removes the reversal. The experiment fixes the law of the class and of a signal that the challenger observes, the class value laws and the cost law. Given the class, the signal tells the challenger nothing more about either value. Its own value depends on the signal only through the class; the incumbent's value is independent of the class and the signal, and the two values are independent given the class and the signal. The cost is independent of the class, both values and the signal. A stronger incumbent then can only lower entry, for every such experiment and every independent cost law (Proposition A.3 in Appendix A.6 gives the formal condition): it lowers gross profit at every belief, so it can only remove a preparation incentive, and entry falls by exactly the probability that the cost lies between gross profit at the two strengths.

In the order slot the experiment fixes λ, β, π, the cost law and both order distributions, and competitive prices then reveal the same posterior after each order at every strength (Appendix A.2).

Panel B of Table 3 holds full orders fixed at both strengths. Only the top posterior can recruit a high-cost challenger, because the preparation threshold $\tau(r)$ exceeds the prior at every strength, so entry is 0.522757 at every strength up to the entry ceiling $r_C = 3.592659$ and ρ above it. Entry falls by the probability that the cost lies between gross profit at the two strengths, which is 0 from r_0 to r_1

Table 3. Equilibrium outcomes and information controls in the benchmark.

	$r_0 = 6/5$	$r_1 = 3$	$r_2 = 18/5$
Panel A. Equilibrium (Proposition 2)			
Informed orders, classes H and L	$(0, 0)$	$(1, -1)$	$(1, -1)$
Posterior after a sale	0.500000	0.268941	0.268941
Posterior after no order	0.500000	0.500000	0.500000
Posterior after a purchase	0.500000	0.731059	0.731059
High-cost preparation	none	after a purchase	none
Price after a sale	0.392708	0.576074	0.610002
Price after no order	0.392708	0.614583	0.664236
Price after a purchase	0.392708	1.362372	0.718471
Entry E	0.250000	0.522757	0.250000
High-class ownership O_H	0.125000	0.324401	0.125000
Expected target proceeds	0.392708	0.872531	0.664236
Panel B. Full orders held fixed (Proposition A.3)			
Entry E	0.522757	0.522757	
High-class ownership O_H	0.324401	0.324401	
Expected target proceeds	0.503998	0.872531	
Panel C. The challenger does not see the price			
Informed orders, classes H and L	$(0, 0)$	$(1, -1)$	
Entry E	0.250000	0.250000	
High-class ownership O_H	0.125000	0.125000	
Expected target proceeds	0.392708	0.614583	

Notes. The benchmark. Panel A: the unique equilibrium at each strength. Panel B: full orders imposed at both strengths; at r_0 this profile is a control, not an equilibrium. Panel C: the unique equilibrium when the challenger decides on its prior. At r_2 the full-order profile is the equilibrium of Panel A.

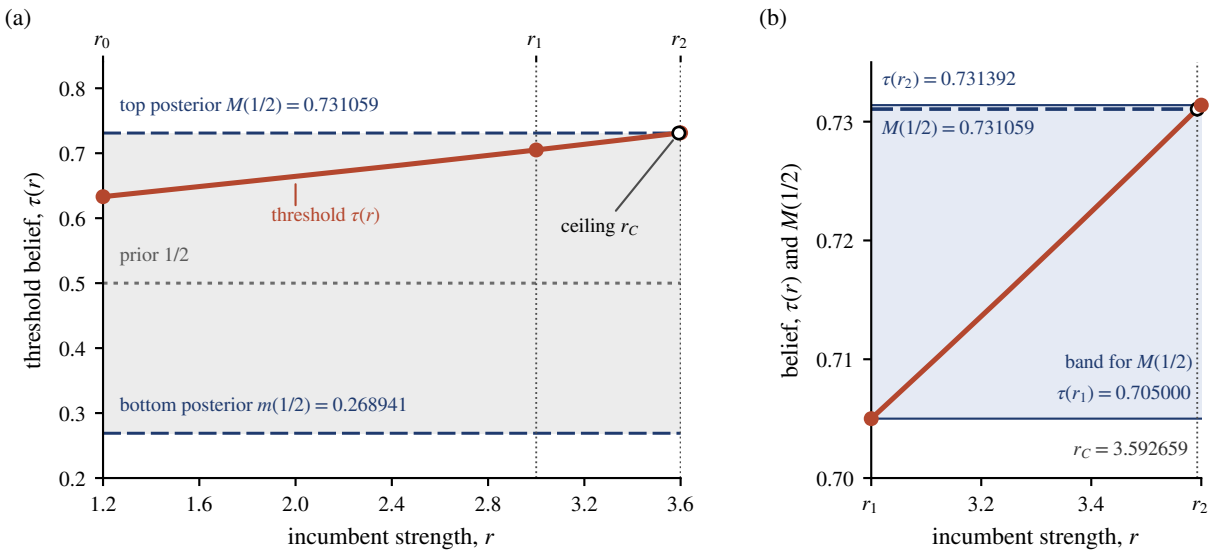


Figure 1. Beliefs and the entry ceiling in the benchmark.

Notes. Panel (a) plots the threshold belief $\tau(r)$ against incumbent strength r. The gray band covers the posteriors that an investor strategy can produce, from $m(1/2)$ to $M(1/2)$. The entry ceiling r_C is the strength at which $\tau(r)$ equals $M(1/2)$. Panel (b) enlarges the strengths from r_1 to r_2. Its blue band holds every top posterior $M(1/2)$ that puts r_C between r_1 and r_2; the benchmark value lies 0.000334 below the upper edge $\tau(r_2)$.

and 0.272757 from r_1 to r_2 (Proposition A.3): with information fixed, entry stays constant from r_0 to r_1 and falls above the entry ceiling.

Panel C hides the price from the challenger, which then decides on its prior and prepares only at zero cost, so entry stays at 0.250000 at both strengths even though the investor trades fully at r_1. The two controls together locate the reversal in the information that trading generates: incumbent strength changes that information, and the entry response to it exceeds the direct deterrence effect of a stronger incumbent at fixed information.

4.3 Coexistence

At the benchmark, Proposition 2 compares the weak and strong economies at r_0 and r_1, and between those strengths an informative equilibrium coexists with an uninformative one. I now describe entry along the whole benchmark path $r \in [r_0, r_2]$ at the prior $1/2$, for every slot in the band of Section 4.2 (Proposition A.1); Appendix A.4 lists the complete equilibrium set.

Coexistence starts at r_A, where a purchase that just reaches the preparation threshold and recruits a high-cost challenger earns exactly the trading cost. Full purchases start at r_F, where a full purchase that recruits earns the trading cost, and the equilibrium without informed trade ends at r_N, where a unit order against an uninformative price earns the trading cost. The entry ceiling r_C of Section 4.2

also enters the proposition: above it even a purchase at the top posterior cannot recruit a high-cost challenger. Figure 2 also marks r_J and the sale threshold r_S, which bound the mixed patterns that the figure draws with the informative branch; r_A, r_N and r_J do not depend on the slot, and Appendix A.4 defines every marked strength and every pattern in the figure.

Proposition 3 (Coexistence on the benchmark path)**.** *Let the auction be the benchmark auction at the prior* $1/2$*, and let the slot lie in the band of Proposition A.1. An equilibrium exists at every strength on the path from* r_0 *to* r_2*, and Appendix A.4 lists the whole equilibrium set.*

(a) The equilibrium without informed trade and an informative equilibrium coexist exactly on $[r_A, r_N]$.

(b) On the informative branch that coexists with no trade, entry is $\rho+(1-\rho)\min\{b\Delta_T(r)/(2k),\ b+\lambda/2\}$*. It is continuous, rises strictly on* $[r_A, r_F]$ *and then stays constant on* $[r_F, r_N]$*, keeps this value along full purchases up to* r_C*, and equals* ρ *above* r_C.

At the benchmark slot the marked strengths are $r_A = 1.393958$, $r_F = 1.467127$, $r_N = 1.747878$, $r_J = 1.966097$, $r_S = 2.124148$ and $r_C = 3.592659$. Entry on the informative branch is 0.454185 at r_A, 0.459588 at 1.40 and 0.506112 at 1.45, and from r_F it stays at 0.522757 until the entry ceiling r_C. The low class abstains up to r_N, mixes on (r_N, r_S), and sells with probability one from r_S.

Figure 2 shows entry along the path, and coexistence comes from the entry response itself. Against an uninformative price with entry ρ, a unit order earns $\rho\Delta_T/2$, which stays below k up to r_N, so the equilibrium without informed trade survives until r_N; an informative equilibrium changes the entry response to a purchase and with it the return to trade, so it can hold at the same strengths.

Entry rises strictly along the informative branch while the high class mixes, because the purchase rate adjusts. At r_A a purchase that just reaches the preparation threshold earns exactly k; above r_A the high class mixes and the posterior after a purchase stays above the threshold, so the purchase earns $\Delta_T[1-\mu(1)] = k$ and a wider target-payoff spread needs a more informative purchase. The purchase rate $b\,[\Delta_T(r)/k - 2]$ therefore rises with r, and entry rises with it, until the purchase rate reaches λ at r_F; from there entry stays constant.

Above r_N the equilibrium without informed trade fails. On (r_N, r_J) the investor can also mix in both directions without recruitment, and from r_J on a full purchase is the only high-class order.

Coexistence also holds away from the even prior. At the benchmark slot the informative branch starts at the same strength r_A for every prior between ≈ 0.396175 and ≈ 0.640739, and it starts later, where the prior reaches the preparation threshold, for priors up to ≈ 0.650059. Below ≈ 0.396175 even the top posterior cannot reach the threshold once a purchase can pay for itself, and from ≈ 0.650059 on, the uninformative equilibrium fails before the prior falls below the threshold; Appendix A.4 gives the coexistence set at every prior (Corollary A.2).

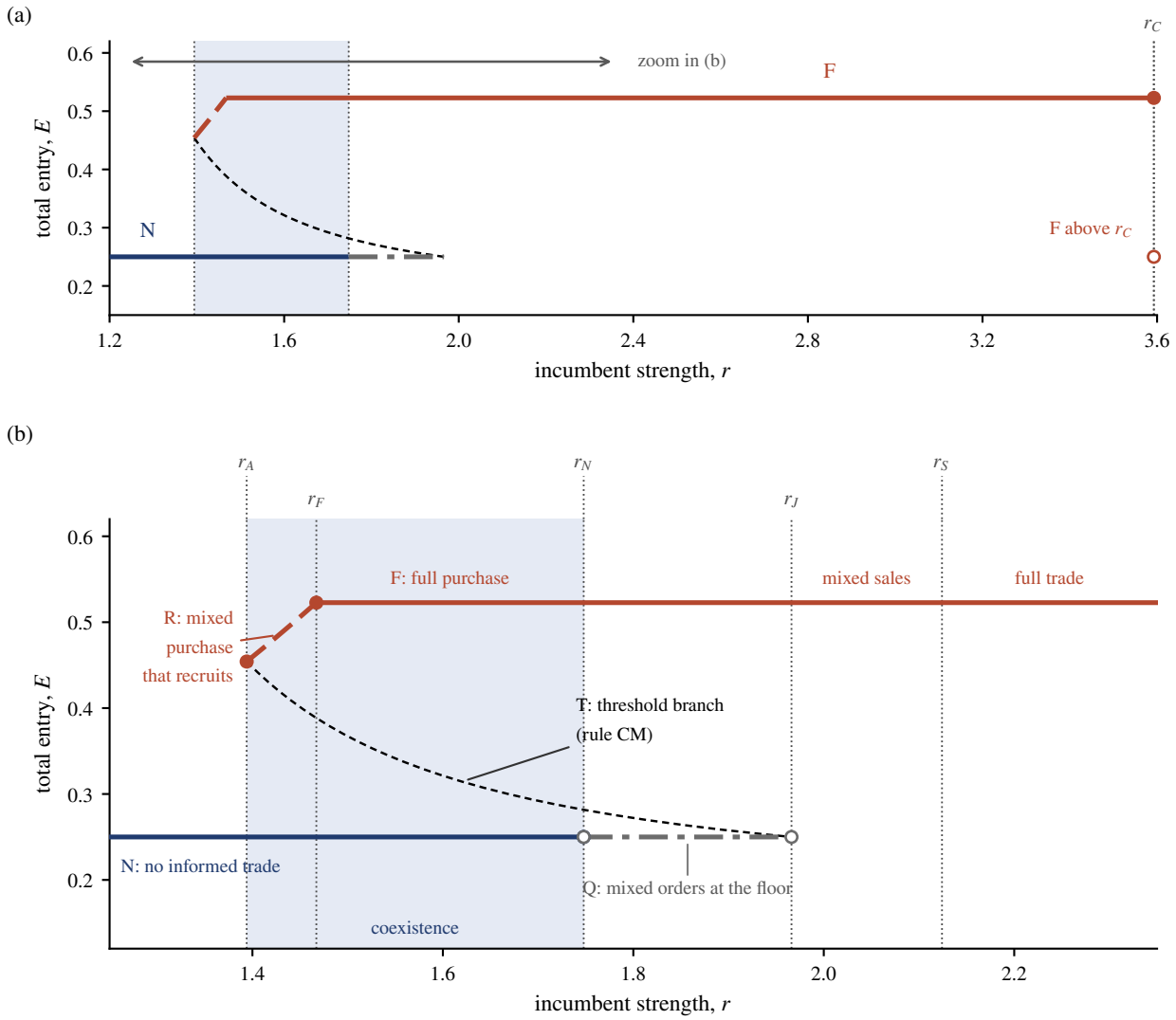


Figure 2. Equilibrium entry along the benchmark path.

Notes. The benchmark at the prior $1/2$, when an indifferent investor may mix over its best orders and an indifferent high-cost challenger prepares. Entry against incumbent strength r in each equilibrium pattern of Proposition 3; Appendix A.4 lists the whole set. The figure labels the equilibrium without informed trade by N, a mixed purchase that recruits by R, a full purchase by F, mixed orders without recruitment by Q and the threshold branch by T. The thin dashed line is the threshold branch, an equilibrium only when an indifferent high-cost challenger may prepare with any probability. Panel (a) covers the path from r_0 to r_2, and panel (b) enlarges the strengths around the coexistence interval, which the blue band marks from r_A to r_N. With a full purchase the low class abstains up to r_N, mixes from r_N to r_S, and sells fully from r_S. Filled markers belong to a branch, and open markers do not, and above the entry ceiling r_C entry with a full purchase equals ρ.

5 Competition creates research

So far the investor is present with a fixed probability λ; now the investor chooses that probability and pays for it. Before it learns the class, the investor chooses a research intensity $\lambda \in [0, \bar{\lambda}]$ at the cost $(a/2)\lambda^2$, where $a > 0$ and the cap satisfies $0 < \bar{\lambda} < 1$. Research succeeds with probability λ, independently of the class, the costs and the incumbent's value. A success reveals the class, and the investor then fills the slot with an order. After a failure, a liquidity trader fills the slot.

Research is therefore the presence probability of Section 2.3, chosen at a cost. It is the specialist's own work on the target's customers, technology and demand, and it must end before the order opportunity. The cap $\bar{\lambda}$ is the probability that this work can reveal the class at all before the challenger decides, because some facts about the match become visible only through the buyer's own diligence. The marginal cost rises because the specialist uses cheap sources first, such as its sector coverage, and costly ones later, such as expert calls. The quadratic cost is the simplest form whose marginal cost rises.

A success takes the place of a liquidity order because the opportunity has room for one order. An investor that knows the class takes it. After a failure, the investor knows no more than the market, so an order would lose at least its cost k. The opportunity then passes to a liquidity trader. More research therefore also removes liquidity orders, and it reduces the camouflage of the investor's own orders.

Market makers and the challenger do not observe the research choice, so they price and decide at a conjectured intensity; throughout this section the prior is $1/2$. In an equilibrium with research, the conjectured intensity is the investor's best research choice at the equilibrium prices and entry rule, and the order after a success is optimal, including after a deviation from zero research. The best research choice is unique: it equals the expected trading payoff after a success divided by a, capped at $\bar{\lambda}$, so research never has a tie. Appendix A.5 states the equilibrium with research formally and gives each result of this section in full.

Research raises the top posterior that a purchase can reach. At intensity λ, I write M_λ for that top posterior and $m_\lambda = 1 - M_\lambda$ for the bottom posterior, the posterior bounds of Section 3.2 at the prior $1/2$ when the presence probability is λ. When $1/2 < \tau(r) < 1$, the *recruitment intensity* $\lambda_E(r)$ is the research level at which a full purchase just reaches the threshold, $M_\lambda = \tau(r)$, so a full purchase recruits exactly when research is at least $\lambda_E(r)$.

Suppose that class H buys after a success, class L abstains, and only a purchase recruits. A purchase then earns $\Delta_T(r)m_\lambda - k$, and research optimality at an interior intensity reads:

$$2a\lambda = \Delta_T(r)\, m_\lambda - k. \tag{9}$$

The left side rises with λ, and the right side falls, because more research removes camouflage. When

$\Delta_T(r) > 2k$, the two sides meet at one intensity $\lambda_R(r) \in (0, 1)$.

On the branch where only a purchase recruits, a wider target-payoff spread raises the return to a purchase, so the investor chooses more research and a purchase becomes more likely; research, entry and high-class ownership then rise together with incumbent strength. At the benchmark strengths, the same force takes the economy from no research and no trade against the weak incumbent, through coexistence of zero research with the research branch at intermediate strengths, to research at the cap with trade in both directions against the strong and the very strong incumbent. The benchmark research parameters are $a = 1/40$ and the cap $\bar{\lambda} = 0.336122$, the presence probability of Section 4.2, so research at the cap reproduces the benchmark slot.

Proposition 4 (Competition creates research)**.** *Let the auction be the benchmark auction at the prior* $1/2$*, and suppose that* $0 < \beta \leq 1/3$ *and* $a > 0$*.*

(a) On the strengths with $\Delta_T(r) > 2k$*, research* $\lambda_R(r)$ *is continuously differentiable and rises strictly in* r*. Wherever the research branch is an equilibrium, entry* $\rho + (1-\rho)\big[\beta + (1/2-\beta)\lambda_R(r)\big]$ *and high-class ownership* $\frac{1}{2}[\rho + (1-\rho)(\beta + (1-\beta)\lambda_R(r))]$ *rise strictly with* r*.*

(b) At the benchmark, with the research cost $a = 1/40$ *and the cap* $\bar{\lambda} = 0.336122$*:*

(i) At r_0*, the unique equilibrium has no research and no trade. Entry is* ρ*.*

(ii) At each $r \in \{1.55, 1.60, 1.65\}$*, there are exactly two equilibria. In the first, the investor does no research, and entry is* ρ*. In the second, the research branch, research is* $\lambda_R(r)$*, with* $\lambda_E(r) < \lambda_R(r) < \bar{\lambda}$*. After a success, class H buys and class L abstains, only a purchase recruits, and entry is* $\rho + (1-\rho)[\beta + (1/2-\beta)\lambda_R(r)]$*.*

(iii) At r_1 *and* r_2*, the unique equilibrium has research at the cap and trade in both directions. It is the equilibrium of Proposition 2 with* $\lambda = \bar{\lambda}$*. Entry is* 0.522757 *at* r_1 *and* ρ *at* r_2*.*

At r_0 every nonzero order after a success loses, so the expected trading payoff after a success is zero and the best research choice is zero. At the intermediate strengths, the equilibrium without research confirms itself: against an uninformative price the best nonzero order after a success earns $\rho\Delta_T/2 - k < 0$, because these strengths lie below r_N; on the research branch, a purchase reveals enough to recruit, and its return pays for the research that makes it informative.

At r_1 and r_2, a bound on every candidate profile makes a full order profitable net of the marginal research cost at the cap. Every trade choice at the six strengths is strict, so the lists do not depend on how an indifferent investor breaks a tie.

Research and entry rise together across the three intermediate strengths in Table 4: research rises from 0.237901 through 0.279037 to 0.316683, and entry rises from 0.507630 through 0.513965 to 0.519763. A stronger incumbent widens the target-payoff spread, so a purchase earns more and

Table 4. Research and entry at the intermediate strengths.

	$r = 1.55$	$r = 1.60$	$r = 1.65$
Target-payoff spread $\Delta_T(r)$	0.097581	0.112500	0.128030
Preparation threshold $\tau(r)$	0.646798	0.648734	0.650670
Recruitment intensity $\lambda_E(r)$	0.196741	0.199698	0.202664
Research $\lambda_R(r)$	0.237901	0.279037	0.316683
Posterior after a purchase M_{λ_R}	0.673142	0.698206	0.720112
Entry on the research branch	0.507630	0.513965	0.519763
High-class ownership on the branch	0.298422	0.309302	0.319260
Entry without research	0.250000	0.250000	0.250000

Notes. The benchmark with $a = 1/40$ and $\bar{\lambda} = 0.336122$. The spread, the threshold and λ_E are exact rationals; research and the values that depend on it are interval enclosures rounded to six places.

the investor chooses more research; the posterior after a purchase then rises above the threshold by more, and a purchase becomes more likely. Part (a) of Proposition 4 shows that this comovement is general along the branch.

The slope of entry is $(1-\rho)(1/2-\beta)\lambda_R'(r)$, which is positive for every $\beta \le 1/3$. A purchase has probability $\beta + (1/2-\beta)\lambda$: research adds informed purchases in class H and removes liquidity orders in both classes, and the first effect dominates.

At the benchmark, the research branch is an equilibrium exactly on $[r_R, \bar{r}_R)$, with $r_R \approx 1.501023$ and $\bar{r}_R \approx 1.677633$. The start r_R is the strength at which the purchase posterior just reaches the threshold and the purchase also pays the marginal research cost: $\Delta_T(1-\tau) = k + 2a\lambda_E(r)$. The end $\bar{r}_R$ is the strength at which research reaches the cap $\bar{\lambda}$. Along the branch, research rises from $\lambda_E(r_R) = 0.193853$ to the cap, and entry rises from 0.500846 to 0.522757. Zero research is an equilibrium along the whole branch, because $\bar{r}_R < r_N = 1.747878$. On $[\bar{r}_R, r_N)$, research at the cap coexists with zero research, and entry equals 0.522757.

Against strong incumbents, research sits at the cap. Above a strength $r_U \approx 2.434046$, a bound on every candidate profile forces full orders and research at the cap (Appendix A.5).

The research branch in Figure 3 starts later than the informative branch of Section 4.3, which starts at $r_A = 1.393958$, because a purchase must now pay for research as well as for trade. The two branches also adjust in different ways: with a fixed presence probability the purchase rate adjusts through a mixture of the high class, while with research the intensity adjusts and the high class buys with probability one after a success.

Proposition A.2 in Appendix A.5 gives the admissible region of the research parameters exactly at $\beta = 0.294655$. The equilibrium patterns of Proposition 4(b) hold with strict margins exactly when the cap lies in the strip $(\lambda_E(r_1), \lambda_E(r_2)) \approx (0.290539, 0.336721)$ and the cost lies in a window that depends on the cap. At the benchmark cap, this window is $(0.021469, 0.036763)$. With $1 + u = \mathrm{e}$,

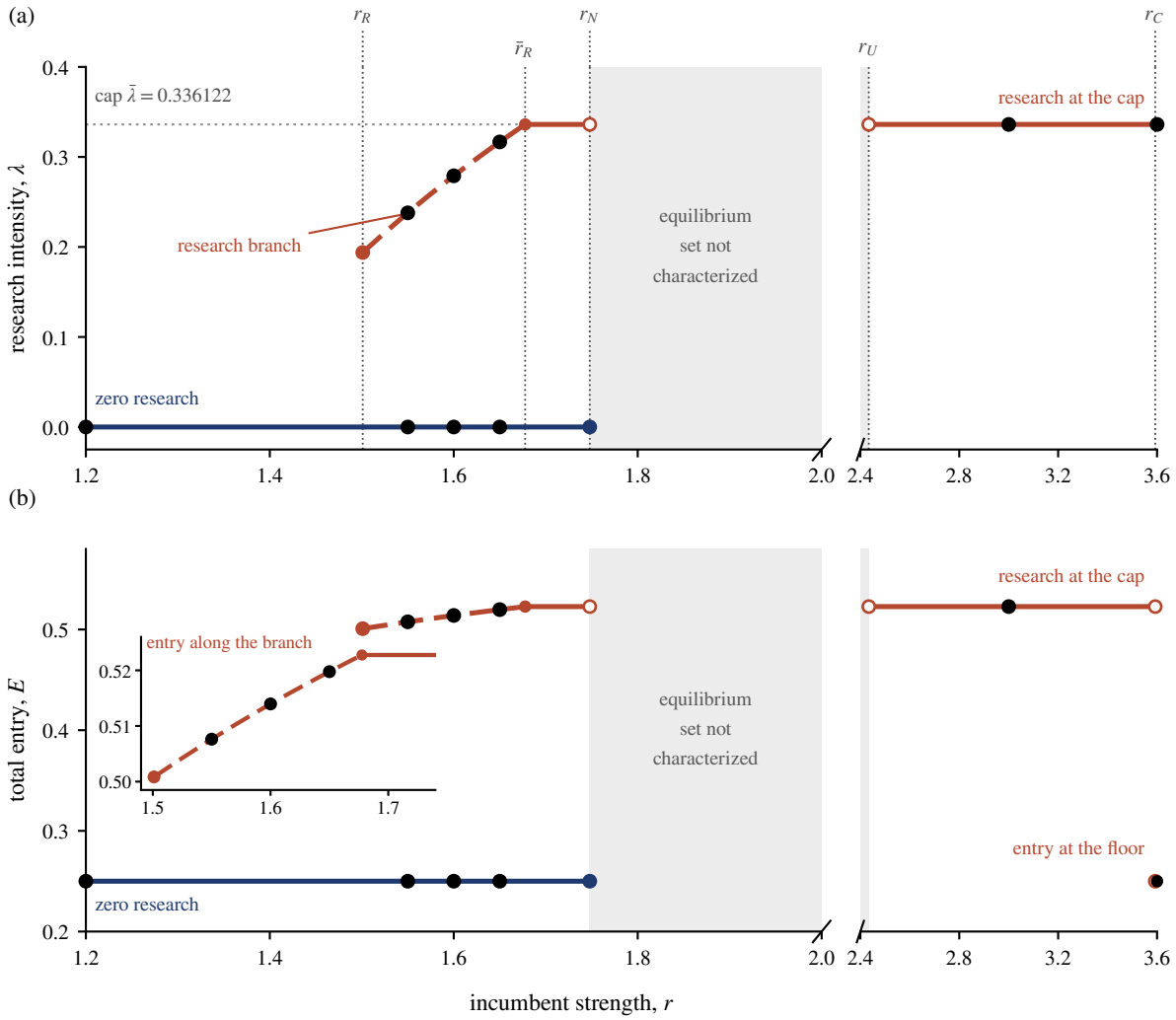


Figure 3. Research and entry along the research branch.

Notes. The benchmark with $a = 1/40$ and $\bar{\lambda} = 0.336122$, where an indifferent high-cost challenger prepares. Panel (a) shows research λ, and panel (b) shows entry; the inset enlarges entry along the research branch. Zero research is an equilibrium from r_0 to r_N. The research branch runs from r_R to $\bar{r}_R$. Research stays at the cap from $\bar{r}_R$ to r_N and above r_U; at the cap, entry is 0.522757 up to the entry ceiling r_C and ρ above. Black dots mark the six strengths of Proposition 4(b). The gray band marks the interval (r_N, r_U) between the two cap ranges, and the horizontal axis is broken there.

the cap $\bar{\lambda} = u\beta/(1+u\beta)$ and $a = 1/40$, these patterns hold for every $\beta \in (0.236157, 1/3]$. Entry at r_1 is $\rho + (1-\rho)[\beta + (1/2-\beta)\bar{\lambda}]$.

6 Robustness

I vary one part of the model at a time: the preparation-cost law, the value classes and the prior, and the challenger's information. Appendix A.6 states each result in full and proves it.

6.1 Preparation costs

The rise of entry in Proposition 2 survives other preparation-cost laws, with or without atoms (Proposition A.4). The cost now mixes a low component with weight ρ and a high component in place of the two-point law of Section 2.2; either component may be atomless or have atoms. The proposition keeps parts (i) and (ii) of Proposition 2 under the trading-cost window (H2) when every cost in the low component lies strictly below $B_{F_1}(m(\pi))$ and every cost in the high component lies strictly between $B_{F_0}(\pi)$ and $B_{F_1}(M(\pi))$. The low component then prepares at every feasible belief, so entry stays at least ρ, and the trading bounds of Section 3.3 apply unchanged.

At the benchmark, the low component runs from zero to $1/5$, and the high component is uniform on $[c_H - \varepsilon, c_H + \varepsilon]$. Every half-width below 0.002688902 keeps all three parts of Proposition 2. With half-width $\varepsilon = 1/10$, entry at r_1 is 0.522757, the value under the two-point law. At r_2, gross profit at the top posterior, which lies 0.002689 below c_H, falls inside this band, and entry at r_2 is 0.382711. With an atomless high component, the drop at the entry ceiling becomes a ramp: full-order entry is continuous and nonincreasing in strength.

6.2 Value classes and priors

The reversal holds for every pair of ranked classes and every prior. The classes satisfy the class ranking (1) and differ above the reserve, and the reserve, the prior, the zero-cost probability and the slot are arbitrary. For uniform incumbents, a nonempty open set of strengths and costs then satisfies (H1)–(H3) and the ownership condition of Proposition 2 (Proposition A.5).

The construction puts the weak and the strong incumbent just above the lowest value v_* at which the classes start to differ above the reserve. The spread vanishes near v_*, so informed trade against the weaker incumbent cannot cover its cost. The profit gap stays near the full class gap there, so a high cost that only a favorable price justifies fits inside it. In the point-mass case $v_* = \ell$, so the reversal holds for every $h > \ell$.

Table A.4 in Appendix A.6 reports six economies with their own costs and strengths. With wide uniform classes, entry rises from 0.250000 to 0.573175, and high-class ownership rises although the

high class can lose to the strong incumbent; point masses at the same means leave entry at 0.250000. A prepared challenger keeps $(\theta - \max\{p, R\})_+$, a convex function of its value, so a mean-preserving spread within a class never lowers gross profit at any belief.

With more than two classes, the reversal holds under a separation condition (Proposition A.6). The price reveals only the posterior mean of class proceeds. Separation splits the classes into a high group and a low group, and it asks for a gap between the proceeds of each high-group class and every feasible posterior mean, and the same gap for the low group. In the strong equilibrium, class proportions within each group then stay at their prior proportions, so the price reveals the whole posterior vector.

A four-class economy near the benchmark satisfies separation with distance 0.178871. Without separation, one price can carry two posterior vectors: a strict three-class equilibrium has two orders with different posterior vectors and one price.

6.3 Complementary private signals

The challenger can gain from the price even when its own information is more accurate than the investor's. A buyer may know its integration plans, while investors who follow the target know its customers or its product market. Proposition A.7 represents the two sources with conditionally independent binary signals. The investor's signal has accuracy a_T, and the challenger's signal has accuracy d.

The zero-cost challenger still prepares after every price and signal, so the price still reveals the public posterior, and the challenger combines that posterior with its own signal. Under strict conditions on the costs, the weak incumbent has zero orders and entry ρ; the strong incumbent has full orders and entry above ρ. The conditions allow $d > a_T$.

The signal example uses its own parameter vector. It keeps the slot, the values, the reserve and the prior of the benchmark, and it sets $\rho = 0.85$, $c_H = 7.14$, $k = 0.015$, $r_0 = 1.1$ and $r_1 = 2.3$. With investor accuracy $a_T = 0.70$ and challenger accuracy $d = 0.75$, entry rises from 0.850000 to 0.879797. A favorable signal alone does not justify high-cost preparation against the weak incumbent; a favorable signal and a favorable price together do so against the strong incumbent. Both sources are needed: without the challenger's signal, entry is 0.850000 at both strengths.

At the benchmark itself, the conditions hold with $(a_T, d) = (0.94, 0.505)$. Entry then rises from 0.250000 to 0.386933, and without the challenger's signal it is 0.250000 at both strengths. At the benchmark the conditions cannot hold with a buyer signal at least as accurate as the investor's: the strong trading bound needs an accurate investor signal, and the weak bound then fails. The signal example meets the conditions with a more accurate buyer signal at other costs and strengths.

7 Welfare and sale terms

7.1 Access to prices

Access to prices improves acquisition outcomes at a fixed level of competition. At the strong incumbent F_1, I compare the equilibrium of Proposition 2 with that of an economy in which the challenger does not see the price. Under the hypotheses of Proposition 2, both economies have unique full orders, and price access raises expected target proceeds and acquisition surplus net of preparation costs (Proposition A.8). Acquisition surplus excludes transfers between bidders, shareholders and traders, and Appendix A.7 states the results of this section in full.

The two economies generate the same orders. The trading bound of Section 3.3 forces full orders even when the challenger cannot use the price and entry stays at ρ, so trading costs coincide. Price access changes only which high-cost challengers prepare: it adds preparation after a purchase. Each added preparation adds its gross profit minus its cost to net surplus, together with the value of any sale that the reserve would otherwise block; a challenger prepares only when its expected gross profit covers its cost, so no added preparation lowers net surplus.

In the benchmark at r_1, expected proceeds rise from 0.614583 to 0.872531, a gain of 0.257948, and net surplus rises by 0.081960. With research at the cap (Section 5), the gains are the same.

Target shareholders gain because a shareholder who holds through the sale receives V_T, not the interim price. The investor's expected gross profit, 0.037665 at r_1, is a transfer from liquidity traders. If one order unit is z shares per share of float, initial shareholders gain in total relative to an economy without the investor at every $z < 6.848394$, even if they bear the whole transfer.

Inside the recruitment region of Proposition 2, where high-cost preparation does not pay at the prior, price access adds preparation. Where it already pays at the prior, an informative price deters some preparation: with the strong benchmark and $c_H = 12/5$, price access changes proceeds by -0.173917 and net surplus by -0.013505 (Appendix A.7).

The information in the price, not its level, separates the two economies. A deterministic external dividend of 0.257948 in the hidden-price economy equalizes the mean prices, and it leaves the orders and entry unchanged.

7.2 The payment rule

The effect of competition on the target-payoff spread depends on the sale payment. Proposition A.9 isolates this dependence at the payment stage, in a verifiable-value institution with zero reserve, point-mass classes and an incumbent distribution on $[0, \bar{r}]$ with $0 < \ell < \bar{r} < h$. After diligence, a sale to the next buyer at its value is an enforceable fallback, and the seller and the winner divide the surplus above the fallback by Nash bargaining, with seller weight $\omega \in [0, 1)$. A stronger incumbent

Table 5. Reserves, preparation and sale in the benchmark.

Reserve	Strength	Entry	Sale	Two admissible bidders	Expected proceeds
$1/2$	r_0	0.250000	0.687500	0.145833	0.392708
	r_1	0.522757	0.920459	0.435631	0.872531
$101/100$	r_0	0.522757	0.431371	0.051364	0.440564
	r_1	0.522757	0.772548	0.215186	0.994384
$31/20$	r_0	0.522757	0.324401	0	0.502822
	r_1	0.250000	0.547917	0.060417	0.893073

Notes. The benchmark, with point-mass classes. At each reserve, full informed orders are the unique outcome at both strengths, except at the reserve $1/2$ and r_0, where the investor does not trade.

never raises the challenger's profit, and it moves the spread only in the direction of the sign of $1-2\omega$; the move is strict when $\mathbb{E}[(R-\ell)_+]$ changes strictly and $\omega \neq 1/2$, and at $\omega = 1/2$ the spread equals $(h-\ell)/2$ at every incumbent distribution.

To see why, take $R > \ell$. A high-class challenger wins and pays $(1-\omega)R + \omega h$, whose sensitivity to R is $1-\omega$. A low-class challenger loses, and the incumbent pays $(1-\omega)\ell + \omega R$, whose sensitivity is ω. Competition widens the difference exactly when the first sensitivity exceeds the second.

In the benchmark, the spread under this rule changes by $13/20\,(1-2\omega)$ from r_0 to r_1. The second-price auction is the case $\omega = 0$: it puts the whole weight on the runner-up's value. Bulow and Klemperer (1996) compare an auction with one more bidder and an optimal negotiation by expected revenue; here the comparison concerns how competition moves the spread.

7.3 Reserves

A reserve affects both the payment required to win and the information in the target's shares. Once the reserve can exclude bidders, preparation, a sale and the meeting of two admissible bidders become different events: *preparation* is the event that the challenger pays its preparation cost, a *sale* occurs when at least one bidder submits an admissible bid, and *two admissible bidders* meet when both the incumbent and a prepared challenger meet the reserve. In the benchmark of Section 4.2, the reserve lies below both challenger values, so every prepared challenger is admissible.

A reserve that excludes the low class can widen the class difference in target proceeds, because the low class then adds nothing to proceeds. Informed trade can then pay against both incumbents, and the price is informative at both strengths. Proposition A.10 gives the consequence. If informed trade is full at both strengths and only a purchase recruits at each, entry is flat in incumbent strength, and strict deterrence needs the stronger incumbent to push gross profit at the top posterior below the high cost.

Table 5 reports three reserves in the benchmark. The reserve $101/100$ excludes the low class.

Full orders are unique at both strengths, only a purchase recruits, and entry is 0.522757 at both r_0 and r_1, while this reserve raises expected proceeds at both strengths. Narrow uniform classes give the same pattern at the reserve $11/10$.

The reserve $31/20$ restores strict deterrence: entry is 0.522757 at r_0 and 0.250000 at r_1, and proceeds exceed their levels at the reserve $1/2$. This reserve excludes the low class and also the weak incumbent, since it exceeds r_0. In the benchmark, strict deterrence after the low class is excluded requires this second exclusion. With a reserve in $(\ell, r_0]$, the top posterior recruits at both strengths, and entry is flat. Strict deterrence holds for reserves between about 1.325270 and 1.792723 (Appendix A.7).

8 Empirical implications

The model gives predictions for publicly visible sale processes in which one bidder is prepared and a further buyer has not yet committed to diligence. I state them for the benchmark auction at the prior $1/2$, with a slot in the band of Section 4.2. Appendix A.8 states each result in full and proves it. Full two-sided trade is the unique outcome at every strength r between the sale threshold r_S and h, where r_S is the exact strength at which a full sale earns the trading cost, $\rho\Delta_T(r)m(1/2) = k$. This interval contains r_1, the entry ceiling r_C and r_2. Under full two-sided trade, the informed share of nonzero orders is $u/(2+u)$.

Entry and incumbent strength. When the price reaches the further buyer before its decision, a stronger prepared bidder can raise the probability that the buyer prepares; in the benchmark, that probability rises by 0.272757 from r_0 to r_1. When the price does not reach the buyer, a stronger prepared bidder can only lower it. Against a strong enough prepared bidder, entry returns to its floor while the price stays informative. Proposition 2 gives the first and third comparisons, and the fixed-information control of Section 4.2 gives the second (Proposition A.3).

The price spread and its entry component. The purchase impact, the price after a purchase minus the price after no order, $P(1) - P(0)$, exceeds the sale impact, the price after no order minus the price after a sale, $P(0) - P(-1)$, exactly when a purchase recruits the buyer, and this test needs no knowledge of the auction objects. The price spread is the difference between the conditional execution prices of a unit purchase and a unit sale. It has a class component, $\rho\Delta_T$ times the informed share, and an entry component, which is positive exactly when a purchase recruits (Proposition A.11). The sale impact contains only half of the class component, so the whole entry component sits in the purchase impact.

In the benchmark at r_1, the price spread is 0.786299, and its entry component is 90.2% of it.

With a hidden price, the price spread would be 0.077020. At r_2 the price spread is 0.108469, the class component alone, and at the entry ceiling r_C it falls by 0.861835.

The informed share. Prices separate slots that order counts cannot. Under full two-sided trade, the order law depends on (λ, β) only through the volume of nonzero orders (Proposition A.12), and the investor's incentives narrow the shares that the counts allow: at r_1, the shares consistent with both form the interval from ≈ 0.250299 to 0.760000.

A slot with share 0.444444 has the benchmark volume and the benchmark entry, but its price spread at r_1 is 0.778935, against 0.786299. Given the entry floor and the target-payoff spread, the sale impact and the volume identify the informed share.

Winners. Orders predict the winner even where they no longer predict entry. In the benchmark, a prepared challenger wins more often after a purchase than after a sale, by 0.308078 at r_1 and by 0.333751 at r_2. The covariance of the order with entry is 0.272757 at r_1 and zero at r_2, while its covariance with a challenger win stays positive. Among prepared challengers, the share of the high class at each price is the price-implied posterior, which rises with the price (Proposition A.13). A prepared high-class challenger always outbids the incumbent, and a low-class one loses whenever the incumbent's value exceeds ℓ.

Announcements. If a public event after the price reveals only whether the challenger prepared, the mean price change in a sample of preparation reports alone is positive: 0.160592 at r_1 in the benchmark (Proposition A.14). The mean change over both reports, with their model probabilities, is zero. Under full two-sided trade below the entry ceiling, on $(r_S, r_C]$, an order opportunity with no order is bad news about entry and no news about the class. At r_1, the price after no order minus the expected price before the opportunity is -0.257948.

Two clusters at intermediate strengths. At one public strength in the range (r_F, r_N) of Section 4.3, processes can form two clusters. In one cluster there is no informed trade, the mean order is zero and the three prices coincide. In the other, the high class buys and the low class abstains; the mean order is then $\lambda/2$ (0.168061 in the benchmark), every challenger prepares after a purchase, and a sale executes above the price after no order, by 0.005363 at 1.60.

Measurement. The counterpart of preparation is the start of substantive diligence or a proposal that requires costly preparation; public offers are a narrower outcome. Incumbent strength is measured from information that is public before that decision, and the price must precede it.

The order is the first prespecified public order opportunity after the sale becomes public, and an opportunity with no trade is recorded as a zero. A scheduled call auction is one example, such as the opening auction on the first trading day after the approach is disclosed; its time is fixed in advance and does not depend on any trader's information, and the auction takes place whether or not an order arrives. The sign of its net order imbalance is the order, and an imbalance below a size fixed in advance records the zero order. A sample of first executed trades omits the zero order, so it cannot identify the volume, the effect of an inactive opportunity, or the informed share.

The announcement formulas need events that reveal only preparation. Target returns before a bid also partly reflect anticipation of that bid (Schwert, 1996), so each test dates the price before the buyer's decision. The trading counterpart is outside trade with a view on the further buyer's fit, such as flow from sector specialists.

9 Conclusion

The bidder pool in a takeover depends on the information that buyers have before they commit to preparation. A stronger incumbent moves the class value gap from the challenger's profit to target proceeds, and because a liquidity order can match the investor's direction in the order slot, an investor who knows the challenger's class keeps a residual advantage after every order. A larger target-payoff spread can make that advantage worth the cost of trade, so the price recruits a challenger that a weaker incumbent could not attract.

The comparison of the weak, strong and very strong incumbent holds with unique outcomes on an open set of parameters. Between the weak and the strong incumbent, informative and uninformative equilibria coexist on an interval of strengths. On the informative branch that coexists with no trade, entry rises strictly while the high class mixes between a purchase and abstention, stays constant from full purchases up to the entry ceiling, and falls to its floor above it. When information is costly, competition creates research: research and entry rise together over a range of strengths, on a branch that coexists with an equilibrium without research.

Sale terms affect entry through the buyer's expected payment and through the information in the stock price. In the recruitment region, access to prices raises target proceeds and acquisition surplus net of preparation costs, and the weight that the payment rule places on the runner-up's value decides how competition moves the target-payoff spread. A reserve that excludes the low class can make the price informative against both incumbents; if informed trade is then full and only a purchase recruits at both strengths, entry is flat in incumbent strength.

The next theoretical step is the seller's choice of terms with the whole pricing and trading continuation retained. The empirical tests ask whether further preparation in visible sale processes responds to incumbent strength when the price precedes the decision, and whether purchase impacts

carry the entry component that Section 8 predicts.

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A Proofs

This appendix proves every result in the paper. Statements use the class model of Section 2. A statement that needs the point-mass benchmark says so. Each equilibrium proof states the equilibrium tuple of Definition A.1, lists its conditions, and checks each condition against the fixed tuple.

The equilibrium tuple. Write $\eta(-1) = \eta(1) = \beta$ and $\eta(0) = 1 - 2\beta$ for the liquidity order distribution of Section 2.3.

Definition A.1 (Equilibrium)**.** An *equilibrium* is a tuple

$$\mathcal{E} = \big(\sigma_H, \sigma_L,\ \mu,\ P,\ \bar{\mu},\ \zeta\big),$$

together with the bids of the prepared bidders. Its components are as follows.

- $\sigma_j = \big(\sigma_j(-1), \sigma_j(0), \sigma_j(1)\big)$ is a probability distribution on $\{-1, 0, 1\}$, the order of the informed investor in class $j \in \{H, L\}$. Mixed and wrong-signed orders are allowed.
- $\mu : \{-1, 0, 1\} \to [0, 1]$ is the market makers' posterior $\mu(x) = \Pr(j = H \mid X = x)$.
- $P : \{-1, 0, 1\} \to \mathbb{R}$ is the price after each order.
- $\bar{\mu}$ assigns to each price $\bar{P}$ in the image of P the challenger's belief $\Pr(j = H \mid P(X) = \bar{P})$.
- $\zeta(\bar{P}) \in [0, 1]$ is the probability that a challenger with the high cost prepares after the price $\bar{P}$. A challenger with the zero cost prepares after every price.

For each order x, write $f_j(x) = (1 - \lambda)\eta(x) + \lambda\sigma_j(x)$ for its probability in class j, and $e(x) = \rho + (1 - \rho)\,\zeta(P(x))$ for entry after it. The tuple satisfies six conditions.

(E1) *Bayes consistency.* For every order x, $\mu(x) = \pi f_H(x)/[\pi f_H(x) + (1 - \pi) f_L(x)]$.

(E2) *Challenger beliefs.* For every price $\bar{P}$ in the image of P, $\bar{\mu}(\bar{P})$ is the posterior of class H given the event $\{x : P(x) = \bar{P}\}$, computed from f_H and f_L.

(E3) *Entry optimality.* For every price $\bar{P}$ in the image, $\zeta(\bar{P}) = 1$ if $B(\bar{\mu}(\bar{P})) > c_H$ and $\zeta(\bar{P}) = 0$ if $B(\bar{\mu}(\bar{P})) < c_H$. At $B(\bar{\mu}(\bar{P})) = c_H$, the challenger's tie rule applies.

(E4) *Competitive prices.* For every order x, $P(x) = \mathbb{E}[V_T \mid X = x]$.

(E5) *Investor optimality.* For each class j, $\sigma_j(q) > 0$ only if q maximizes

$$U_j(q) = q\left[t_0 + e(q)(t_j - t_0) - P(q)\right] - k|q|$$

over $q \in \{-1, 0, 1\}$, with $P(\cdot)$ and $e(\cdot)$ held at their values in $\mathcal{E}$.

(E6) *Truthful bids.* Each prepared bidder whose value meets the reserve bids its value.

The gross profit of a zero-cost challenger is an expected auction rent, which is never negative. Preparation is therefore optimal for it after every price, and the definition fixes this choice. The probability $f_j(x)$ is at least $(1 - \lambda)\eta(x) > 0$ under every strategy. Every order therefore occurs with positive probability in both classes. Bayes' rule defines μ after every order, and no off-path belief enters the definition.

The two indifferences in (E3) and (E5) follow the default tie conventions of Section 2.4: an indifferent investor may mix over its best orders (rule IM), and an indifferent high-cost challenger prepares (rule C1). Appendix A.4 states the alternatives I0 and CM.

A.1 Acquisition payoffs

Truthful bids. Fix a bidder with value v, and let $Y \geq 0$ be the highest competing admissible bid, with $Y = 0$ when there is none. Any admissible bid that wins pays $\max\{p, Y\}$, so a bid yields either $v - \max\{p, Y\}$ or zero. Suppose $v \geq p$, and consider the bid v. If $v > \max\{p, Y\}$, it wins and earns $v - \max\{p, Y\} > 0$. If $v < \max\{p, Y\}$, then $Y > v$, and the bid loses and earns zero. If $v = \max\{p, Y\}$, a win pays v, so the bid earns zero.

The bid v therefore earns $(v - \max\{p, Y\})_+$, the larger of the two possible payoffs, at every Y. Suppose $v < p$. Every bid that wins pays at least $p > v$, so no bid earns more than zero, the payoff of no bid. Truthful bids, and no bid below the reserve, are therefore weakly dominant at every competing bid. This verifies condition (E6) of Definition A.1 for every tuple. No payoff below rests on a belief.

Proof of Proposition 1. Without entry the target receives $p\mathbf{1}\{R \geq p\}$. A prepared challenger with $\theta \geq p$ wins at $\theta > R$ and pays $\max\{p, R\}$. A loss makes the incumbent pay θ. The target receives $\max\{p, \min(R, \theta)\}$, and the challenger keeps $(\theta - \max\{p, R\})_+$. At $\theta < p$ the challenger does not bid. The target again receives $p\mathbf{1}\{R \geq p\}$, and the challenger keeps nothing.

The benchmark. Let R be uniform on $[0, r]$ with $0 < p < \ell < r < h$, as in (2). Then $t_0 = \Pr(R \geq p)\, p = p(1 - p/r)$. Since $R \leq r < h$, the high-value challenger always wins and pays $\max\{p, R\}$. The low-value challenger wins and pays $\max\{p, R\}$ when $R < \ell$; it loses when $R \geq \ell$.

Integration over R gives

$$\begin{aligned}
t_H &= \frac{1}{r}\Big[\int_0^p p\,dy + \int_p^r y\,dy\Big] = \frac{1}{r}\Big[p^2 + \frac{r^2-p^2}{2}\Big] = \frac{r}{2} + \frac{p^2}{2r},\\
t_L &= \frac{1}{r}\Big[\int_0^p p\,dy + \int_p^\ell y\,dy + \int_\ell^r \ell\,dy\Big] = \frac{1}{r}\Big[p^2 + \frac{\ell^2-p^2}{2} + \ell(r-\ell)\Big] = \ell - \frac{\ell^2-p^2}{2r},\\
g_H &= \frac{1}{r}\int_0^r \big(h - \max\{p,y\}\big)\,dy = h - t_H = h - \frac{r}{2} - \frac{p^2}{2r},\\
g_L &= \frac{1}{r}\Big[\int_0^p (\ell-p)\,dy + \int_p^\ell (\ell-y)\,dy\Big] = \frac{1}{r}\Big[p(\ell-p) + \frac{(\ell-p)^2}{2}\Big] = \frac{\ell^2-p^2}{2r}.
\end{aligned} \tag{A.1}$$

The spread is

$$\Delta_T(r) = t_H - t_L = \frac{r}{2} + \frac{p^2}{2r} - \ell + \frac{\ell^2-p^2}{2r} = \frac{r^2-2\ell r+\ell^2}{2r} = \frac{(r-\ell)^2}{2r}.$$

The sums $t_H + g_H = h$ and $t_L + g_L = \ell$ check the split. These are the expressions in (6).

Differentiation in r gives

$$\Delta_T'(r) = \frac{2(r-\ell)\cdot 2r - 2(r-\ell)^2}{4r^2} = \frac{(r-\ell)(r+\ell)}{2r^2} = \frac{1}{2} - \frac{\ell^2}{2r^2},$$
$$g_H'(r) = -\frac{1}{2} + \frac{p^2}{2r^2}, \qquad g_L'(r) = -\frac{\ell^2-p^2}{2r^2}.$$

The signs in (7) follow from $r > \ell$, $r > p$, and $\ell > p$. For every $\mu \in [0,1]$, $\partial B_r(\mu)/\partial r = (1-\mu)g_L'(r) + \mu g_H'(r) < 0$.

General classes and a continuous F. Write the payments as integrals of indicators. For $\theta \geq p$,

$$\max\{p, \min(R,\theta)\} = p + \int_p^\theta \mathbf{1}\{R > y\}\,dy, \qquad (\theta - \max\{p,R\})_+ = \int_p^\theta \mathbf{1}\{R < y\}\,dy. \tag{A.2}$$

The two indicators sum to one except at the single point $y = R$, which gives $t(\theta) + g(\theta) = \theta$.

Every integrand in (A.2) is nonnegative and jointly measurable in (R, y). Tonelli's theorem therefore moves the expectation over R inside each integral. Since F is continuous, $\Pr(R > y) = 1 - F(y)$. This gives $t(\theta) = p + \int_p^\theta [1 - F(y)]\,dy$ and $g(\theta) = \int_p^\theta F(y)\,dy$ for $\theta \geq p$. For $\theta < p$, $t(\theta) = t_0 = p[1 - F(p)]$ and $g(\theta) = 0$.

At $\theta = p$, t jumps by $pF(p)$. When the incumbent fails the reserve, an admissible challenger buys at p, and an inadmissible one leaves the target unsold. Hence, for every θ,

$$t(\theta) = t_0 + pF(p)\mathbf{1}\{\theta \geq p\} + \int_p^\infty \mathbf{1}\{y \leq \theta\}[1 - F(y)]\,dy, \qquad g(\theta) = \int_p^\infty \mathbf{1}\{y \leq \theta\}F(y)\,dy.$$

Take the class expectation and apply Tonelli's theorem again, with $\mathbb{E}_j \mathbf{1}\{\theta \geq p\} = 1 - G_j(p^-)$ and $\Pr_j(\theta \geq y) = 1 - G_j(y^-)$. The functions $G_j(y^-)$ and $G_j(y)$ differ at countably many y, a Lebesgue-null set, so $G_j(y)$ can replace $G_j(y^-)$ inside the integral. This gives

$$t_j = t_0 + pF(p)[1 - G_j(p^-)] + \int_p^\infty [1 - G_j(y)][1 - F(y)]\, dy$$

and g_j in (8). The difference of the two classes gives Δ_T in (8). Each integral is finite, since $0 \leq \int_p^\infty [1 - G_j(y)]\, dy \leq \mathbb{E}_j \theta < \infty$ and $0 \leq F \leq 1$.

Since $\mathbb{E}_j[\max\{\theta, p\}] = p + \int_p^\infty [1 - G_j(y)]\, dy$, the sum of the two lines of (8) gives the split in its general form,

$$\Delta_T(F) + g_H(F) - g_L(F) = pF(p)D(p^-) + \mathbb{E}_H[\max\{\theta, p\}] - \mathbb{E}_L[\max\{\theta, p\}].$$

The reserve term vanishes under equal admissibility, $G_H(p^-) = G_L(p^-)$. It also vanishes when the incumbent always meets the reserve, $F(p) = 0$.

Part (a). In (8), the weight $1 - G_j$ on F is nonnegative. The pointwise decrease from F to $\tilde{F}$ therefore lowers g_j by $\int_p^\infty [1 - G_j(y)][F(y) - \tilde{F}(y)]\, dy \geq 0$. The decrease is strict exactly when this integral is positive. Gross profit at a fixed belief μ is $(1 - \mu)g_L + \mu g_H$, an average with fixed weights, so it falls as well.

Part (b). The difference of (8) at the two distributions is

$$\Delta_T(\tilde{F}) - \Delta_T(F) = \int_p^\infty [F(y) - \tilde{F}(y)]\, D(y)\, dy - p[F(p) - \tilde{F}(p)]\, D(p^-).$$

Under the class ranking (1), $D(p^-) = 0$, so the second term is zero. Also $D(y) \geq 0$ and $F(y) - \tilde{F}(y) \geq 0$ for every y, so the integrand is nonnegative. The spread therefore rises, strictly exactly when the integral is positive.

Equal admissibility also makes the second term zero. Together with $D(y) \geq 0$ for $y > p$, it gives the same conclusion. □

Proposition 1(c) (the weakest restriction). *Let G_H and G_L have finite means. The spread weakly increases under every first-order stochastic strengthening of every continuous F if and only if $D(y) \geq 0$ for every $y > p$ and $D(p^-) \leq 0$. When neither class has an atom at p, this condition is exactly first-order dominance above the reserve together with equal admissibility, $G_H(p^-) = G_L(p^-)$.*

Proof of Proposition 1(c). Fix the incumbent's value R. Suppose $R \geq p$. The target receives $\min\{R, \max\{\theta, p\}\}$ for every challenger value. An admissible challenger and the incumbent both

bid at least p, and an inadmissible challenger leaves the incumbent to pay p. Suppose $R < p$. The target receives $p\mathbf{1}\{\theta \geq p\}$.

For $R \geq p$, $\mathbb{E}_j[\min\{R, \max\{\theta, p\}\}] = p + \int_p^R [1 - G_j(y)]\, dy$. The expected difference in proceeds between the classes at incumbent value R is therefore

$$\xi(R) = \begin{cases} pD(p^-), & R < p, \\ \int_p^R D(y)\, dy, & R \geq p, \end{cases}$$

and $\Delta_T(F) = \mathbb{E}_F[\xi(R)]$.

The function ξ is bounded, since $\int_p^\infty |D(y)|\, dy \leq \mathbb{E}_H\theta + \mathbb{E}_L\theta < \infty$. It is constant below p, continuous on $[p, \infty)$, and has derivative D almost everywhere above p. It moves from $pD(p^-)$ to $\xi(p) = 0$ at p. Thus ξ is nondecreasing if and only if $D \geq 0$ almost everywhere above p and $D(p^-) \leq 0$.

If ξ is nondecreasing, every first-order stochastic strengthening raises $\mathbb{E}_F[\xi(R)]$. If it is not, continuity on each side of p gives values $0 < R_1 < R_2$, both different from p, with $\xi(R_1) > \xi(R_2)$. Let F be uniform on $[R_1 - \delta, R_1 + \delta]$, and let $\tilde{F}$ be the same law shifted to R_2. Then $\tilde{F} \leq F$, and as $\delta \to 0$, $\Delta_T(\tilde{F}) - \Delta_T(F) \to \xi(R_2) - \xi(R_1) < 0$.

Suppose the support must start at zero. Mix the same small uniform component on $[0, \bar{r}]$ into both laws. Both mixtures are continuous, and the mixture scales the difference by a positive factor and keeps its sign.

Distribution functions are right-continuous. Thus $D \geq 0$ almost everywhere above p is the same as $D(y) \geq 0$ for every $y > p$, and it implies $D(p) \geq 0$. When neither class has an atom at p, $D(p^-) = D(p) \geq 0$. The condition then holds exactly when $D(p^-) = 0$, which with the ranking above the reserve is equal admissibility. □

Scope of the class ranking. The proofs in this subsection use the class ranking (1) only through $D(y) \geq 0$ for $y > p$ and $D(p^-) = 0$. These two facts give $g_H \geq g_L$ and $\Delta_T \geq 0$ in (8), the comparison in part (b), and the split identity of Section 3.1. The same results therefore hold under the ranking above the reserve together with equal admissibility.

For the point masses with $p < \ell$, $D(y) = \mathbf{1}\{\ell \leq y < h\}$ and $D(p^-) = 0$. Also $\xi(R) = (R - \ell)_+$ for $R < h$. Equation (8) then gives $\Delta_T = \int_\ell^h [1 - F(y)]\, dy$, which equals $\int_\ell^{\bar{r}} [1 - F(y)]\, dy$ at $\bar{r} < h$. It also gives $g_j = \int_p^\theta F(y)\, dy$ at $\theta = h$ and $\theta = \ell$.

For the uniform law on $[0, r]$, these integrals reproduce (A.1). The kernel ξ belongs to the cash second-price rule with reserve. Section 7.2 changes the rule and the kernel with it.

A.2 Beliefs and trading

This subsection states and proves the results behind Sections 3.2 and 3.3. Each result holds for every order profile, in or out of equilibrium. Every later proof uses them in that form.

Conventions for Appendices A.2 to A.9. The conditions (E1) to (E6) are those of Definition A.1.

- A *profile* is a pair (σ_H, σ_L) of order distributions, pure or mixed, with wrong-signed orders allowed. A *candidate* is a tuple $\mathcal{E}$ that satisfies (E1) to (E4) and (E6) for some profile. An equilibrium is a candidate that also satisfies (E5).
- A deviation by the investor holds the candidate's price function P and preparation rule ζ fixed. It changes only the order that reaches them.
- Proofs use the deviation comparison to test a candidate equilibrium, and they solve the equilibrium again in each economy to compare outcomes across economies.
- A uniqueness proof derives its bounds for every candidate. It does not use the proposed tuple.
- A verification checks each condition by direct computation against the stated tuple. It does not cite a lemma whose premises include the condition under check.
- Write $w_j = t_j - t_0$ for the gain in expected proceeds that entry brings in class j, and $w(\mu) = (1-\mu)w_L + \mu w_H = t_L - t_0 + \Delta_T \mu$ for its average at the belief μ. Proposition 1 and Section 3.1 give $w_H \geq w_L \geq 0$.
- A step that uses a script names the script and its run log. Every script path in Appendix A is relative to the folder `audit/slot_manuscript_2026_10_23/scripts/` of the project repository.

Orders and posteriors. Write δ_q for the point mass at the order q, and $b_0 = (1-\lambda)(1-2\beta)$ for the probability of a liquidity abstention. Under any order distributions σ_H, σ_L, Bayes' rule in odds form reads

$$\frac{\mu(x)}{1-\mu(x)} = \frac{\pi}{1-\pi} \cdot \frac{f_H(x)}{f_L(x)}. \tag{A.3}$$

Lemma A.1 (Order probabilities and beliefs)**.** *Fix any order distributions σ_H and σ_L.*

(a) Full support. $f_j(x) \geq (1-\lambda)\eta(x) > 0$ *for every order x and class j.*

(b) Nonzero orders. *For $x \in \{-1, 1\}$, $b \leq f_j(x) \leq b + \lambda$. Hence $m(\pi) \leq \mu(x) \leq M(\pi)$.*

(c) The zero order. $b_0 \leq f_j(0) \leq b_0 + \lambda$, *and $b_0 \geq b$. Hence $m(\pi) \leq \mu(0) \leq M(\pi)$.*

(d) Full orders. *If $\sigma_H = \delta_1$ and $\sigma_L = \delta_{-1}$, the posteriors after the orders $(-1, 0, 1)$ are $\big(m(\pi), \pi, M(\pi)\big)$.*

Proof of Lemma A.1. Fix any profile (σ_H, σ_L). By (A.3), the posterior $\mu(x)$ increases strictly in the likelihood ratio $f_H(x)/f_L(x)$. The odds of $M(\pi)$ are $\pi(1+u)/(1-\pi)$, and the odds of $m(\pi)$ are $\pi/[(1-\pi)(1+u)]$.

Part (a). The liquidity trader fills the slot with probability $1-\lambda > 0$. It places the order x with probability $\eta(x)$, and $\eta(x) > 0$ for every x because $0 < \beta < 1/2$. The informed term $\lambda\sigma_j(x)$ is nonnegative. Hence $f_j(x) \geq (1-\lambda)\eta(x) > 0$ for every order and class. Every order has positive probability in both classes, so (E1) defines $\mu(x)$ at every order, and no off-path belief enters any condition. Also $0 < \mu(x) < 1$, since $0 < \pi < 1$.

Part (b). For $x = \pm 1$, $(1-\lambda)\eta(x) = b$ and $0 \leq \lambda\sigma_j(x) \leq \lambda$. Hence $b \leq f_j(x) \leq b + \lambda$, and

$$\frac{1}{1+u} = \frac{b}{b+\lambda} \leq \frac{f_H(x)}{f_L(x)} \leq \frac{b+\lambda}{b} = 1+u.$$

By (A.3), $m(\pi) \leq \mu(x) \leq M(\pi)$. This step uses only $0 < \beta < 1/2$.

Part (c). For $x = 0$, $(1-\lambda)\eta(0) = b_0$, so $b_0 \leq f_j(0) \leq b_0 + \lambda$, and the likelihood ratio lies in $[(1+\lambda/b_0)^{-1}, 1+\lambda/b_0]$. The inequality $b_0 \geq b$ is $1 - 2\beta \geq \beta$, that is $\beta \leq 1/3$. Then $\lambda/b_0 \leq \lambda/b = u$, and the argument of part (b) gives $m(\pi) \leq \mu(0) \leq M(\pi)$. If $\beta > 1/3$, then $b_0 < b$. The profile $\sigma_H = \delta_0$ with $\sigma_L(0) = 0$ gives $f_H(0) = b_0 + \lambda$ and $f_L(0) = b_0$. The ratio is $1 + \lambda/b_0 > 1 + u$, so $\mu(0) > M(\pi)$, and the bound fails.

Part (d). The profile $\sigma_H = \delta_1$, $\sigma_L = \delta_{-1}$ gives $f_H = (b,\ b_0,\ b+\lambda)$ and $f_L = (b+\lambda,\ b_0,\ b)$ on the orders $(-1, 0, 1)$. The likelihood ratios are $(1+u)^{-1}$, 1 and $1+u$, so $(\mu(-1), \mu(0), \mu(1)) = (m(\pi), \pi, M(\pi))$.

Beliefs at a price. Fix a price $\bar{P}$ in the image of P, and let $X(\bar{P}) = \{x : P(x) = \bar{P}\}$. By (E2), the challenger's belief $\bar{\mu}(\bar{P})$ has the odds

$$\frac{\bar{\mu}(\bar{P})}{1-\bar{\mu}(\bar{P})} = \frac{\pi}{1-\pi} \cdot \frac{\sum_{x \in X(\bar{P})} f_H(x)}{\sum_{x \in X(\bar{P})} f_L(x)}.$$

A ratio of two sums of positive terms lies between the smallest and the largest ratio of its terms. Hence $\bar{\mu}(\bar{P})$ lies between the smallest and the largest $\mu(x)$ over $X(\bar{P})$. Every bound of parts (b) and (c) therefore holds for $\bar{\mu}(\bar{P})$.

Only part (c) uses $\beta \leq 1/3$. Parts (a), (b) and (d) hold for every $\beta < 1/2$. □

Corollary A.1 (High-cost preparation under every profile)**.** *Let $g_H \geq g_L$. If $B(M(\pi)) < c_H$, then no profile that satisfies (E1)–(E3), in or out of equilibrium, has high-cost preparation after any order.*

Proof of Corollary A.1. Let $\beta \leq 1/3$, $g_H \geq g_L$ and $B(M(\pi)) < c_H$. Fix any profile and any tuple

that satisfies (E1) to (E3). By Lemma A.1(b)–(c), $\mu(x) \le M(\pi)$ at every order, and by the step on beliefs at a price, $\bar{\mu}(\bar{P}) \le M(\pi)$ at every price. Since $g_H \ge g_L$, $B(\mu) = g_L + \mu(g_H - g_L)$ is nondecreasing in μ. Hence $B(\bar{\mu}(\bar{P})) \le B(M(\pi)) < c_H$ at every price, and (E3) gives $\zeta(\bar{P}) = 0$ with no tie. The conclusion holds under both challenger rules. The bound at the zero order is the only use of $\beta \le 1/3$ in this proof. □

Lemma A.2 (Entry and prices)**.** *Let $\Delta_T > 0$ and $g_H > g_L$. Fix a tuple that satisfies (E1)–(E4) and (E6).*

(a) Floor. *$e(x) \ge \rho$ for every order x.*

(b) Price. *$P(x) = t_0 + e(x)\big[t_L - t_0 + \Delta_T\, \mu(x)\big]$ for every order x.*

(c) Sufficiency. *If $P(x) = P(x')$, then $\mu(x) = \mu(x')$. The challenger's belief at the price $P(x)$ is $\mu(x)$.*

(d) Entry. *$e(x) = \rho$ if $\mu(x) < \tau$, and $e(x) = 1$ if $\mu(x) \ge \tau$.*

(e) Monotonicity. *If $\mu(x) > \mu(x')$, then $e(x) \ge e(x')$ and $P(x) - P(x') \ge \rho\Delta_T\,[\mu(x) - \mu(x')] > 0$.*

Under rule CM, part (d) at $\mu(x) = \tau$ allows any $e(x) \in [\rho, 1]$; the other parts hold under both challenger rules.

Proof of Lemma A.2. Fix a candidate, and let $\Delta_T > 0$ and $g_H > g_L$. The proof has a preliminary step and five parts.

Step 0: the terminal payoff given the class and the order. The challenger's choice depends only on the price $P(x)$ and its cost C. The cost is independent of the class, the values, the incumbent's value and the slot draws. Given the class j and the order x, the challenger therefore prepares with probability $e(x) = \rho + (1 - \rho)\zeta(P(x))$, and this event is independent of (θ, R). The order depends on the class and the slot draws only, so it carries no information about (θ, R) given j. Bids are truthful by (E6). With entry, expected proceeds in class j are t_j; without entry, they are t_0. Hence

$$\mathbb{E}[V_T \mid j, X = x] = t_0 + e(x)\, w_j. \tag{A.4}$$

Part (a): the floor. The zero-cost challenger prepares after every price (Definition A.1), and it has probability ρ. Hence $e(x) \ge \rho$ at every order.

Part (b): the price. Average (A.4) over the class with the weights $\mu(x)$ and $1 - \mu(x)$ of (E1). Condition (E4) gives $P(x) = t_0 + e(x)\, w(\mu(x))$, which is (5). This step uses only (E1), (E4), the truthful bids (E6) of Step 0, and the independence of the cost.

Part (c): sufficiency. Let $P(x) = P(x')$. Entry depends on the order only through the price, so $e(x) = e(x') =: \bar{e}$, and $\bar{e} \ge \rho > 0$ by part (a). Part (b) gives $\bar{e}\,\Delta_T\,[\mu(x) - \mu(x')] = 0$, so

$\mu(x) = \mu(x')$. By the step on beliefs at a price in the proof of Lemma A.1, $\bar{\mu}(P(x))$ lies between the smallest and the largest posterior of the orders with the price $P(x)$. These posteriors are equal, so $\bar{\mu}(P(x)) = \mu(x)$. The price therefore reveals the posterior.

Part (d): entry. Since $g_H > g_L$, $B(\mu) > c_H$ exactly when $\mu > \tau$. By part (c), the challenger's gross profit at $P(x)$ is $B(\mu(x))$. Condition (E3) gives $e(x) = \rho$ if $\mu(x) < \tau$ and $e(x) = 1$ if $\mu(x) > \tau$. At $\mu(x) = \tau$, $e(x) = 1$ under rule C1, and $e(x)$ can take any value in $[\rho, 1]$ under rule CM.

Part (e): monotonicity. Let $\mu(x) > \mu(x')$. If $\mu(x') < \tau$, then $e(x') = \rho \leq e(x)$. If $\mu(x') \geq \tau$, then $\mu(x) > \tau$ and $e(x) = 1 \geq e(x')$. Part (b) gives

$$P(x) - P(x') = e(x)\,\Delta_T\,[\mu(x) - \mu(x')] + [e(x) - e(x')]\,w(\mu(x')). \tag{A.5}$$

The second term is nonnegative, because $w(\mu) \geq w_L \geq 0$. The first term is at least $\rho\Delta_T[\mu(x) - \mu(x')] > 0$. Prices therefore rise strictly with the posterior. □

Constructed tuples. The identity (A.5) uses only the price formula. It therefore applies to any prices of the form $P(x) = t_0 + e(x)\,w(\mu(x))$ with $\Delta_T > 0$ and with entry in $[\rho, 1]$ that is nondecreasing in the posterior. The verifications below use it in this form, before any condition is checked: a stated tuple has distinct prices at distinct posteriors, and equal prices at equal posteriors. The belief $\bar{\mu}(P(x)) = \mu(x)$ then follows from (E2) by direct computation.

Lemma A.3 (Informed trading)**.** *Let $\Delta_T \geq 0$. Fix a tuple that satisfies (E1), (E4) and (E6), with entry $e(x) \in [\rho, 1]$ after every order. Conditional on its arrival, the investor in each class earns*

$$U_H(1) = e(1)\Delta_T[1 - \mu(1)] - k, \qquad U_H(0) = 0, \qquad U_H(-1) = -e(-1)\Delta_T[1 - \mu(-1)] - k,$$
$$U_L(-1) = e(-1)\Delta_T\,\mu(-1) - k, \qquad U_L(0) = 0, \qquad U_L(1) = -e(1)\Delta_T\,\mu(1) - k.$$

Write $K_H = e(1)\Delta_T[1 - \mu(1)]$ and $K_L = e(-1)\Delta_T\,\mu(-1)$ for the gross returns of the two correct orders.

(a) A wrong-signed order earns at most $-k$, so abstention strictly beats it.

(b) Class H buys if $K_H > k$ and abstains if $K_H < k$. At $K_H = k$, the purchase and abstention are its two best orders. Class L sells if $K_L > k$ and abstains if $K_L < k$. At $K_L = k$, the sale and abstention are its two best orders.

(c) Under every candidate profile, $\rho\Delta_T[1 - M(\pi)] \leq K_H \leq \Delta_T$ and $\rho\Delta_T\,m(\pi) \leq K_L \leq \Delta_T$. Hence $\rho\,\kappa(\pi)\Delta_T \leq K_j \leq \Delta_T$, where $\kappa(\pi) = \min\{m(\pi), 1 - M(\pi)\}$.

(d) K_H depends on the candidate only through $\mu(1)$ and $e(1)$, and K_L only through $\mu(-1)$ and $e(-1)$.

Proof of Lemma A.3. Fix a tuple that satisfies (E1), (E4) and (E6), with $e(x) \in [\rho, 1]$ after every order, and let $\Delta_T \geq 0$. A deviation to the order q reaches the tuple's price $P(q)$ and entry $e(q)$. By (A.4) and part (b) of the proof of Lemma A.2,

$$\mathbb{E}[V_T \mid H, X = q] - P(q) = e(q)\,[w_H - w(\mu(q))] = e(q)\,\Delta_T\,[1 - \mu(q)],$$
$$P(q) - \mathbb{E}[V_T \mid L, X = q] = e(q)\,[w(\mu(q)) - w_L] = e(q)\,\Delta_T\,\mu(q).$$

The proceeds t_0 and the term $e(q)w_L$ cancel. Multiply by q and subtract $k|q|$. This gives the six payoffs of the lemma, with $K_H = e(1)\Delta_T[1 - \mu(1)]$ and $K_L = e(-1)\Delta_T\mu(-1)$.

Part (a). Since $e \geq 0$, $\Delta_T \geq 0$ and $0 \leq \mu \leq 1$, the payoffs $U_H(-1) = -e(-1)\Delta_T[1 - \mu(-1)] - k$ and $U_L(1) = -e(1)\Delta_T\mu(1) - k$ are at most $-k < 0 = U_j(0)$. No class puts weight on a wrong-signed order under (E5).

Part (b). By part (a), class H compares the purchase with abstention, and $U_H(1) - U_H(0) = K_H - k$. Class H buys with probability one if $K_H > k$ and abstains if $K_H < k$. If $K_H = k$, the purchase and abstention are its two maximizers: rule IM allows any mixture of them, and rule I0 selects abstention. Class L compares the sale with abstention, and $U_L(-1) - U_L(0) = K_L - k$. Class L sells with probability one if $K_L > k$ and abstains if $K_L < k$. If $K_L = k$, the sale and abstention are its two maximizers: rule IM allows any mixture of them, and rule I0 selects abstention.

Part (c). By Lemma A.1(b), $\mu(1) \leq M(\pi)$ and $\mu(-1) \geq m(\pi)$ under every profile, for every $\beta < 1/2$. Also $\rho \leq e \leq 1$ and $0 \leq \mu \leq 1$. Hence

$$\rho\,\kappa(\pi)\,\Delta_T \leq \rho\,[1 - M(\pi)]\,\Delta_T \leq K_H \leq \Delta_T, \qquad \rho\,\kappa(\pi)\,\Delta_T \leq \rho\,m(\pi)\,\Delta_T \leq K_L \leq \Delta_T. \tag{A.6}$$

The upper bounds use only $e \leq 1$ and $0 \leq \mu \leq 1$.

Part (d). The return K_H depends on the tuple only through $(\mu(1), e(1))$, and K_L only through $(\mu(-1), e(-1))$. Entry after the zero order enters no payoff. The ex ante payoff of the investor is $\lambda\,U_j(q)$, and the factor $\lambda > 0$ changes no comparison. □

Where $\beta \leq 1/3$ enters. The bound $\beta \leq 1/3$ enters only at the zero order, and through it only Corollary A.1 and the bound $\bar{\mu} \leq M(\pi)$ at a price that pools the zero order. The bounds (A.6) use the nonzero orders only. The proofs of Propositions 2 and 3 and of Corollary A.2 use parts (a), (b) and (d) of Lemma A.1 only, so these results hold for every $\beta < 1/2$.

A.3 Competition creates competition

This subsection proves Proposition 2 for general classes and priors. It then states and proves Proposition A.1, the admissible region at the benchmark. Write F_1 for a strengthening of F_0 and F_2

for a strengthening of F_1. For $i \in \{0, 1, 2\}$, write $B_i = B_{F_i}$, $\Delta_i = \Delta_T(F_i)$, and $t_{0,i}$, $t_{j,i}$, $g_{j,i}$, $w_i(\mu)$ for the objects at F_i. Also write $M = M(\pi)$, $m = m(\pi)$ and $\kappa = \kappa(\pi)$. Under the class ranking (1) every value meets the reserve, so $W_H(F) = \mathbb{E}_H[F(\theta)]$, the probability that a prepared high-class challenger outbids the incumbent. Value ties have probability zero because F is continuous.

Proof of Proposition 2. The proof has seven steps. Steps 1, 3 and 5 state the equilibrium tuple of each economy, check every condition against it, and prove uniqueness from bounds on every candidate.

Step 0: implied inequalities. Proposition 1(a) gives $B_1(\pi) \le B_0(\pi)$, so (H1) gives $B_1(\pi) < c_H < B_1(M)$. Since $M > \pi$, $B_1(M) - B_1(\pi) = (M - \pi)(g_{H,1} - g_{L,1}) > 0$. Hence $g_{H,1} > g_{L,1}$, the threshold τ_1 at F_1 is defined, and $\pi < \tau_1 < M$. By (H2), $\Delta_1 > k/(\rho\kappa) > 0$. Proposition 1(b) gives $\Delta_2 \ge \Delta_1$, so $\rho\kappa\Delta_2 > k$. Proposition 1 also gives $g_{H,2} \ge g_{L,2}$, so $B_2(\mu) \le B_2(M) < c_H$ for every $\mu \le M$. The zero-cost challenger prepares after every price, so Lemma A.2(a) gives $e \ge \rho$ in every candidate of each economy.

Step 1: the weak incumbent. Fix any candidate at F_0. By Lemma A.3(a) and the upper bounds in (A.6), each correctly signed order earns at most $\Delta_0 - k < 0$, and each wrong-signed order earns at most $-k$. Abstention is therefore the unique maximizer of U_H and of U_L, and (E5) gives $\sigma_H = \sigma_L = \delta_0$ under both investor rules. Then $f_H(x) = f_L(x) = (1 - \lambda)\eta(x) + \lambda\mathbf{1}\{x = 0\}$, so (E1) gives $\mu(x) = \pi$ at every order, and (E2) gives $\bar{\mu}(\bar{P}) = \pi$ at every price. Since $B_0(\pi) < c_H$, (E3) gives $\zeta(\bar{P}) = 0$ with no tie, so $e(x) = \rho$. Condition (E4) and (5) then fix one price for every order. Every equilibrium at F_0 is therefore the tuple

$$\mathcal{E}_0 : \quad \begin{aligned} &\sigma_H = \sigma_L = \delta_0, \quad \mu(x) = \pi, \quad P(x) = \bar{P}_0 := t_{0,0} + \rho\,[t_{L,0} - t_{0,0} + \Delta_0\pi], \\ &\bar{\mu}(\bar{P}_0) = \pi, \quad \zeta(\bar{P}_0) = 0, \end{aligned}$$

with truthful bids.

Verification of $\mathcal{E}_0$. (E1): $f_H = f_L$, so the posterior at every order is π. (E2): the single price pools all orders, and the odds of $\bar{\mu}(\bar{P}_0)$ are $\pi \sum_x f_H(x)/[(1 - \pi)\sum_x f_L(x)] = \pi/(1 - \pi)$. (E3): $B_0(\pi) < c_H$, strictly. (E4): by (A.4) with $e \equiv \rho$, $\mathbb{E}[V_T \mid X = x] = t_{0,0} + \rho\, w_0(\pi) = \bar{P}_0$. (E5): at $\mathcal{E}_0$, $K_H = \rho\Delta_0(1 - \pi) < k$ and $K_L = \rho\Delta_0\pi < k$, so $U_H(1) < 0 = U_H(0)$ and $U_L(-1) < 0 = U_L(0)$, and the wrong-signed payoffs are at most $-k$. (E6): Appendix A.1. Entry is ρ, and high-class ownership is $\pi\rho W_H(F_0)$.

Step 2: the strong incumbent, orders in every candidate. Fix any candidate at F_1. By (A.6), $K_H \ge \rho\kappa\Delta_1 > k$ and $K_L \ge \rho\kappa\Delta_1 > k$. By Lemma A.3(a)–(b), the purchase is the unique maximizer of class H, and the sale is the unique maximizer of class L. Every equilibrium at F_1 has $\sigma_H = \delta_1$ and $\sigma_L = \delta_{-1}$, under both investor rules. These bounds use only the nonzero orders, so they hold for every $\beta < 1/2$.

Step 3: the strong incumbent, the tuple. Lemma A.1(d) gives the posteriors (m, π, M) after the orders $(-1, 0, 1)$. Step 0 gives $\Delta_1 > 0$ and $g_{H,1} > g_{L,1}$, so Lemma A.2 applies. Since $B_1(M) > c_H$, $B_1(\pi) < c_H$ and $B_1(m) < B_1(\pi)$, Lemma A.2(d) gives $e(1) = 1$ and $e(0) = e(-1) = \rho$, with no tie. Every equilibrium at F_1 is therefore the tuple

$$\mathcal{E}_1: \quad \begin{aligned} &\sigma_H = \delta_1, \quad \sigma_L = \delta_{-1}, \quad (\mu(-1), \mu(0), \mu(1)) = (m, \pi, M), \\ &P(x) = t_{0,1} + e(x)\, w_1(\mu(x)), \quad \bar{\mu}(P(x)) = \mu(x), \quad \zeta(P(x)) = \mathbf{1}\{x = 1\}, \end{aligned}$$

with $(e(-1), e(0), e(1)) = (\rho, \rho, 1)$ and truthful bids.

Verification of $\mathcal{E}_1$. (E1): the likelihoods $f_H = (b, b_0, b + \lambda)$ and $f_L = (b + \lambda, b_0, b)$ give (m, π, M). (E2): the stated entry is nondecreasing in the posterior and lies in $[\rho, 1]$, and $\Delta_1 > 0$. By the paragraph on constructed tuples in Appendix A.2, the three prices are strictly increasing in the posterior, so $P(-1) < P(0) < P(1)$. Each price has one order in its preimage, so $\bar{\mu}(P(x)) = \mu(x)$. (E3): $B_1(M) > c_H$, $B_1(\pi) < c_H$ and $B_1(m) < c_H$, all strict. (E4): (A.4) with the stated entry. (E5): at $\mathcal{E}_1$, $K_H = \Delta_1(1 - M) \geq \rho\kappa\Delta_1 > k$ and $K_L = \rho\Delta_1 m \geq \rho\kappa\Delta_1 > k$. Hence $U_H(1) > 0 = U_H(0) > U_H(-1)$ and $U_L(-1) > 0 = U_L(0) > U_L(1)$. (E6): Appendix A.1.

Step 4: entry and ownership at F_1. Class H places the order 1 with probability $f_H(1) = b + \lambda$, through a liquidity purchase or an informed purchase. Class L places it with probability $f_L(1) = b$. Only the order 1 recruits the high-cost challenger. Hence

$$e_H = \rho + (1-\rho)(b+\lambda), \qquad e_L = \rho + (1-\rho)b, \qquad \mathsf{E}_1 = \pi e_H + (1-\pi)e_L = \rho + (1-\rho)(b + \pi\lambda) > \rho,$$

since $b > 0$. Preparation depends on the order and the cost, and both are independent of (θ, R) given the class. A prepared high-class challenger acquires the target when $\theta > R$, with probability $W_H(F_1)$. So $\mathsf{O}_{H,1} = \pi e_H W_H(F_1)$, and Step 1 gives $\mathsf{O}_{H,0} = \pi\rho W_H(F_0)$. Division by $\pi > 0$ shows that ownership rises exactly when $[\rho + (1 - \rho)(b + \lambda)]\, W_H(F_1) > \rho\, W_H(F_0)$. In the point-mass benchmark $h > r$ at every strength, so $W_H = 1$, and the condition reads $e_H > \rho$, which holds.

Step 5: the very strong incumbent. Fix any candidate at F_2. By (A.6), $K_H \geq \rho\kappa\Delta_2 > k$ and $K_L \geq \rho\kappa\Delta_2 > k$, so every equilibrium has full orders, and Lemma A.1(d) gives the posteriors (m, π, M). By Step 0, $B_2(\mu) \leq B_2(M) < c_H$ at each posterior, and every price belief is one of these posteriors or an average of them. So (E3) gives $\zeta \equiv 0$ with no tie, and $e \equiv \rho$. Every equilibrium at F_2 is therefore the tuple

$$\mathcal{E}_2: \quad \begin{aligned} &\sigma_H = \delta_1, \quad \sigma_L = \delta_{-1}, \quad (\mu(-1), \mu(0), \mu(1)) = (m, \pi, M), \\ &P(x) = t_{0,2} + \rho\, w_2(\mu(x)), \quad \bar{\mu}(P(x)) = \mu(x), \quad \zeta \equiv 0, \end{aligned}$$

with truthful bids. *Verification.* (E1): the likelihoods $f_H = (b, b_0, b + \lambda)$ and $f_L = (b + \lambda, b_0, b)$

give (m, π, M). (E2): since $\Delta_2 > 0$ and $m < \pi < M$, the three prices $t_{0,2} + \rho\, w_2(\mu(x))$ differ, so each price has one order in its preimage. (E3): $B_2(\mu(x)) < c_H$ at each posterior. (E4): (A.4) with $e \equiv \rho$. (E5): at $\mathcal{E}_2$, $K_H = \rho\Delta_2(1 - M) > k$ and $K_L = \rho\Delta_2 m > k$. (E6): Appendix A.1. The prices differ, so the price stays informative while entry equals ρ.

Step 6: tie rules. Every investor comparison in Steps 1, 2 and 5 is strict. Every high-cost comparison at an equilibrium belief is strict. No tie rule is used, so the four pairs of rules give the same unique outcome in each economy. □

Each hypothesis is used once.

- $B_0(\pi) < c_H$ gives entry ρ at F_0. Through Proposition 1(a), it also stops recruitment after the zero order at F_1.
- $c_H < B_1(M)$ gives recruitment after a purchase at F_1.
- $\Delta_0 < k$ defeats every informed order at F_0, against every candidate.
- $k < \rho\kappa\Delta_1$ forces full orders at F_1, and through Proposition 1(b) at F_2, against every candidate.
- $B_2(M) < c_H$ stops recruitment at F_2.

The bound at the zero order, Lemma A.1(c), is the only use of $\beta \leq 1/3$, and it enters no step.

The weaker condition $B_1(\pi) < c_H$ cannot replace $B_0(\pi) < c_H$. Keep calibration B and the benchmark values, but set $c_H = 47/10$ and $r_2 = 8$. Then $k - \Delta_0 = 0.003333$, $c_H - B_1(1/2) = 0.408333$, $B_1(M) - c_H = 1.517155$, $\rho\kappa\Delta_1 - k = 0.024824$ and $c_H - B_2(M) = 0.312465$ are positive, but $c_H - B_0(1/2) = -0.104167$. At r_0, Step 1 gives zero orders and the posterior $1/2$, and $B_0(1/2) = 1153/240 > c_H$. Both cost types prepare, so entry is 1.000000 at r_0. Entry then falls to 0.522757 at r_1. On the benchmark path the left side of (H2) also guards a conclusion. At $r = 1.55$ it fails, since $\Delta_T = 121/1240 > k$ there, and Proposition 3 gives two equilibria.

The tuples at calibration B. Table 3 lists $\mathcal{E}_0$, $\mathcal{E}_1$ and $\mathcal{E}_2$ at calibration B and the prior $\pi = 1/2$. In $\mathcal{E}_1$ the posteriors are $(0.268941,\ \pi,\ 0.731059)$ and the prices are 0.576074, 0.614583 and 1.362372. In $\mathcal{E}_2$ the prices are 0.610002, 0.664236 and 0.718471, and in $\mathcal{E}_0$ every order has the price 0.392708. The values come from `P1_core/p1_numbers.py` (run log `P1_core/p1_numbers.log`). This script evaluates every quantity that depends on e in outward interval arithmetic (`mpmath.iv`, 50 digits). The independent recomputation `XP1/xp1_numbers.py` (run log `XP1/xp1_numbers.log`) bounds e by its series through the term $1/110!$ with a rational bound on the tail. It records exact rational endpoints for every printed value in `XP1/xp1_registry_recompute.csv`. These rational endpoints are the certificate of each display.

The open set. For uniform incumbents, (H1) to (H3) are strict inequalities between continuous functions of $(r_0, r_1, r_2, c_H, k, \rho, \pi, \lambda, \beta)$. They define an open set in the domain $0 < \beta < 1/2$. The model adds the structural bound $\beta \leq 1/3$, and in that domain the set is relatively open. Proposition A.1 gives its section through the benchmark. The sufficient condition $(1+\rho)W_H(F_1) \geq 2\rho W_H(F_0) > 0$ for higher ownership implies the exact condition of Step 4 whenever $b + \lambda > 1/2$, because then $e_H > (1+\rho)/2$. At calibration B, $b + \lambda = 0.531737$.

Proposition A.1 (The admissible region)**.** *Take the benchmark auction and cost primitives, the strengths r_0, r_1, r_2, and a slot with $0 < \lambda < 1$ and $0 < \beta \leq 1/3$.*

(a) At the prior $1/2$, (H1)–(H3) hold if and only if

$$82/59 < u < 2686/1559, \quad \textit{that is,} \quad \tau(r_1) = 141/200 < M(1/2) < 4245/5804 = \tau(r_2).$$

Since $\lambda = u\beta/(1+u\beta)$ rises with u, the band is an open interval of λ at each β.

(b) The open band is exactly the set on which the strict hypotheses hold. If $u < 82/59$, entry at r_1 is ρ. At $u = 82/59$, orders are uniquely zero at r_0 and uniquely full at r_1 and r_2. Rule C1 gives every conclusion of Proposition 2. Under rule CM, entry after a purchase at r_1 can take any value in $[\rho, 1]$. If $2686/1559 \leq u < 19/3$, entry after a purchase at r_2 is one under rule C1. If $u \geq 19/3$, the right side of (H2) fails at r_1.

(c) With $1 + u = \mathrm{e}$ and the other benchmark inputs fixed, (H1)–(H3) hold if and only if

$$\frac{\tau(r_1)}{\tau(r_1) + [1 - \tau(r_1)]\,\mathrm{e}} < \pi < \frac{\tau(r_2)}{\tau(r_2) + [1 - \tau(r_2)]\,\mathrm{e}},$$

an interval with endpoints ≈ 0.467850 and ≈ 0.500424. Hypotheses (H1) and (H2) give parts (i) and (ii) of Proposition 2. They hold if and only if π lies above the same lower endpoint and below $\tau(r_0) = 195/308 \approx 0.633117$.

Proof of Proposition A.1. Take the benchmark primitives $(h, \ell, p) = (10, 1, 1/2)$, $(\rho, c_H, k) = (1/4, 6, 1/50)$ and the strengths $(r_0, r_1, r_2) = (6/5, 3, 18/5)$. Then $\tau(r) = (48r-3)/[4(20r-r^2-1)]$, with $\tau(r_0) = 195/308$, $\tau(r_1) = 141/200$ and $\tau(r_2) = 4245/5804$. Also $\Delta_T(r_0) = 1/60$, $\Delta_T(r_1) = 2/3$ and $\Delta_T(r_2) = 169/180$ by (6).

Part (a). At $\pi = 1/2$, $M = (1+u)/(2+u)$ and $\kappa = m = 1 - M = 1/(2+u)$. The left side of (H1) has the margin $c_H - B_{r_0}(1/2) = 1.195833$, and the left side of (H2) has the margin $k - \Delta_T(r_0) = 0.003333$. Neither depends on the slot. The right side of (H1) is $M > \tau(r_1)$, that is $200(1+u) > 141(2+u)$, that is $u > 82/59$. Hypothesis (H3) is $M < \tau(r_2)$, that is $5804(1+u) < 4245(2+u)$, that is $u < 2686/1559$. The right side of (H2) is $\rho(1-M)\Delta_T(r_1) > k$,

that is $(1 - M)/6 > 1/50$, that is $M < 22/25$, that is $u < 19/3$. The bound $u < 2686/1559$ implies it; at the benchmark slot $19/3 - u = 4.615052$. Hence (H1) to (H3) hold if and only if $82/59 < u < 2686/1559$, that is $\tau(r_1) < M < \tau(r_2)$. At fixed β, $\lambda = u\beta/(1 + u\beta)$ increases strictly in u, so the band is an open interval of λ.

Part (b). For $u < 19/3$, Step 2 of the proof of Proposition 2 holds at r_1, and at r_2 because $\Delta_T(r_2) > \Delta_T(r_1)$. Orders are full at both strengths, and the posteriors are $(m, 1/2, M)$. If $u < 82/59$, then $M < \tau(r_1)$, no order recruits at r_1, and entry at r_1 is ρ. If $2686/1559 \le u < 19/3$, then $M \ge \tau(r_2)$, and a purchase recruits at r_2 under rule C1. For $u \ge 19/3$, the right side of (H2) fails at r_1.

At $u = 82/59$, the left side of (H1), both sides of (H2) and (H3) still hold strictly. At r_0, every correct order earns at most $\Delta_T(r_0) - k = -1/300 < 0$, so orders are uniquely zero. At r_1 and r_2, orders are uniquely full by Step 2, and parts (i) and (iii) hold. A purchase at r_1 leaves the high-cost challenger indifferent, and rule C1 gives entry E_1. Rule CM allows entry after a purchase anywhere in $[\rho, 1]$, so entry at r_1 can take any value in $[\rho, \mathsf{E}_1]$.

Part (c). Fix $1 + u = \mathrm{e}$ and the other benchmark inputs, and vary π. The inequality $M(\pi) > \tau$ is $\pi/(1-\pi) > \tau/[(1-\tau)(1+u)]$, that is $\pi > \tau/[\tau + (1-\tau)\mathrm{e}]$. The right side of (H1) gives the lower endpoint of the proposition, and (H3) gives its upper endpoint. The left side of (H1) is $B_{r_0}(\pi) < c_H$, that is $\pi < \tau(r_0) = 195/308$. The left side of (H2) does not depend on π. The right side of (H2) is $\kappa(\pi) > k/(\rho\Delta_T(r_1)) = 3/25$: $m(\pi) > 3/25$ is $\pi > 3\mathrm{e}/(22 + 3\mathrm{e})$, and $1 - M(\pi) > 3/25$ is $\pi < 22/(22 + 3\mathrm{e})$, that is $\pi \in (0.270432, 0.729568)$ approximately. This interval contains the other two. The intersection is the interval of the proposition, with endpoints ≈ 0.467850 and ≈ 0.500424. Without (H3), the upper endpoint is $\tau(r_0) \approx 0.633117$. □

The region as a result. Table 2 gives the region in decimals, with the margin of the benchmark to each boundary. Two further prior windows complete it at calibration B. The trade bound $\kappa(\pi) > 3/25$ at $k = 1/50$ holds on $(0.270432, 0.729568)$. Some trade cost satisfies (H2) when $\Delta_T(r_0) < \rho\kappa\Delta_T(r_1)$, that is $\kappa(\pi) > 1/10$, on $(0.231969, 0.768031)$ (`P2_auction/derive.py`, run log `P2_auction/run.log`). The margins of every hypothesis at calibration B are in Table A.6.

A.4 Coexistence

This subsection states in full and proves Proposition 3, and it states and proves Corollary A.2. Throughout, the auction is the benchmark: point masses h and ℓ, $R \sim U[0, r]$, and the benchmark values of $(h, \ell, p, \rho, c_H, k)$. The path is $r \in [r_0, r_2]$. The slot is any pair (λ, β) with $0 < \lambda < 1$, $0 < \beta < 1/2$ and $\tau(r_1) < M(1/2) < \tau(r_2)$, the band of Proposition A.1. On the path, $g_H > g_L$ and $\Delta_T > 0$ because $\ell < r < h$, so Lemma A.2 applies at every strength.

The path and its thresholds

Two functions organize the path. The first is the gross return of a purchase that just reaches the posterior $\tau(r)$ and recruits,

$$J(r) = \Delta_T(r)\,[1-\tau(r)] = \frac{(r-\ell)^2\,[2r(h-c_H)-r^2-p^2]}{2r\,(2rh-r^2-\ell^2)}.$$

The second is $\mathfrak{r}(d) = \ell + d + \sqrt{d^2+2\ell d}$, the root above ℓ of $\Delta_T(r) = d$. Since $\Delta_T'(r) = \frac{1}{2} - \ell^2/(2r^2) > 0$ for $r > \ell$, $\Delta_T(r) \le d$ holds exactly when $r \le \mathfrak{r}(d)$.

Lemma A.4 (The path)**.** *(a) τ is strictly increasing on $[r_0, r_2]$, with $\tau(r_0) = 195/308 > 1/2$ and $\tau(r_2) = 4245/5804 < 3/4$.*

(b) J is strictly increasing on $[r_0, r_2]$, with $J(r_0) = 113/18480 < k$ and $\rho J(r_2) = 263471/4178880 > k$.

(c) $\Delta_T(r)\,[1-\tau(r)]^2$ is strictly increasing on $[r_0, 2]$.

(d) $\Delta_T(r)\,\tau(r)$ is strictly increasing on $[r_0, r_2]$.

Proof. (a) Write $\tau = N_\tau/D_\tau$ with $N_\tau = 2rc_H - \ell^2 + p^2$ and $D_\tau = 2rh - r^2 - \ell^2$. The derivative of τ has the sign of

$$N_\tau' D_\tau - N_\tau D_\tau' = 2c_H r^2 - 2(\ell^2 - p^2)\,r + 2h(\ell^2 - p^2) - 2c_H\ell^2,$$

which at the benchmark is $12r^2 - \frac{3}{2}r + 3$. Its discriminant $\frac{9}{4} - 144$ is negative, so it is positive for every r. Also $D_\tau = 20r - r^2 - 1 > 0$ on the path. The two values follow from the closed form $\tau(r) = (48r-3)/[4(20r-r^2-1)]$.

(b) The derivative J' has the sign of

$$Q_J(r) = 4r^6 - 160r^5 + 743r^4 - 12r^3 - 614r^2 + 40r - 1.$$

The script `P1_core/p1_certificates.py` (run log `P1_core/p1_certificates.log`) builds Q_J from the primitives in exact rational arithmetic. A Sturm sequence counts no real root of Q_J in $(r_0, r_2]$, and $Q_J(r_0) = 4634399/15625 > 0$. The two values follow from the closed form of J.

(c) The derivative of $\Delta_T(1-\tau)^2$ has the sign of the degree-10 polynomial

$$16r^{10} - 1280r^9 + 29256r^8 - 229344r^7 + 562917r^6 - 32808r^5 - 374607r^4 + 48456r^3 - 2685r^2 + 80r - 1.$$

The same script counts no real root in $(r_0, 2]$ with a Sturm sequence in exact rational arithmetic, and

it evaluates the polynomial at r_0 with a positive value. The independent script `XP1/xp1_signs.py` (run log `XP1/xp1_signs.log`) derives both polynomials again and repeats both Sturm counts.

(d) Δ_T and τ are positive and strictly increasing on the path, by (a) and $\Delta_T' > 0$.

Parts (a) and (d) are proved by hand. Parts (b) and (c) rest on the exact Sturm counts. □

At the prior $1/2$, with $M = (1+u)/(2+u)$ and $m = 1 - M$, define six strengths.

- r_A: the root of $J(r) = k$ in (r_0, r_2). A purchase that just reaches τ and recruits earns k.
- $r_F = \mathfrak{r}\big(k/(1-M)\big)$. A full purchase that recruits earns $\Delta_T(1-M)$; this equals k at r_F.
- $r_N = \mathfrak{r}(2k/\rho)$. Against no informed trade, a unit order earns $\rho\Delta_T/2$; this equals k at r_N.
- r_J: the root of $\rho J(r) = k$ in (r_0, r_2). A purchase that just reaches τ without recruitment earns ρJ; this equals k at r_J.
- $r_S = \mathfrak{r}\big(k/(\rho m)\big)$. A full sale earns $\rho\Delta_T m$; this equals k at r_S.
- r_C: the root of $\tau(r) = M$, that is of $B_r(M) = c_H$. In closed form,

$$r_C = \frac{Mh - c_H + \sqrt{(Mh - c_H)^2 + M[(1-M)\ell^2 - p^2]}}{M}.$$

Lemma A.4(a)–(b) make r_A, r_J and r_C unique, and Lemma A.4(d) makes the root r_* of $\rho\Delta_T(r)\tau(r) = k$ unique. The strengths r_A, r_N and r_J do not depend on the slot.

Lemma A.5 (The order of the thresholds)**.** *For every slot in the band,* $r_0 < r_A < r_F < r_N < r_J < r_S < r_1 < r_C < r_2$.

Proof. Each inequality reduces to a fact of Lemma A.4 or to the band.

- $r_0 < r_A < r_1$: $J(r_0) < k < \rho J(r_1) < J(r_1)$, with $\rho J(r_1) = 59/1200$, and J is increasing.
- $r_A < r_F$: $\Delta_T(r_A) < k/(1-M)$ is $(1-M)\Delta_T(r_A) < k = (1-\tau(r_A))\Delta_T(r_A)$, that is $M > \tau(r_A)$. This holds because $M > \tau(r_1) > \tau(r_A)$, since $r_A < r_1$.
- $r_F < r_N$: $k/(1-M) < 2k/\rho$ is $M < 1 - \rho/2$. This holds because $M < \tau(r_2) < 3/4 < 1 - \rho/2$.
- $r_N < r_J$: $\rho J(r_N) = \rho\Delta_T(r_N)[1 - \tau(r_N)] = 2k[1 - \tau(r_N)] < k$, because $\tau(r_N) > 1/2$, with $\tau(r_N) - 1/2 = 0.154463$.
- $r_J < r_S$: $\Delta_T(r_J) < k/(\rho m)$ is $\rho(1-M)\Delta_T(r_J) < k = \rho(1-\tau(r_J))\Delta_T(r_J)$, that is $M > \tau(r_J)$. The value $\rho J(r_1) > k$ gives $r_J < r_1$, so $M > \tau(r_1) > \tau(r_J)$.

- $r_S < r_1$: $\rho(1-M)\Delta_T(r_1) > k$ is $(1-M)/6 > 1/50$, that is $M < 22/25$. This holds because $M < 3/4$.
- $r_1 < r_C < r_2$: this is $\tau(r_1) < M < \tau(r_2)$, the band, since τ is increasing. □

Table A.1 gives the thresholds at calibration B. The script `P1_core/p1_certificates.py` brackets r_A and r_J by exact bisection, with the signs at both ends. The script `XP1/xp1_numbers.py` brackets every threshold by rational bisection to a width below 10^{-95} and records the exact endpoints in `XP1/xp1_registry_recompute.csv`.

Table A.1. Thresholds of the benchmark path at calibration B, prior $1/2$.

Threshold	Definition	Value at B	Depends on the slot
r_A	$J(r) = k$	1.393958	no
r_F	$\Delta_T(r)(1-M) = k$	1.467127	yes
r_N	$\rho\Delta_T(r)/2 = k$	1.747878	no
r_J	$\rho J(r) = k$	1.966097	no
r_S	$\rho\Delta_T(r)m = k$	2.124148	yes
r_C	$\tau(r) = M$	3.592659	yes

The two investor problems

Fix a strength r on the path and a prior $\pi < \tau(r)$. By Lemma A.3(a), no equilibrium puts weight on a wrong-signed order. Write $s = s_+ = \lambda\sigma_H(1)$ and $t = s_- = \lambda\sigma_L(-1)$ for the probabilities of an informed purchase and an informed sale. On the orders $(-1, 0, 1)$,

$$f_H = (b,\ b_0 + \lambda - s,\ b + s), \qquad f_L = (b + t,\ b_0 + \lambda - t,\ b).$$

Bayes' rule gives

$$\mu_+(s) := \mu(1) = \frac{\pi(b+s)}{b+\pi s}, \qquad 1 - \mu_+(s) = \frac{(1-\pi)b}{b+\pi s}, \qquad \mu_-(t) := \mu(-1) = \frac{\pi b}{b + (1-\pi)t},$$

and $\mu_0(s,t) := \mu(0)$ has the odds $\frac{\pi}{1-\pi} \cdot \frac{b_0+\lambda-s}{b_0+\lambda-t}$. On $[0,\lambda]$, μ_+ is continuous and rises strictly from π to $M(\pi)$, and μ_- is continuous and falls strictly from π to $m(\pi)$.

On the path $\tau(r) < 3/4$ by Lemma A.4(a). For $\pi < \tau < 1$, define

$$s^* = \frac{b(\tau - \pi)}{\pi(1-\tau)} > 0.$$

The formula for $\mu_+(s)$ rises strictly in $s \geq 0$, and $\mu_+(s) = \tau$ is the linear equation $\pi(b+s) = \tau(b+\pi s)$, so s^* is its unique root. Since $\mu_+(\lambda) = M(\pi)$, $s^* < \lambda$ exactly when $\tau < M(\pi)$, and $s^* = \lambda$ when

$\tau = M(\pi)$. If $\tau \geq M(\pi)$, no rate in $(0, \lambda)$ reaches the threshold, and the forms (R) and (T) below are empty. Entry after each order follows from Lemma A.2(d).

Lemma A.6 (The high class)**.** *Fix r and π with $\pi < \tau(r)$. In a candidate with no wrong-signed order, the pair $(s, e(1))$ satisfies (E1) to (E3) at the order 1, and (E5) for class H, exactly when it is one of these five pairs.*

(N) $s = 0$ *and* $e(1) = \rho$*, with* $\rho\Delta_T(1 - \pi) \leq k$*.*

(Q) $0 < s < \min\{s^*, \lambda\}$*,* $e(1) = \rho$ *and* $\rho\Delta_T[1 - \mu_+(s)] = k$*. Only under rule IM.*

(R) $s^* < s < \lambda$*,* $e(1) = 1$ *and* $\Delta_T[1 - \mu_+(s)] = k$*. Only under rule IM.*

(T) $s = s^* \in (0, \lambda)$ *and* $e(1) = e^* := k/J(r)$*. Under rule C1 this requires* $e^* = 1$*, that is* $J(r) = k$*. Under rule CM it requires* $\rho \leq e^* \leq 1$*. Only under rule IM.*

(F) $s = \lambda$*. Here* $e(1) = 1$ *if* $M(\pi) > \tau$ *and* $e(1) = \rho$ *if* $M(\pi) < \tau$*. If* $M(\pi) = \tau$*, then* $e(1) = 1$ *under rule C1, and* $e(1) \in [\rho, 1]$ *under rule CM. The return must satisfy* $e(1)\Delta_T[1-M(\pi)] \geq k$ *under rule IM, and* $> k$ *under rule I0.*

Each equation in (Q) and (R) has at most one solution.

Proof. By Lemma A.3(b), (E5) for class H requires $K_H \leq k$ at $s = 0$. At $s = \lambda$ it requires $K_H \geq k$ under rule IM, and $K_H > k$ under rule I0, because rule I0 selects abstention at a tie. At $0 < s < \lambda$ it requires $K_H = k$. An interior s is a mixture at indifference, so it requires rule IM. Here $K_H = e(1)\Delta_T[1 - \mu_+(s)]$, and $e(1)$ follows from Lemma A.2(d) at the posterior $\mu_+(s)$.

- At $s = 0$, $\mu_+(0) = \pi < \tau$, so $e(1) = \rho$. This is (N).
- At $0 < s < \lambda$ with $\mu_+(s) < \tau$: $e(1) = \rho$, and $K_H = k$ is the equation of (Q).
- At $0 < s < \lambda$ with $\mu_+(s) > \tau$: $e(1) = 1$, and $K_H = k$ is the equation of (R).
- At $0 < s < \lambda$ with $\mu_+(s) = \tau$: then $s = s^*$, and $K_H = e(1)\Delta_T(1 - \tau) = e(1)J(r) = k$ gives $e(1) = e^*$. Rule C1 sets $e(1) = 1$. Rule CM allows any $e(1) \in [\rho, 1]$. This is (T).
- At $s = \lambda$, $\mu_+(\lambda) = M(\pi)$. Lemma A.2(d) gives $e(1)$, and the condition on K_H is that of (F).

The function $1 - \mu_+$ falls strictly, so each equation has at most one root. □

Lemma A.7 (The low class)**.** *Fix r and π with $\pi < \tau(r)$. Then $\mu_-(t) \leq \pi < \tau$ for every t, so $e(-1) = \rho$ in every candidate. In a candidate with no wrong-signed order, the rate t satisfies (E1) to (E3) at the order -1, and (E5) for class L, exactly when it is one of these:*

(N) $t = 0$*, with* $\rho\Delta_T\pi \le k$*;*

(Q) $0 < t < \lambda$ *with* $\rho\Delta_T\mu_-(t) = k$*; only under rule IM;*

(F) $t = \lambda$*, with* $\rho\Delta_T m(\pi) \ge k$ *under rule IM, and* $> k$ *under rule I0.*

Under rule IM exactly one rate qualifies at each r*:*

$$s_-^*(r) = \begin{cases} 0, & \rho\Delta_T\pi \le k, \\ \dfrac{b\,(\pi\rho\Delta_T/k - 1)}{1-\pi}, & \rho\Delta_T m(\pi) < k < \rho\Delta_T\pi, \\ \lambda, & \rho\Delta_T m(\pi) \ge k. \end{cases}$$

Under rule I0 no rate qualifies when $\rho\Delta_T m(\pi) \le k < \rho\Delta_T\pi$*.*

Proof. Since μ_- is decreasing, $\mu_-(t) \le \mu_-(0) = \pi < \tau$, so Lemma A.2(d) gives $e(-1) = \rho$. Then $K_L = \rho\Delta_T\mu_-(t)$ is continuous and strictly decreasing in t, from $\rho\Delta_T\pi$ at $t = 0$ to $\rho\Delta_T m(\pi)$ at $t = \lambda$. By Lemma A.3(b), (E5) for class L requires $K_L \le k$ at $t = 0$; $K_L \ge k$ at $t = \lambda$ under rule IM, and $> k$ under rule I0; and $K_L = k$ at $0 < t < \lambda$, under rule IM only. If $\rho\Delta_T\pi \le k$, then $K_L < k$ for every $t > 0$, so only $t = 0$ qualifies. If $\rho\Delta_T m(\pi) \ge k$, then $K_L > k$ for every $t < \lambda$, so only $t = \lambda$ qualifies. Otherwise $\rho\Delta_T m(\pi) < k < \rho\Delta_T\pi$, and the intermediate value theorem gives exactly one interior root. The equation $\pi b/[b + (1-\pi)t] = k/(\rho\Delta_T)$ gives the formula. Under rule I0 with $\rho\Delta_T m(\pi) \le k < \rho\Delta_T\pi$, the rate $t = 0$ fails because $K_L > k$ there; the rate $t = \lambda$ fails because $K_L \le k$ there; and an interior rate requires rule IM. □

Lemma A.8 (Product structure)**.** *Fix* r *and* π *with* $\pi < \tau(r)$*, and rates with* $s \ge t$*. Then* $\mu_0(s,t) \le \pi < \tau$*. The tuple with the rates* (s,t) *is an equilibrium if and only if these four conditions hold:*

(i) $(s, e(1))$ *is a pair of Lemma A.6;*

(ii) t *is a rate of Lemma A.7;*

(iii) $e(0) = e(-1) = \rho$*;*

(iv) the prices are $P(x) = t_0 + e(x)\,w(\mu(x))$*, and* $\bar{\mu}(P(x)) = \mu(x)$*.*

Proof. If $s \ge t$, then $b_0 + \lambda - s \le b_0 + \lambda - t$, so the odds of μ_0 are at most $\pi/(1-\pi)$. Hence $\mu_0 \le \pi < \tau$.

Necessity. In any equilibrium, Lemma A.2(d) gives $e(0) = \rho$ at the posterior $\mu_0 < \tau$, and Lemma A.2(b)–(c) give condition (iv). Lemma A.3(d) shows that (E5) for class H involves only $(s, \mu(1), e(1))$, and (E5) for class L only $(t, \mu(-1), e(-1))$. So (E5) gives (i) and (ii).

Sufficiency. Take rates and entry that satisfy (i) to (iv), and check each condition by direct computation. (E1): the posteriors are computed from f_H and f_L. (E2): the stated entry is ρ at the posteriors $\mu_-(t)$ and μ_0, both at most $\pi < \tau$, and it is $e(1) \geq \rho$ at $\mu_+(s) \geq \pi$, with $e(1) = 1$ when $\mu_+(s) > \tau$. So entry lies in $[\rho, 1]$ and is nondecreasing in the posterior. By the paragraph on constructed tuples in Appendix A.2, orders with distinct posteriors have distinct prices, and orders with equal posteriors have equal entry and equal prices. Hence the orders with one price share one posterior, and (E2) gives $\bar{\mu}(P(x)) = \mu(x)$. (E3): at each price the belief is a posterior of an order, and the stated entry is the one that Lemma A.2(d) allows at that posterior under the tie rule in use; condition (i) states it at the order 1, and the posteriors $\mu_-(t)$ and μ_0 lie below τ. (E4): (A.4) with the stated entry. (E5): by Lemma A.3(d), the payoffs of class H involve only $(s, \mu(1), e(1))$, and those of class L only $(t, \mu(-1), e(-1))$; conditions (i) and (ii) state optimality for each. (E6): Appendix A.1. □

The equilibrium set at the prior $1/2$

At $\pi = 1/2$, $1 - \mu_+(s) = b/(2b + s)$, $\mu_-(t) = b/(2b + t)$ and $s^* = b(2\tau - 1)/(1 - \tau)$. Five forms of the high class's pair appear. In each, the low class uses the rate of Lemma A.7, which at $\pi = 1/2$ is

$$s^*_-(r) = 0 \text{ on } [r_0, r_N], \qquad s^*_-(r) = b\,[\rho\Delta_T(r)/k - 2] \text{ on } (r_N, r_S), \qquad s^*_-(r) = \lambda \text{ on } [r_S, r_2].$$

- **N** (no informed trade): $s = 0$, $e(1) = \rho$.
- **R** (a mixed purchase that recruits): $s = s_R(r) := b[\Delta_T(r)/k - 2]$, $e(1) = 1$.
- **F** (a full purchase): $s = \lambda$; $e(1) = 1$ for $r \leq r_C$, and $e(1) = \rho$ for $r > r_C$.
- **Q** (mixed orders at the entry floor): $s = s_Q(r) := b[\rho\Delta_T(r)/k - 2] = s^*_-(r)$, $e(1) = \rho$.
- **T** (the threshold pair): $s = s^*(r)$, $e(1) = e^*(r) = k/J(r)$.

Each form defines the equilibrium tuple completely:

- orders: $\sigma_H(1) = s/\lambda$, $\sigma_H(0) = 1 - s/\lambda$, $\sigma_L(-1) = s^*_-(r)/\lambda$, $\sigma_L(0) = 1 - s^*_-(r)/\lambda$;
- posteriors: $\mu_-(s^*_-)$, $\mu_0(s, s^*_-)$ and $\mu_+(s)$ after the orders -1, 0 and 1;
- challenger beliefs: $\bar{\mu}(P(x)) = \mu(x)$ at each price;
- high-cost preparation: $\zeta(P(1)) = (e(1) - \rho)/(1 - \rho)$, and $\zeta(P(x)) = 0$ at the prices of the orders 0 and -1;
- prices: $P(x) = t_0 + e(x)\, w(\mu(x))$;

- bids: truthful.

If two orders share a price, they share the posterior and the entry (Lemma A.8), so ζ is well defined as a function of the price.

Proposition 3 (full statement). *Take the benchmark auction, the prior* $1/2$ *and any slot in the band of Proposition A.1(a). Then* $r_0 < r_A < r_F < r_N < r_J < r_S < r_1 < r_C < r_2$*, and the equilibrium set at each strength is*

Strength	*Equilibria*
$r_0 \le r < r_A$	N
$r = r_A$	N; T *with entry one after a purchase (the left end of* R*)*
$r_A < r < r_F$	N; R
$r_F \le r \le r_N$	N; F
$r_N < r < r_J$	F; Q
$r_J \le r \le r_2$	F

Entry is ρ *in* N *and* Q*,* $\mathsf{E}_R(r) = \rho + (1-\rho)\, b\Delta_T(r)/(2k)$ *in* R*, and* $\mathsf{E}_F = \rho + (1-\rho)(b + \lambda/2)$ *in* F *up to* r_C*, with* $\mathsf{E}_F = \rho$ *above* r_C*.*

(a) No informed trade and an informative equilibrium coexist exactly on $[r_A, r_N]$*.*

(b) On the informative branch that coexists with no trade, entry is $\rho + (1-\rho)\min\{b\Delta_T(r)/(2k),\ b + \lambda/2\}$*. It is continuous, rises strictly on* $[r_A, r_F]$*, and is constant on* $[r_F, r_N]$*. It keeps this value along* F *up to* r_C *and equals* ρ *above* r_C*.*

(c) The equilibrium set is nonempty at every strength.

Proof of Proposition 3. Fix any slot in the band and the prior $1/2$. The proof has eight steps.

Step 1: the frame. By Lemma A.4(a), $\tau(r) \ge \tau(r_0) > 1/2 = \pi$ on the path, so Lemmas A.6 to A.8 apply at every r. Every equilibrium has a high-class pair from Lemma A.6 and a low-class rate from Lemma A.7. Step 4 shows $s \ge t$ for every such combination. Lemma A.8 then gives the equilibrium set as their product. Recall $1 - M = 1/(2+u)$ and $ub = \lambda$.

Step 2: the high-class pairs. Use Lemma A.4, the definitions of the thresholds, and Lemma A.5.

- (N) holds if and only if $\rho\Delta_T/2 \le k$, that is $r \le r_N$.
- (R): $\Delta_T b/(2b+s) = k$ gives $s = s_R(r)$. Then $\mu_+(s_R) > \tau$ if and only if $k/\Delta_T < 1 - \tau$, that is $J(r) > k$, that is $r > r_A$. Also $s_R < \lambda$ if and only if $\Delta_T/k < 2 + u$, that is $\Delta_T(1-M) < k$, that is $r < r_F$. So (R) holds if and only if $r_A < r < r_F$, under rule IM.

- (Q): $\rho\Delta_T b/(2b+s) = k$ gives $s = s_Q(r)$. Then $s_Q > 0$ if and only if $r > r_N$; $\mu_+(s_Q) < \tau$ if and only if $\rho J(r) < k$, that is $r < r_J$; and $s_Q < \lambda$ if and only if $\rho\Delta_T(1-M) < k$, that is $r < r_S$. Since $r_J < r_S$, (Q) holds if and only if $r_N < r < r_J$, under rule IM.

- (T): $s^* \in (0, \lambda)$ if and only if $1/2 < \tau(r) < M$, that is $r < r_C$. Also $\rho \le e^* \le 1$ if and only if $\rho J(r) \le k \le J(r)$, that is $r_A \le r \le r_J$, and $e^* = 1$ if and only if $r = r_A$. So (T) holds at r_A under rule C1, and on $[r_A, r_J]$ under rule CM, under rule IM in both cases.

- (F): for $r < r_C$, $M > \tau(r)$, so $e(1) = 1$, and the condition is $\Delta_T(1-M) \ge k$, that is $r \ge r_F$ ($r > r_F$ under rule I0). For $r > r_C$, $M < \tau(r)$, so $e(1) = \rho$, and $\rho\Delta_T(1-M) \ge k$ holds because $r > r_C > r_S$. At $r = r_C$, rule C1 gives $e(1) = 1$; rule CM allows any $e(1) \in [\rho, 1]$, and every such value satisfies the condition because $\rho\Delta_T(r_C)(1-M) > k$. So (F) holds if and only if $r \ge r_F$ (rule IM) or $r > r_F$ (rule I0).

Step 3: the low-class rate. At $\pi = 1/2$, Lemma A.7 gives $t = 0$ if and only if $r \le r_N$, $t = \lambda$ if and only if $r \ge r_S$, and $s^*_-(r) = b(\rho\Delta_T/k - 2)$ on (r_N, r_S), under rule IM. Under rule I0 no rate qualifies if and only if $\rho\Delta_T m \le k < \rho\Delta_T/2$, that is on $(r_N, r_S]$.

Step 4: $s \ge t$. For $r \le r_N$, $t = 0 \le s$. For $r > r_N$, the pairs are F with $s = \lambda \ge t$; Q with $s = s_Q(r) = s^*_-(r)$; and T with $r_N < r \le r_J$. For T, $s^* \ge s^*_-(r)$ if and only if $(2\tau - 1)/(1-\tau) \ge \rho\Delta_T/k - 2$, that is $1/(1-\tau) \ge \rho\Delta_T/k$, that is $\rho J(r) \le k$, which holds for $r \le r_J$. By Lemma A.8, entry after the zero order and after a sale is ρ in every equilibrium.

Step 5: the set. Combine Steps 2 to 4 with the order of Lemma A.5. Under rules IM and C1 the set is: N on $[r_0, r_A)$; N and T with $e^* = 1$ at r_A; N and R on (r_A, r_F); N and F on $[r_F, r_N]$; F and Q on (r_N, r_J); F on $[r_J, r_2]$. Under rule CM, T also exists on $(r_A, r_J]$, and at r_C entry after a purchase in F can be any $e(1) \in [\rho, 1]$. Under rule I0, (Q), (R) and (T) drop out, (F) needs $r > r_F$, and Step 3 removes every equilibrium on $(r_N, r_S]$. The set under rule I0 is N on $[r_0, r_F]$; N and F on $(r_F, r_N]$; empty on $(r_N, r_S]$; and F on $(r_S, r_2]$.

Step 6: verification of each tuple. Fix a form and a strength at which Steps 2 and 3 admit it. Step 4 gives $s \ge t$, so the sufficiency part of Lemma A.8 applies: it checks (E1) to (E6) against the tuple by direct computation. In particular, if two orders have equal posteriors, one of them is 0 or -1, so the common posterior is at most $1/2 < \tau$, and both orders have entry ρ and the same price.

Step 7: entry. Entry is $\mathsf{E} = \sum_x \Pr(X = x)\, e(x)$. Since $e(0) = e(-1) = \rho$ and $\Pr(X = 1) = \frac{1}{2}(b+s) + \frac{1}{2}b = b + s/2$, $\mathsf{E} = \rho + [e(1) - \rho](b + s/2)$.

- R: $b + s_R/2 = b\Delta_T/(2k)$ and $e(1) = 1$, so $\mathsf{E}_R(r) = \rho + (1-\rho)\, b\Delta_T(r)/(2k)$. It rises strictly in r because Δ_T does. At r_F, $\Delta_T(r_F) = k/(1-M) = k(2+u)$, so $b\Delta_T(r_F)/(2k) = b + ub/2 = b + \lambda/2$: the branch joins F continuously.

- F: $b + \lambda/2$, with $e(1) = 1$ below r_C and $e(1) = \rho$ above r_C. So $\mathsf{E}_F = \rho + (1-\rho)(b + \lambda/2)$ on $[r_F, r_C)$ and $\mathsf{E}_F = \rho$ on $(r_C, r_2]$. At r_C, rule C1 gives $e(1) = 1$ and the upper value; rule CM gives every value $\mathsf{E}_F(r_C) = \rho + (e(1) - \rho)(b + \lambda/2) \in [\rho,\ \rho + (1-\rho)(b + \lambda/2)]$.

- N and Q: $e(1) = \rho$, so $\mathsf{E} = \rho$.

- T: $e^* - \rho = (k - \rho J)/J$ and $b + s^*/2 = b/[2(1-\tau)]$, so

$$\mathsf{E}_T(r) = \rho + \frac{b}{2} \cdot \frac{k - \rho J(r)}{J(r)\,[1 - \tau(r)]} = \rho + \frac{b}{2} \cdot \frac{k - \rho J(r)}{\Delta_T(r)\,[1 - \tau(r)]^2}.$$

On the informative branch that coexists with N, entry is $\rho + (1-\rho)\min\{b\Delta_T(r)/(2k),\ b + \lambda/2\}$. It is continuous, rises strictly on $[r_A, r_F]$, and is constant on $[r_F, r_N]$. The strict rise needs rule IM, because R is a mixture at indifference. At r_A, $J = k$ and $\Delta_T(r_A)/k = 1/(1 - \tau(r_A))$, so $\mathsf{E}_T(r_A) = \rho + \frac{1}{2}(1-\rho)\, b/(1 - \tau(r_A)) = \mathsf{E}_R(r_A)$. At r_J, $\rho J = k$, so $\mathsf{E}_T(r_J) = \rho$. On $[r_A, r_J]$, $k - \rho J$ is positive except at r_J and strictly decreasing, by Lemma A.4(b). The term $\Delta_T(1-\tau)^2$ is positive and strictly increasing, by Lemma A.4(c) and $r_J < 2$; the last inequality holds because $\rho J(2) - k = 11/11200 > 0$ and J rises. So E_T falls strictly from $\mathsf{E}_R(r_A)$ to ρ. For $r_A < r \le r_N$: $r < r_J$ gives $e^* > \rho$, so $\mathsf{E}_T > \rho$; and $J > k$ gives $e^* < 1$ with $s^* < s$ in R or F, so $\mathsf{E}_T < \mathsf{E}_R$ on (r_A, r_F) and $\mathsf{E}_T < \mathsf{E}_F$ on $[r_F, r_N]$.

Step 8: existence. Under rule IM, N exists on $[r_0, r_N]$ and F on $[r_F, r_2]$, and $r_F < r_N$. The set is nonempty at every strength under rule IM, with either challenger rule. □

Calibration B. Table A.2 gives the set at calibration B under rules IM and C1. On branch R, entry is 0.459588 at $r = 1.40$ and 0.506112 at $r = 1.45$. The low class abstains on $[r_0, r_N]$, mixes on (r_N, r_S) and sells fully on $[r_S, r_2]$. The strengths 1.55, 1.60 and 1.65 lie in $[r_F, r_N]$, where the equilibria are N and F under rule C1. Under rule CM the threshold pair T adds entry 0.341462, 0.321311 and 0.305208 at these three strengths. The values come from `P1_core/p1_numbers.py` in outward interval arithmetic. The script `P1_core/p1_enumerate.py` (run log `P1_core/p1_enumerate.log`) rebuilds the equilibrium set from the flow law at 241 strengths, for the four pairs of tie rules. It finds no difference from the proposition. This cross-check enters no step of the proof.

Coexistence at other priors

Coexistence also holds at other priors. For a prior π, write $r_N(\pi) = \mathfrak{r}\big(k/(\rho \max\{\pi, 1-\pi\})\big)$. When some strength on the path has $\tau(r) \le M(\pi)$, let $r_C(\pi)$ be the largest one. When $\tau(r_0) \le \pi \le \tau(r_2)$, let r_π solve $\tau(r_\pi) = \pi$. Three priors divide the cases: π_A^{lo} solves $M(\pi) = \tau(r_A)$; $\pi_A^{\mathrm{hi}} = \tau(r_A)$; and $\pi_{\max} = \tau(r_*)$, where r_* solves $\rho\Delta_T(r)\tau(r) = k$.

Table A.2. The equilibrium set on the benchmark path at calibration B (rules IM and C1).

Strengths	Equilibria	Entry
$[r_0,\ 1.393958)$	N	0.250000
1.393958	N; T with $e^* = 1$	0.250000; 0.454185
(1.393958, 1.467127)	N; R	0.250000; from 0.454185 up to 0.522757
[1.467127, 1.747878]	N; F	0.250000; 0.522757
(1.747878, 1.966097)	F; Q	0.522757; 0.250000
[1.966097, 3.592659]	F	0.522757
$(3.592659,\ r_2]$	F	0.250000

Corollary A.2 (Coexistence at every prior)**.** *Take the benchmark auction, any slot, and any prior $\pi \in (0,1)$. The set $C(\pi)$ of strengths on the path at which no informed trade and an informative equilibrium coexist is*

$$C(\pi) = \big\{r \in [r_0, r_2] :\ r \geq r_A,\ \ \pi < \tau(r) \leq M(\pi),\ \ \rho\Delta_T(r)\max\{\pi, 1-\pi\} \leq k\big\}.$$

(a) If $\pi < \pi_A^{\text{lo}}$, $C(\pi)$ is empty. At $\pi = \pi_A^{\text{lo}}$, $C(\pi) = \{r_A\}$.

(b) If $\pi_A^{\text{lo}} < \pi < \pi_A^{\text{hi}}$, then $C(\pi) = [r_A, \min\{r_N(\pi), r_C(\pi)\}]$, a nondegenerate interval. Coexistence starts at r_A for every prior in this window and every slot.

(c) If $\pi_A^{\text{hi}} \leq \pi < \pi_{\max}$, then $C(\pi) = (r_\pi, \min\{r_N(\pi), r_C(\pi)\}]$, a nonempty interval with $r_\pi \geq r_A$. Coexistence starts where the prior reaches the threshold.

(d) If $\pi \geq \pi_{\max}$, $C(\pi)$ is empty.

At each $r \in C(\pi)$, the informative equilibrium is R or F (the threshold pair at r_A in case (b)). Its entry is $\rho + (1-\rho)\min\{(1-\pi)\, b\Delta_T(r)/k,\ b + \pi\lambda\}$, which rises strictly while the purchase is mixed and is then constant.

Proof of Corollary A.2. Fix any slot with $0 < \lambda < 1$ and $0 < \beta < 1/2$, any prior $\pi \in (0,1)$, and the rules IM and C1. For a prior π, write $r_N(\pi) = \mathfrak{r}\big(k/(\rho\max\{\pi, 1-\pi\})\big)$. When some strength on the path has $\tau(r) \leq M(\pi)$, let $r_C(\pi)$ be the largest one; otherwise the condition $\tau(r) \leq M(\pi)$ holds nowhere on the path. When $\tau(r_0) \leq \pi \leq \tau(r_2)$, let r_π solve $\tau(r_\pi) = \pi$; this root is unique by Lemma A.4(a). For $\pi > \tau(r_2)$, no strength on the path satisfies $\pi < \tau(r)$. Write $\tau_A = \tau(r_A)$, and let r_* solve $\rho\Delta_T(r_*)\tau(r_*) = k$ on the path; it is unique by Lemma A.4(d). Define

$$\pi_A^{\text{lo}} = \frac{\tau_A}{\tau_A + (1-\tau_A)(1+u)}, \qquad \pi_A^{\text{hi}} = \tau_A, \qquad \pi_{\max} = \tau(r_*).$$

So $M(\pi_A^{\text{lo}}) = \tau_A$. The numbers π_A^{hi} and $\pi_{\max}$ do not depend on the slot. The claim is that the set of

strengths on the path at which no informed trade and an informative equilibrium coexist is

$$C(\pi) = \big\{r \in [r_0, r_2] : \; r \geq r_A, \;\; \pi < \tau(r) \leq M(\pi), \;\; \rho\Delta_T(r) \max\{\pi, 1-\pi\} \leq k\big\},$$

with the four cases of the corollary. The proof has three steps.

Step 1: strengths with $\pi < \tau(r)$. Lemmas A.6 to A.8 apply. No informed trade is an equilibrium if and only if (N) holds in both lemmas, that is $\rho\Delta_T \max\{\pi, 1-\pi\} \leq k$; then $\mu_0 = \pi < \tau$ and $e(0) = \rho$. Suppose this holds. Lemma A.7 then gives $t = 0$ in every equilibrium, so an informative equilibrium has $s > 0$. Pair (Q) is impossible, because $\rho\Delta_T[1 - \mu_+(s)] < \rho\Delta_T(1-\pi) \leq k$ for $s > 0$. Pair (F) with $M(\pi) < \tau$ is impossible for the same reason. The other pairs, (R), (T) and (F) with $M(\pi) \geq \tau$, need $\mu_+(s) \geq \tau$ for some $s \leq \lambda$, so $\tau \leq M(\pi)$. They also need $k \leq e(1)\Delta_T[1 - \mu_+(s)] \leq \Delta_T(1-\tau) = J(r)$, so $r \geq r_A$ by Lemma A.4(b).

Conversely, let $\tau(r) \leq M(\pi)$ and $r \geq r_A$, so $J(r) \geq k$. If $\Delta_T(1 - M(\pi)) \geq k$, pair (F) holds with $e(1) = 1$. Otherwise $\tau < M(\pi)$, and $s \mapsto \Delta_T[1 - \mu_+(s)]$ is continuous and strictly decreasing on $[s^*, \lambda]$, from $J(r) \geq k$ to $\Delta_T(1 - M(\pi)) < k$. It equals k at exactly one $s \in [s^*, \lambda)$. At $s = s^*$ this is (T) with $J(r) = k$ and $e(1) = 1$; at $s > s^*$ it is (R). With $t = 0 \leq s$, Lemma A.8 completes the tuple. So a strength with $\pi < \tau(r)$ lies in $C(\pi)$ if and only if it satisfies the three conditions in the display.

Step 2: strengths with $\pi \geq \tau(r)$. Under rule C1, no informed trade gives the posterior $\pi \geq \tau$ after every order, so $e \equiv 1$. It is an equilibrium if and only if $\Delta_T \max\{\pi, 1-\pi\} \leq k$. Suppose this holds, and take any candidate with $s > 0$. Then $\mu_+(s) > \pi \geq \tau$, so $e(1) = 1$ and $K_H = \Delta_T[1 - \mu_+(s)] < \Delta_T(1-\pi) \leq k$: class H strictly prefers abstention, a contradiction. Take any candidate with $t > 0$. Then $\mu_-(t) < \pi$ and $e(-1) \leq 1$, so $K_L \leq \Delta_T\mu_-(t) < \Delta_T\pi \leq k$: again a contradiction. So no strength with $\pi \geq \tau(r)$ lies in $C(\pi)$.

Step 3: the four cases. By Lemma A.4(a), $\pi < \tau(r)$ holds for every r if $\pi < \tau(r_0)$, for $r > r_\pi$ if $\tau(r_0) \leq \pi \leq \tau(r_2)$, and for no r if $\pi > \tau(r_2)$. The condition $\tau(r) \leq M(\pi)$ holds for $r \leq r_C(\pi)$ when $r_C(\pi)$ exists, and the last condition holds for $r \leq r_N(\pi)$. Also $\rho\Delta_T(r_A)\max\{\pi, 1-\pi\} < \rho\Delta_T(r_A) = \rho k/(1-\tau_A) < k$, because $\tau_A < 3/4 = 1 - \rho$. So $r_A < r_N(\pi)$ for every prior; at the benchmark $3/4 - \tau_A = 0.109261$.

(a) M is increasing in π, so $\pi < \pi_A^{\text{lo}}$ gives $M(\pi) < \tau_A \leq \tau(r)$ for every $r \geq r_A$. No strength qualifies. At $\pi = \pi_A^{\text{lo}}$ only r_A qualifies, and there $M(\pi) = \tau_A$: the informative equilibrium is the full purchase F, with $K_H = \Delta_T(r_A)(1-\tau_A) = k$.

(b) Let $\pi_A^{\text{lo}} < \pi < \pi_A^{\text{hi}}$. For $r \geq r_A$, $\tau(r) \geq \tau_A > \pi$. Also $\tau_A < M(\pi)$ gives $r_A < r_C(\pi)$, and $r_A < r_N(\pi)$. So $C(\pi) = [r_A, \min\{r_N(\pi), r_C(\pi)\}]$, a nondegenerate interval. At r_A, $\Delta_T(r_A)(1 - M(\pi)) < \Delta_T(r_A)(1-\tau_A) = k$, so Step 1 gives the threshold pair with $J = k$. There $\mu_+ = \tau$, so the challenger is indifferent, and $K_H = k$, so the investor is indifferent. The start r_A is the same for every prior in this window and every slot.

(c) Let $\pi_A^{\text{hi}} \le \pi < \pi_{\max}$. Now $r_\pi \ge r_A$, and $\pi < \tau(r)$ holds exactly for $r > r_\pi$. The interval $C(\pi) = (r_\pi, \min\{r_N(\pi), r_C(\pi)\}]$ is nonempty: $r_\pi < r_C(\pi)$ because $\tau(r_\pi) = \pi < M(\pi)$; and $r_\pi < r_N(\pi)$ if and only if $\rho\Delta_T(r_\pi)\pi < k$ (as $\pi > 1/2$), that is $\rho\Delta_T(r_\pi)\tau(r_\pi) < k$, that is $r_\pi < r_*$ by Lemma A.4(d), that is $\pi < \pi_{\max}$.

(d) Let $\pi \ge \pi_{\max}$. Then $\pi > 1/2$, and any r with $\pi < \tau(r)$ has $r > r_\pi \ge r_*$, so $\rho\Delta_T(r)\pi > \rho\Delta_T(r_*)\tau(r_*) = k$. No informed trade is not an equilibrium there. Step 2 covers the other strengths, so $C(\pi)$ is empty.

Entry. Entry after the orders 0 and -1 is ρ, and $\Pr(X = 1) = \pi(b + s) + (1 - \pi)b = b + \pi s$. In R the equation $\Delta_T(1 - \pi)b/(b + \pi s) = k$ gives $b + \pi s = (1 - \pi)b\Delta_T/k$; in F, $\Pr(X = 1) = b + \pi\lambda$, and $\Delta_T(1 - M(\pi)) \ge k$ is $(1 - \pi)b\Delta_T/k \ge b + \pi\lambda$. Entry on the informative branch is therefore $\rho + (1 - \rho)\min\{(1 - \pi)b\Delta_T(r)/k,\ b + \pi\lambda\}$. It rises strictly while the purchase is mixed and then stays constant. At calibration B, $\pi_A^{\text{lo}} \approx 0.396175$, $\pi_A^{\text{hi}} \approx 0.640739$ and $\pi_{\max} \approx 0.650059$, from `P1_core/p1_numbers.py`. □

Other tie rules. The results use the default rules IM and C1. Two alternatives show which conclusions depend on these choices:

- under rule I0, an investor indifferent between abstention and a nonzero order abstains;
- under rule CM, an indifferent high-cost challenger may prepare with any probability.

This paragraph states each place on the benchmark path where I0 or CM changes a conclusion; Remark A.1 in Appendix A.5 does the same with research.

At the prior $1/2$: under the challenger rule CM, the threshold pair T is also an equilibrium on $(r_A, r_J]$. Its entry falls strictly from $\mathsf{E}_R(r_A)$ at r_A to ρ at r_J; at the benchmark it is 0.341462, 0.321311 and 0.305208 at 1.55, 1.60 and 1.65. At r_C, entry after a purchase can then take any value in $[\rho, 1]$.

Under the investor rule I0, the equilibrium set is N on $[r_0, r_F]$; N and F on $(r_F, r_N]$; empty on $(r_N, r_S]$; and F on $(r_S, r_2]$. On $(r_N, r_S]$ the low class has no admissible sale rate under I0. Abstention fails, because a sale earns more than k. A full sale fails, because it earns at most k. Under the investor rule IM, the equilibrium set is nonempty along the whole path. The strict rise of entry along R also needs IM.

At a general prior, under rule CM, a challenger tie at the prior adds exactly one strength. At $r = r_\pi$, the prior equals $\tau(r_\pi)$, and rule CM lets no informed trade keep entry ρ. That tuple is an equilibrium exactly when $\rho\Delta_T(r_\pi)\max\{\pi, 1 - \pi\} \le k$. An informative purchase at r_π has $\mu_+(s) > \pi = \tau$, so it recruits. It needs $\Delta_T(r_\pi)[1 - \mu_+(s)] = k$ for some $s \in (0, \lambda)$, or $\Delta_T(r_\pi)[1 - M(\pi)] \ge k$ at $s = \lambda$; one of the two holds exactly when $\Delta_T(r_\pi)(1 - \pi) > k$, that is $J(r_\pi) > k$, that is $r_\pi > r_A$. In this

tuple the low class abstains, so the sale keeps the posterior $\pi = \tau$, and rule CM lets entry stay ρ after the sale as well; the zero order has a posterior below π. The two conditions hold exactly when $r_A < r_\pi \le r_*$, that is $\tau_A < \pi \le \pi_{\max}$. Hence

$$C_{\mathrm{CM}}(\pi) = C(\pi) \cup \{r_\pi\} \quad \text{for } \tau_A < \pi \le \pi_{\max}, \qquad C_{\mathrm{CM}}(\pi) = C(\pi) \text{ otherwise.}$$

At $\pi = \pi_{\max}$, the point $r_\pi = r_* = 1.634218$ is the whole set: there $M(\pi_{\max}) = 0.834698$, the zero-order posterior is 0.454140, the full purchase earns $\Delta_T(r_*)[1 - M(\pi_{\max})] - k = 0.000343$, and entry is 0.560586. The pair T adds equilibria inside $C(\pi)$, since it needs $J(r) \ge k$. Under rule I0, R and T drop out, and F needs $\Delta_T(1 - M(\pi)) > k$. Coexistence then holds on the strengths of $C(\pi)$ above $\mathfrak{r}\big(k/(1 - M(\pi))\big)$; under rules I0 and CM, the point r_π above joins it when $\Delta_T(r_\pi)[1 - M(\pi)] > k$. Table A.3 records the dependence at the prior $1/2$.

Table A.3. Which conclusions depend on the tie rules (prior $1/2$).

Result	IM, C1	IM, CM	I0, C1	I0, CM
Proposition 2	holds	holds	holds	holds
No informed trade N	$[r_0, r_N]$	$[r_0, r_N]$	$[r_0, r_N]$	$[r_0, r_N]$
Mixed purchase R	(r_A, r_F)	(r_A, r_F)	none	none
Threshold pair T	r_A only	$[r_A, r_J]$	none	none
Full purchase F	$[r_F, r_2]$	$[r_F, r_2]$	$(r_F, r_N] \cup (r_S, r_2]$	$(r_F, r_N] \cup (r_S, r_2]$
Set on (r_N, r_J)	F; Q	F; Q; T	empty	empty
Set at r_J	F	F; T	empty	empty
Set on $(r_J, r_S]$	F	F	empty	empty
Entry after a purchase at r_C	1	any in $[\rho, 1]$	1	any in $[\rho, 1]$
Coexistence	$[r_A, r_N]$	$[r_A, r_N]$	$(r_F, r_N]$	$(r_F, r_N]$

Under rule I0, F on $(r_N, r_S]$ is not an equilibrium, because the low class has no admissible rate there (Lemma A.7). Rule IM is therefore the investor rule under which the equilibrium set is nonempty along the whole path. The strict rise of entry along R also needs rule IM. The challenger rule matters only through T and the point r_C.

A.5 Research

This subsection proves both parts of Proposition 4. It also states and proves Remark A.1 and Proposition A.2, the admissible cap and cost.

Equilibrium with research. Market makers and the challenger do not observe the research choice; they use a conjectured intensity $\hat{\lambda}$. An equilibrium with research is a tuple $(\hat{\lambda}, \sigma_H, \sigma_L, \mu, P, \bar{\mu}, \zeta)$, together with the bids. Conditions (E1)–(E6) of Definition A.1 hold at the intensity $\hat{\lambda}$. Condition

(E5) holds also at $\hat{\lambda} = 0$: a deviation in research makes a success possible, and the order after it must be optimal. A seventh condition completes the definition.

(E7) *Research optimality.* Hold P and ζ at their values in the tuple. Then $\hat{\lambda}$ is the research choice of a best response among all joint deviations in research and orders.

A joint deviation changes the true order probabilities, but not the prices or the entry rule. With the research value V below, the expected trading payoff after a success at the candidate's prices, Lemma A.9 shows that the best research choice is unique and equals $\min\{\bar{\lambda}, V/a\}$. Condition (E7) therefore reads $\hat{\lambda} = \min\{\bar{\lambda}, V/a\}$, and research never has a tie.

Standing assumptions. The auction is the benchmark, the prior is $1/2$, and $0 < \beta \leq 1/3$. Fix a candidate with conjectured intensity $\hat{\lambda} \in [0, \bar{\lambda}]$ and order distributions σ_H, σ_L after a success. Write $\hat{b} = (1-\hat{\lambda})\beta$ and $\hat{b}_0 = (1-\hat{\lambda})(1-2\beta)$, and $f_j(q) = (1-\hat{\lambda})\eta(q) + \hat{\lambda}\sigma_j(q)$. The conditions (E1) to (E6) hold at the intensity $\hat{\lambda}$, and (E7) is research optimality. A candidate satisfies (E1) to (E4) and (E6). Lemma A.3 applies at $\hat{\lambda}$ and gives the payoffs of an order after a success,

$$U_H(q) = q\, e(q)\Delta_T[1-\mu(q)] - k|q|, \qquad U_L(q) = -q\, e(q)\Delta_T\mu(q) - k|q|. \tag{A.7}$$

Let $V = \frac{1}{2}\max_q U_H(q) + \frac{1}{2}\max_q U_L(q) \geq 0$ be the research value. Since $r > \ell$, $\Delta_T > 0$ and $g_H > g_L$ at every strength of this subsection.

Derived functions. For $\lambda \in [0,1)$, write $u_\lambda = \lambda/[(1-\lambda)\beta]$, $M_\lambda = (1+u_\lambda)/(2+u_\lambda)$ and $m_\lambda = 1 - M_\lambda$; these are the posterior bounds of Section 3.2 at the prior $1/2$ and the presence λ. Then $m_\lambda = (1-\lambda)\beta/[2\beta + \lambda(1-2\beta)]$ falls strictly from $1/2$ to 0, because $1-2\beta > 0$. At the cap of calibration B, $u_{\bar{\lambda}} = u$, so $M_{\bar{\lambda}} = \mathrm{e}/(1+\mathrm{e})$ and $m_{\bar{\lambda}} = 1/(1+\mathrm{e})$. For $1/2 < \tau(r) < 1$, the recruitment intensity $\lambda_E(r)$ of Section 5 solves $M_\lambda = \tau(r)$:

$$\lambda_E(r) = \frac{\beta\,[2\tau(r)-1]}{\beta + (1-2\beta)[1-\tau(r)]},$$

and $d\lambda_E/d\tau = \beta/\{\beta + (1-2\beta)(1-\tau)\}^2 > 0$. Clear the denominator in (9):

$$\begin{aligned} F_r(\lambda) &= 2a(1-2\beta)\lambda^2 + \big[4a\beta + k(1-2\beta) + \beta\Delta_T(r)\big]\lambda + \beta\big[2k - \Delta_T(r)\big] \\ &= 2\big[2\beta + \lambda(1-2\beta)\big]\big[a\lambda - \tfrac{1}{2}\big(\Delta_T(r)m_\lambda - k\big)\big]. \end{aligned} \tag{A.8}$$

Every coefficient except the constant is positive, so F_r rises strictly on $\lambda \geq 0$. If $\Delta_T(r) > 2k$, then $F_r(0) < 0 < F_r(1)$, and $\lambda_R(r)$ is the unique root of F_r in $(0,1)$. Define the recruitment margin and

the threshold entry

$$\varphi_a(r) = \Delta_T(r)[1-\tau(r)] - k - 2a\lambda_E(r), \qquad e^*(r) = \frac{2a\lambda_E(r)+k}{\Delta_T(r)[1-\tau(r)]}.$$

Since $m_{\lambda_E} = 1-\tau$, (A.8) gives $F_r(\lambda_E(r)) = -[2\beta + \lambda_E(1-2\beta)]\,\varphi_a(r)$. Hence $\lambda_R(r) \geq \lambda_E(r)$ exactly when $\varphi_a(r) \geq 0$, and $e^*(r) < 1$ exactly when $\varphi_a(r) > 0$. At calibration B, $B_r(1/2) = 5 - r/4 + 1/(8r) < c_H$ for $1 < r < 10$, so $\tau(r) > 1/2$ at every strength below.

Lemma A.9 (Joint deviations)**.** *For every candidate and every $z \in [0, \bar{\lambda}]$, the best payoff of a joint deviation $(z, \sigma'_H, \sigma'_L)$ is $zV - (a/2)z^2$. The unique best research choice is $z^* = \min\{\bar{\lambda}, V/a\}$. Hence (E7) holds if and only if $\hat{\lambda} = \min\{\bar{\lambda}, V/a\}$, and research never has a tie.*

Proof. A joint deviation changes the true order law to $(1-z)\eta + z\sigma'_j$. It does not change the prices or the preparation rule, because market makers and the challenger use the candidate's $\hat{\lambda}$ and σ_j. The investor trades only after a success, so its expected payoff is

$$\Pi(z, \sigma') = z\left[\tfrac{1}{2}\sum_q \sigma'_H(q)U_H(q) + \tfrac{1}{2}\sum_q \sigma'_L(q)U_L(q)\right] - \frac{a}{2}z^2,$$

with U_j from (A.7) at the candidate's prices and entry. The bracket is at most V, with equality exactly when each σ'_j puts weight only on maximizers of U_j. So $\sup_{\sigma'} \Pi(z, \sigma') = zV - (a/2)z^2$. This function of z is strictly concave with derivative $V - az$. If $V \geq a\bar{\lambda}$, the derivative is positive on $[0, \bar{\lambda})$, and the maximizer is $\bar{\lambda}$. If $V < a\bar{\lambda}$, the maximizer is $V/a \in [0, \bar{\lambda})$. Strict concavity makes the maximizer unique, and it also excludes a nondegenerate mixture over research intensities. □

Lemma A.10 (Bounds under every profile)**.** *Let $0 \leq \hat{\lambda} \leq \bar{\lambda} < 1$. For every profile and every candidate:*

(a) every posterior satisfies $m_{\hat{\lambda}} \leq \mu(q) \leq M_{\hat{\lambda}}$, and $M_{\hat{\lambda}} \leq M_{\bar{\lambda}}$;

(b) the price reveals the posterior: if $P(q) = P(q')$, then $\mu(q) = \mu(q')$, and $\bar{\mu}(P(q)) = \mu(q)$;

(c) $U_H(-1) \leq -k$ and $U_L(1) \leq -k$, so every equilibrium has $\sigma_H(-1) = \sigma_L(1) = 0$;

(d) if $\sigma_H(-1) = \sigma_L(1) = 0$, then $\mu(-1) \leq 1/2 \leq \mu(1)$ and $\mu(0) \leq 1/2$ when also $\sigma_H(1) = 1$; and $\mu(1) = M_{\hat{\lambda}}$ when $\sigma_H(1) = 1$.

Proof. (a) This is Lemma A.1(b)–(c) at the presence $\hat{\lambda}$ and the prior $1/2$; the zero order uses $\beta \leq 1/3$. The map $\lambda \mapsto u_\lambda$ rises, so $M_{\hat{\lambda}} \leq M_{\bar{\lambda}}$. (b) This is Lemma A.2(c). (c) This is Lemma A.3(a). (d) Here $f_H(1) = \hat{b} + \hat{\lambda}\sigma_H(1) \geq \hat{b} = f_L(1)$ and $f_H(-1) = \hat{b} \leq \hat{b} + \hat{\lambda}\sigma_L(-1) = f_L(-1)$. With $\sigma_H(1) = 1$, $f_H(0) = \hat{b}_0 \leq \hat{b}_0 + \hat{\lambda}\sigma_L(0) = f_L(0)$, and $\mu(1) = (\hat{b} + \hat{\lambda})/(2\hat{b} + \hat{\lambda}) = M_{\hat{\lambda}}$. □

Lemma A.11 (Positive research needs a strict full purchase)**.** *Let $\tau(r) > 1/2$, and take an equilibrium with $\hat{\lambda} > 0$. Then $U_H(1) > 0$ and $\sigma_H(1) = 1$. Also $\mu(1) = M_{\hat{\lambda}}$, $\mu(0) \le 1/2$, $\mu(-1) \le 1/2$, and $\zeta(P(0)) = \zeta(P(-1)) = 0$.*

Proof. By Lemma A.9, $\hat{\lambda} = \min\{\bar{\lambda}, V/a\} > 0$, so $V > 0$. By Lemma A.10(c) and $U_j(0) = 0$, either $U_H(1) > 0$ or $U_L(-1) > 0$. Suppose $U_L(-1) > 0$. Then the sale is the unique maximizer of class L, and (E5) gives $\sigma_L(-1) = 1$. With $\sigma_H(-1) = 0$, $f_H(-1) = \hat{b}$ and $f_L(-1) = \hat{b} + \hat{\lambda}$, so $\mu(-1) = m_{\hat{\lambda}} \le 1/2 < \tau(r)$. Hence $e(-1) = \rho$ and $U_L(-1) = \rho\Delta_T m_{\hat{\lambda}} - k > 0$. The purchase then earns $U_H(1) = e(1)\Delta_T[1 - \mu(1)] - k \ge \rho\Delta_T(1 - M_{\hat{\lambda}}) - k = \rho\Delta_T m_{\hat{\lambda}} - k > 0$, by Lemma A.10(a). So $U_H(1) > 0$ in both cases. The purchase is the unique maximizer of class H, and $\sigma_H(1) = 1$ under both investor rules. Lemma A.10(d) gives the three posteriors. Since $\tau(r) > 1/2$, Lemma A.2(d) gives $\zeta(P(0)) = \zeta(P(-1)) = 0$ under both challenger rules. □

Lemma A.12 (The equilibrium set at intermediate strengths)**.** *Let $1/2 < \tau(r) < 1$, $\rho\Delta_T(r) < 2k$ and $\lambda_E(r) \neq \bar{\lambda}$. The equilibria are exactly these tuples. Each has $\sigma_L = \delta_0$, $\zeta(P(0)) = \zeta(P(-1)) = 0$, prices $P(q) = t_0 + e(q)\, w(\mu(q))$, $\bar{\mu}(P(q)) = \mu(q)$, and truthful bids.*

- *(Z)* $\hat{\lambda} = 0$, $\sigma_H = \delta_0$, $\mu \equiv 1/2$, $\zeta \equiv 0$. *It always exists.*
- *(R)* $\hat{\lambda} = \lambda_R(r)$, $\sigma_H = \delta_1$, $(\mu(-1), \mu(0), \mu(1)) = \big(1/2,\ \hat{b}_0/(2\hat{b}_0 + \hat{\lambda}),\ M_{\hat{\lambda}}\big)$, $\zeta(P(1)) = 1$. *It exists if and only if $\varphi_a(r) \ge 0$ and $F_r(\bar{\lambda}) > 0$.*
- *(R̄)* $\hat{\lambda} = \bar{\lambda}$, *with the orders, posteriors and entry of (R). It exists if and only if $F_r(\bar{\lambda}) \le 0$ and $\lambda_E(r) \le \bar{\lambda}$.*
- *(T) Rule CM only:* $\hat{\lambda} = \lambda_E(r)$, $\sigma_H = \delta_1$, $\mu(1) = \tau(r)$, $\zeta(P(1)) = (e^* - \rho)/(1 - \rho)$. *It exists if and only if $\lambda_E(r) < \bar{\lambda}$ and $\varphi_a(r) > 0$.*

Every trade choice in these equilibria is strict, so the investor's tie rule plays no role. Entry is ρ in (Z), $\rho + (1 - \rho)[\beta + (1/2 - \beta)\hat{\lambda}]$ in (R) and (R̄), and $\rho + (e^* - \rho)[\beta + (1/2 - \beta)\lambda_E]$ in (T).

Proof. Step 1: (Z). (E1): $\hat{\lambda} = 0$ gives $f_H = f_L = \eta$, so $\mu \equiv 1/2$. (E2): every price belief is $1/2$. (E3): $\tau(r) > 1/2$ gives $B_r(1/2) < c_H$, so $\zeta \equiv 0$ with no tie. (E4): the price $t_0 + \rho\, w(1/2)$ at every order. (E5): after a hypothetical success, $U_H(1) = U_L(-1) = \rho\Delta_T/2 - k < 0$, and the wrong-signed orders earn at most $-k$, so abstention is the strict best order of each class. (E6): truthful bids. (E7): $V = 0$, so $\min\{\bar{\lambda}, V/a\} = 0 = \hat{\lambda}$.

Step 2: (R). Let $\hat{\lambda} = \lambda_R(r)$, $\sigma_H = \delta_1$, $\sigma_L = \delta_0$. Then $f_H = (\hat{b}, \hat{b}_0, \hat{b} + \hat{\lambda})$ and $f_L = (\hat{b}, \hat{b}_0 + \hat{\lambda}, \hat{b})$ on $(-1, 0, 1)$, which give the stated posteriors with $\mu(0) < 1/2 = \mu(-1) < \mu(1)$. Suppose $\varphi_a(r) \ge 0$ and $F_r(\bar{\lambda}) > 0$. Then $\Delta_T(1 - \tau) \ge k + 2a\lambda_E > k$, and $1 - \tau < 1/2$ gives $\Delta_T > 2k$, so $\lambda_R(r)$ exists. (E1) holds by the likelihoods. (E2): the stated entry is ρ at the posteriors $\mu(0)$ and $\mu(-1)$, and 1 at

$\mu(1)$, so it is nondecreasing in the posterior; by the paragraph on constructed tuples in Appendix A.2, the three prices differ, and $\bar{\mu}(P(q)) = \mu(q)$. (E3): $F_r(\lambda_E) = -[2\beta + \lambda_E(1-2\beta)]\varphi_a \le 0 = F_r(\lambda_R)$, so $\lambda_R \ge \lambda_E$ and $\mu(1) = M_{\lambda_R} \ge \tau(r)$; rule C1 gives $\zeta(P(1)) = 1$, and rule CM forces it if $\mu(1) > \tau$ and allows it if $\mu(1) = \tau$; the posteriors $\mu(0), \mu(-1) \le 1/2 < \tau$ give $\zeta = 0$ there. (E4): (A.4). (E5): $U_H(1) = \Delta_T m_{\hat{\lambda}} - k$, and $F_r(\lambda_R) = 0$ gives $\Delta_T m_{\lambda_R} - k = 2a\lambda_R > 0$; $U_L(-1) = \rho\Delta_T\mu(-1) - k = \rho\Delta_T/2 - k < 0$; the wrong-signed orders earn at most $-k$. So both choices are strict. (E6): truthful bids. (E7): $V = \frac{1}{2}(2a\lambda_R) + \frac{1}{2}\cdot 0 = a\lambda_R$, and $F_r(\bar{\lambda}) > 0$ gives $\lambda_R < \bar{\lambda}$, so $\min\{\bar{\lambda}, V/a\} = \lambda_R = \hat{\lambda}$. Conversely, (R) needs $\mu(1) \ge \tau$, that is $\lambda_R \ge \lambda_E$, that is $\varphi_a \ge 0$, and it needs $\lambda_R < \bar{\lambda}$, that is $F_r(\bar{\lambda}) > 0$.

Step 3: ($\bar{\text{R}}$). The orders and posteriors are those of Step 2 with $\hat{\lambda} = \bar{\lambda}$. If $\lambda_E \le \bar{\lambda}$, then $\mu(1) = M_{\bar{\lambda}} \ge \tau$; rule C1 gives $\zeta(P(1)) = 1$, and rule CM forces it if $\mu(1) > \tau$ and allows it if $\mu(1) = \tau$. Then $V = \frac{1}{2}(\Delta_T m_{\bar{\lambda}} - k)$, and $F_r(\bar{\lambda}) \le 0$ is $V \ge a\bar{\lambda}$, so $\min\{\bar{\lambda}, V/a\} = \bar{\lambda}$. The purchase earns $\Delta_T m_{\bar{\lambda}} - k \ge 2a\bar{\lambda} > 0$, and the sale earns $\rho\Delta_T/2 - k < 0$, so the choices are strict. The other conditions hold by direct computation: (E1) by the likelihoods of Step 2 at $\hat{\lambda} = \bar{\lambda}$; (E2) because the stated entry is nondecreasing in the posterior, so the three prices differ; (E4) by (A.4); (E6) by truthful bids. Both conditions are also necessary.

Step 4: (T), rule CM. Let $\hat{\lambda} = \lambda_E < \bar{\lambda}$, $\sigma_H = \delta_1$, $\sigma_L = \delta_0$ and $\zeta(P(1)) = (e^* - \rho)/(1-\rho)$. Then $\mu(1) = M_{\lambda_E} = \tau(r)$, so $B_r(\mu(1)) = c_H$, and rule CM allows any $\zeta(P(1))$. By the definition of e^*, $U_H(1) = e^*\Delta_T(1-\tau) - k = 2a\lambda_E > 0$, so the purchase is strict. The sale earns $\rho\Delta_T\mu(-1) - k = \rho\Delta_T/2 - k < 0$, so the low class abstains strictly. Then $V = a\lambda_E$, and $\min\{\bar{\lambda}, V/a\} = \lambda_E$. The preparation probability lies in $(0,1)$ if and only if $\rho < e^* < 1$. Here $e^* < 1$ is $\varphi_a(r) > 0$, and $e^* > \rho$ always holds, because $\rho\Delta_T(1-\tau) < \rho\Delta_T/2 < k < 2a\lambda_E + k$. At $e^* = 1$ the tuple is (R) with $\lambda_R = \lambda_E$.

Step 5: no other equilibrium. Take any equilibrium. If $\hat{\lambda} = 0$, then $f_H = f_L = \eta$, $\mu \equiv 1/2$ and $e \equiv \rho$, so $U_H(1) = U_L(-1) = \rho\Delta_T/2 - k < 0$, and (E5) gives $\sigma_H = \sigma_L = \delta_0$: this is (Z). If $\hat{\lambda} > 0$, Lemma A.11 gives $\sigma_H = \delta_1$, $\mu(1) = M_{\hat{\lambda}}$, $\mu(-1) \le 1/2$ and $e(-1) = \rho$. Then $U_L(-1) \le \rho\Delta_T/2 - k < 0$, so $\sigma_L = \delta_0$, and $V = \frac{1}{2}\big(e(1)\Delta_T m_{\hat{\lambda}} - k\big)^+$. Three cases remain.

- $\hat{\lambda} < \lambda_E$: $\mu(1) < \tau$, so $e(1) = \rho$, and $\rho\Delta_T m_{\hat{\lambda}} < \rho\Delta_T/2 < k$, so $V = 0$. Lemma A.9 then gives $\hat{\lambda} = 0$, a contradiction.

- $\hat{\lambda} > \lambda_E$: $\mu(1) > \tau$, so $e(1) = 1$ under both rules. If $\hat{\lambda} < \bar{\lambda}$, then $a\hat{\lambda} = V > 0$, which is $F_r(\hat{\lambda}) = 0$, so $\hat{\lambda} = \lambda_R$: this is (R). If $\hat{\lambda} = \bar{\lambda}$, then $V \ge a\bar{\lambda}$, which is $F_r(\bar{\lambda}) \le 0$: this is ($\bar{\text{R}}$).

- $\hat{\lambda} = \lambda_E$: under rule C1, $e(1) = 1$, and the previous case applies with $\lambda_R = \lambda_E$. Under rule CM, $e(1) = e \in [\rho, 1]$. Since $\lambda_E \ne \bar{\lambda}$, research is interior, and $a\lambda_E = \frac{1}{2}\big(e\Delta_T(1-\tau) - k\big)^+$, so $e = e^*$: this is (T), or (R) when $e^* = 1$.

In each listed equilibrium with positive research, the purchase earns $2a\hat{\lambda} > 0$ or more. The sale earns $\rho\Delta_T/2 - k < 0$. Step 1 gives strict abstention in (Z), so every trade choice is strict. A purchase has probability $\frac{1}{2}(\hat{b} + \hat{\lambda}) + \frac{1}{2}\hat{b} = \beta + (1/2 - \beta)\hat{\lambda}$, and only the purchase can raise entry above ρ. This gives the three entry formulas. □

Lemma A.13 (Strong incumbents)**.** *Let $\tau(r) > 1/2$, $M_{\bar{\lambda}} \neq \tau(r)$ and*

$$\rho\Delta_T(r)\, m_{\bar{\lambda}} - k > a\bar{\lambda}. \tag{U}$$

Under every pair of tie rules the equilibrium is unique: $\hat{\lambda} = \bar{\lambda}$, $\sigma_H = \delta_1$, $\sigma_L = \delta_{-1}$, $(\mu(-1), \mu(0), \mu(1)) = (m_{\bar{\lambda}}, 1/2, M_{\bar{\lambda}})$, $\zeta(P(0)) = \zeta(P(-1)) = 0$, and $\zeta(P(1)) = \mathbf{1}\{M_{\bar{\lambda}} > \tau(r)\}$, with $P(q) = t_0 + e(q)\, w(\mu(q))$, $\bar{\mu}(P(q)) = \mu(q)$ and truthful bids. Entry is $\rho + (1 - \rho)(\bar{b} + \bar{\lambda}/2)$, with $\bar{b} = (1 - \bar{\lambda})\beta$, if $M_{\bar{\lambda}} > \tau(r)$, and ρ if $M_{\bar{\lambda}} < \tau(r)$. This is the equilibrium of Proposition 2 with presence $\bar{\lambda}$.

Proof. Uniqueness. Take any candidate. By Lemma A.10(a), every posterior lies in $[m_{\bar{\lambda}}, M_{\bar{\lambda}}]$ whatever the orders, and $e \geq \rho$ after every price. Hence $U_H(1) \geq \rho\Delta_T(1 - M_{\bar{\lambda}}) - k$ and $U_L(-1) \geq \rho\Delta_T m_{\bar{\lambda}} - k$. Both lower bounds equal $\rho\Delta_T m_{\bar{\lambda}} - k > a\bar{\lambda} \geq 0$ by (U). So the purchase and the sale are the strict best orders after a success, under both investor rules. Then $V = \frac{1}{2}[U_H(1) + U_L(-1)] > a\bar{\lambda}$, and Lemma A.9 gives $\hat{\lambda} = \bar{\lambda}$. With full orders, $f_H = (\bar{b}, \bar{b}_0, \bar{b} + \bar{\lambda})$ and $f_L = (\bar{b} + \bar{\lambda}, \bar{b}_0, \bar{b})$, so the posteriors are $(m_{\bar{\lambda}}, 1/2, M_{\bar{\lambda}})$. Since $\tau(r) > 1/2$, $\zeta(P(0)) = \zeta(P(-1)) = 0$. Since $M_{\bar{\lambda}} \neq \tau(r)$, $\zeta(P(1))$ is 1 or 0 with no tie, under both challenger rules. Every component of the tuple is fixed.

Verification. (E1): the likelihoods above. (E2): the stated entry is nondecreasing in the posterior, so the three prices differ, and each price reveals its order. (E3): $B_r(M_{\bar{\lambda}})$ lies strictly on one side of c_H, and $B_r(1/2), B_r(m_{\bar{\lambda}}) < c_H$. (E4): (A.4). (E5): the strict bounds above, read at the tuple. (E6): truthful bids. (E7): $V > a\bar{\lambda}$. The purchase has probability $\frac{1}{2}(\bar{b} + \bar{\lambda}) + \frac{1}{2}\bar{b} = \bar{b} + \bar{\lambda}/2$, and it recruits exactly when $M_{\bar{\lambda}} > \tau(r)$. □

Proof of Proposition 4(b). Throughout, $a = 1/40$ and $\bar{\lambda} = 0.336122$.

Part (b)(i). Take any candidate at r_0. The gross return of a nonzero order after a success is at most $e(q)\Delta_T \max\{\mu(q), 1 - \mu(q)\} \leq \Delta_T$, and $\Delta_T(r_0) = 1/60 < k$, with margin 0.003333. So $U_j(q) < 0$ for $q \neq 0$ against every price function and entry rule. Then $V = 0$, Lemma A.9 gives $\hat{\lambda} = 0$, and (E5) gives $\sigma_H = \sigma_L = \delta_0$. Every posterior is $1/2$, and $B_{r_0}(1/2) < c_H$, with margin 1.195833, so $\zeta \equiv 0$ and entry is ρ. Step 1 of the proof of Lemma A.12 verifies this tuple. The general statement behind this part needs both $\Delta_T(r) < k$ and $B_r(1/2) < c_H$: with c_H small enough that $B_r(1/2) > c_H$, zero research is still unique, but both cost types prepare and entry is one.

Part (b)(ii). At each $r \in \{1.55, 1.60, 1.65\}$, the hypotheses of Lemma A.12 hold: $1/2 < \tau(r) < 1$;

$k - \rho\Delta_T(r)/2$ equals 0.007802, 0.005938 and 0.003996; and $\lambda_E(r) < \bar{\lambda}$. Also $\lambda_R(r) - \lambda_E(r)$ equals 0.041160, 0.079340 and 0.114019, so $\varphi_a(r) > 0$; and $\bar{\lambda} - \lambda_R(r)$ equals 0.098221, 0.057085 and 0.019439, so $F_r(\bar{\lambda}) > 0$. Lemma A.12 therefore gives the equilibrium set {Z, R} under rule C1, and {Z, R, T} under rule CM, under either investor rule. Table 4 gives the values. The research values come from bisection of F_r with exact signs; Δ_T, τ, λ_E and e^* are exact rationals.

Part (b)(iii). At calibration B, $M_{\bar{\lambda}} = \mathrm{e}/(1+\mathrm{e})$, and $\tau(r_1) < M_{\bar{\lambda}} < \tau(r_2)$, with margins 0.026059 and 0.000334 in posterior. Condition (U) is $\Delta_T(r) > (k + a\bar{\lambda})(1+\mathrm{e})/\rho$, which holds for every $r > r_U = \mathfrak{r}\big((k + a\bar{\lambda})(1+\mathrm{e})/\rho\big) = 2.434046$; its margin is 0.016421 at r_1 and 0.034723 at r_2. Lemma A.13 gives the unique equilibrium at both strengths, with entry 0.522757 at r_1 and 0.250000 at r_2.

Every trade choice at the six strengths is strict, by Step 5 of the proof of Lemma A.12 and by the bounds of Lemma A.13. The investor's tie rule therefore plays no role. □

Proof of Proposition 4(a). The polynomial $F_r(\lambda)$ is smooth in (λ, r), since its coefficients are affine in $\Delta_T(r)$. For $\lambda \geq 0$,

$$\partial_\lambda F_r = 4a(1-2\beta)\lambda + 4a\beta + k(1-2\beta) + \beta\Delta_T > 0, \qquad \partial_r F_r = -\beta(1-\lambda)\Delta_T'(r) < 0,$$

because $\lambda < 1$ and $\Delta_T'(r) = \frac{1}{2}(1 - \ell^2/r^2) > 0$. On $\{r : \Delta_T(r) > 2k\}$, the implicit function theorem gives

$$\lambda_R'(r) = \frac{\beta\,[1-\lambda_R(r)]\,\Delta_T'(r)}{\partial_\lambda F_r(\lambda_R(r))} > 0.$$

On the branch (R), Lemma A.12 gives $\mathsf{E}_R = \rho + (1-\rho)[\beta + (1/2-\beta)\lambda_R]$. Since $\beta < 1/2$, E_R rises strictly with λ_R and hence with r: $\mathsf{E}_R' = (1-\rho)(1/2-\beta)\lambda_R' > 0$. Class H buys with probability $\hat{b} + \lambda_R = \beta + (1-\beta)\lambda_R$, and $W_H = 1$ because $h > r$. So ownership is $\frac{1}{2}[\rho + (1-\rho)(\beta + (1-\beta)\lambda_R)]$, which also rises strictly. These signs hold for every $0 < \beta \leq 1/3$, wherever the branch is an equilibrium. □

The branch at calibration B. At $a = 1/40$, (R) exists exactly on $[r_R, \bar{r}_R) = [1.501023, 1.677633)$. The proof has four steps.

1. For $r \leq r_0$, $\Delta_T(r) \leq \Delta_T(r_0) < k$, so part (b)(i) applies, and (R) does not exist.

2. The margin φ_a rises on $[r_0, 7/4]$. By the formula for $d\lambda_E/d\tau$, $\varphi_a' = \Delta_T'(1-\tau) - \Delta_T\tau' - 2a\beta\tau'/\{\beta + (1-2\beta)(1-\tau)\}^2$, with $\tau'(r) = 3(8r^2 - r + 2)/[2(20r - r^2 - 1)^2] > 0$. On this interval Δ_T', Δ_T and τ rise, so $\Delta_T' \geq 11/72$, $1 - \tau \geq 1 - \tau(7/4) = 19/55$, $\Delta_T \leq \Delta_T(7/4) = 9/56$ and $\tau' \leq 16875/211288$. These give $(\Delta_T(1-\tau))' \geq 0.039942$ and, at $a = 1/40$, $\varphi_a' \geq 0.033767 > 0$. The bound stays positive for every $a < a_{\text{mono}} = 0.161713$. Exact rational

evaluation gives $\varphi_a(r_0) < 0 < \varphi_a(7/4)$. So φ_a has the unique root r_R on $[r_0, 7/4]$, and bisection with exact signs encloses it.

3. For $r_R \le r < \bar{r}_R$: $r < r_N$, so $\rho\Delta_T < 2k$, and $1/2 < \tau < 1$; also $\varphi_a(r) \ge 0$. Since $m_{\bar{\lambda}} = 1/(1+\mathrm{e})$, $F_r(\bar{\lambda}) = 0$ is $\Delta_T(r) = (2a\bar{\lambda} + k)(1 + \mathrm{e})$, and $\bar{r}_R = \mathfrak{r}\big((2a\bar{\lambda} + k)(1 + \mathrm{e})\big)$. Since λ_R rises and $\lambda_R(\bar{r}_R) = \bar{\lambda}$, $F_r(\bar{\lambda}) > 0$ for $r < \bar{r}_R$, and (R) exists by Lemma A.12. On (r_0, r_R), $\varphi_a < 0$, and (R) does not exist.

4. For $r \ge \bar{r}_R$, take any equilibrium with $e(1) = 1$ and $\hat{\lambda} < \bar{\lambda}$. Lemma A.11 gives $\sigma_H = \delta_1$ and $\mu(1) = M_{\hat{\lambda}}$. The low class earns at least zero, so $a\hat{\lambda} = V \ge \frac{1}{2}(\Delta_T m_{\hat{\lambda}} - k)$, that is $F_r(\hat{\lambda}) \ge 0$. Hence $\hat{\lambda} \ge \lambda_R(r) \ge \lambda_R(\bar{r}_R) = \bar{\lambda}$, a contradiction. This step also covers $r > r_N$, where the low class may sell.

Zero research is an equilibrium exactly when $\rho\Delta_T \le 2k$, that is $r \le r_N$, and $\bar{r}_R < r_N = 1.747878$, with margin 0.070245. Along the branch, research rises from $\lambda_E(r_R) = 0.193853$ to the cap, and entry rises from 0.500846 to 0.522757. On $[\bar{r}_R, r_N)$, $F_r(\bar{\lambda}) \le 0$ and $\lambda_E(r) \le \lambda_E(7/4) = 0.208635 < \bar{\lambda}$, so ($\bar{\mathrm{R}}$) exists by Lemma A.12, with entry 0.522757. Under rule CM, (T) exists on (r_R, r_N), since $\varphi_a > 0$ and $\lambda_E < \bar{\lambda}$ there.

Proposition A.2 (The admissible cap and cost)**.** *Fix $\beta = 0.294655$ and the other primitives of B, and vary the cap $\bar{\lambda}$ and the cost a. The equilibrium patterns of Proposition 4(b), with $\lambda_E(r) < \lambda_R(r) < \bar{\lambda}$ at each intermediate strength, hold with strict margins if and only if*

$$\lambda_E(r_1) < \bar{\lambda} < \lambda_E(r_2) \qquad \textit{and} \qquad a_{\mathrm{lo}}(\bar{\lambda}) < a < a_{\mathrm{up}},$$

where

$$a_{\mathrm{lo}}(\bar{\lambda}) = \frac{\Delta_T(1.65)\, m_{\bar{\lambda}} - k}{2\bar{\lambda}}, \qquad a_{\mathrm{up}} = \frac{\Delta_T(1.55)\,[1 - \tau(1.55)] - k}{2\lambda_E(1.55)}.$$

The cap strip is $(0.290539, 0.336721)$ to six places. At the cap $\bar{\lambda} = 0.336122$, the cost window is $(0.021469, 0.036763)$; at $a = 1/40$, the cap range is $(0.316683, 0.336721)$. Entry at r_1 is $\rho + (1-\rho)[\beta + (1/2 - \beta)\bar{\lambda}]$. With $1 + u = \mathrm{e}$, the cap $\bar{\lambda} = u\beta/(1 + u\beta)$ and $a = 1/40$, these patterns hold for every $\beta \in (0.236157, 1/3]$.

The lower cost bound keeps research below the cap at 1.65. The upper cost bound keeps recruitment after a purchase at 1.55. The cap strip gives strict recruitment at r_1 and strict exclusion at r_2.

Proof of Proposition A.2. Fix $\beta = 0.294655$ and the other primitives of calibration B. The required equilibrium patterns are those of Proposition 4(b) with $\lambda_E(r) < \lambda_R(r) < \bar{\lambda}$ at each intermediate strength.

Sufficiency. Let the two conditions hold. Part (b)(i) holds for every $(\bar{\lambda}, a)$, since $\Delta_T(r_0) < k$. For part (b)(ii), the hypotheses $1/2 < \tau < 1$ and $\rho\Delta_T < 2k$ of Lemma A.12 do not involve $(\bar{\lambda}, a)$ and hold at the three strengths. The condition $a < a_{\text{up}}$ is $\varphi_a(1.55) > 0$. Since $a_{\text{up}} < a_{\text{mono}}$, φ_a rises on $[r_0, 7/4]$, so $\varphi_a > 0$ at 1.60 and 1.65 too. The condition $a > a_{\text{lo}}(\bar{\lambda})$ is $F_{1.65}(\bar{\lambda}) > 0$, that is $\lambda_R(1.65) < \bar{\lambda}$. Since λ_R rises in r, $\lambda_R < \bar{\lambda}$ at the two lower strengths too. Then $\lambda_E < \lambda_R < \bar{\lambda}$, and Lemma A.12 gives the sets. For part (b)(iii), M_λ rises in λ and $M_{\lambda_E(r)} = \tau(r)$, so the strip gives $\tau(r_1) < M_{\bar{\lambda}} < \tau(r_2)$. Condition (U) at r_1 is $a < a_U(r_1; \bar{\lambda}) := [\rho\Delta_T(r_1)m_{\bar{\lambda}} - k]/\bar{\lambda}$, and at r_2 it is $a < a_U(r_2; \bar{\lambda}) := [\rho\Delta_T(r_2)m_{\bar{\lambda}} - k]/\bar{\lambda}$. Both bounds fall in $\bar{\lambda}$, so on the strip they exceed their values at $\bar{\lambda} = \lambda_E(r_2)$, which are 0.073556 and 0.127845. Both exceed $a_{\text{up}} = 0.036763$, so Lemma A.13 applies at both strengths. Its entry formula gives $\rho + (1-\rho)[\beta + (1/2 - \beta)\bar{\lambda}]$ at r_1.

Necessity. Each failure of a condition breaks one claim.

- $\bar{\lambda} \le \lambda_E(r_1)$: every posterior at r_1 is at most $M_{\bar{\lambda}} \le \tau(r_1)$, so entry above ρ needs a challenger tie, and part (b)(iii) fails with strict margins.
- $\bar{\lambda} \ge \lambda_E(r_2)$: research at the cap gives $\mu(1) = M_{\bar{\lambda}} \ge \tau(r_2)$, so rule C1 raises entry above ρ at r_2, and part (b)(iii) fails.
- $a \ge a_{\text{up}}$: $\varphi_a(1.55) \le 0$, so (R) is absent at 1.55, or has $\mu(1) = \tau$ with a challenger tie.
- $a \le a_{\text{lo}}(\bar{\lambda})$: $F_{1.65}(\bar{\lambda}) \le 0$, so research at 1.65 sits at the cap.

The cost window is nonempty on the whole strip: a_{lo} falls in $\bar{\lambda}$, because its numerator falls and stays positive there, and $a_{\text{lo}}(\lambda_E(r_1)) = 0.030579 < a_{\text{up}}$. The cap strip is $(0.290539, 0.336721)$; the margins of $\bar{\lambda} = 0.336122$ are 0.045583 and 0.000599, and the margins of $a = 1/40$ in its window are 0.003531 and 0.011763. The upper cap margin corresponds to the posterior margin $\tau(r_2) - M_{\bar{\lambda}} = 0.000334$; the two numbers measure one boundary in two coordinates.

General β. Set $\bar{\lambda} = \lambda_B(\beta) = u\beta/(1 + u\beta)$ with $1 + u = \mathrm{e}$, and keep $a = 1/40$. Then $u_{\bar{\lambda}} = u$ for every β, so $M_{\bar{\lambda}} = \mathrm{e}/(1 + \mathrm{e})$ and $m_{\bar{\lambda}} = 1/(1 + \mathrm{e})$ do not depend on β, and the endpoint conditions $\tau(r_1) < M_{\bar{\lambda}} < \tau(r_2)$ hold for every β. Condition (U) at r_1 holds for every $\beta \le 1/3$, because $\lambda_B(\beta)$ rises in β and $\rho\Delta_T(r_1)/(1 + \mathrm{e}) - k - a\lambda_B(1/3) > 0$. At the intermediate strengths, $a_{\text{lo}} = [\Delta_T(1.65)/(1 + \mathrm{e}) - k]/[2\lambda_B(\beta)]$ falls in β, so $a_{\text{lo}} < a$ exactly when $\beta > \beta_{\text{lo}} = 0.236157$. The upper bound $a_{\text{up}}(\beta)$ falls in β, because $\partial\lambda_E/\partial\beta = (2\tau - 1)(1 - \tau)/\{\beta + (1 - 2\beta)(1 - \tau)\}^2 > 0$, and it exceeds a for every $\beta < 0.4897$. The monotonicity bound $a_{\text{mono}}(\beta)$ falls in β on $(0, 1/3]$ and equals $0.150885 > a$ at $\beta = 1/3$. The steps above then apply at each $\beta \in (0.236157, 1/3]$. Strict growth of entry across the three strengths alone holds on a wider set, because research can reach the cap at 1.65 while entry still rises. □

Remark A.1 (Tie rules with research). Research never has a tie, and Proposition 4(b) needs no investor tie rule. Under the challenger rule CM, each intermediate strength has one more equilibrium. In it, research is $\lambda_E(r)$, and the purchase posterior equals the threshold. The indifferent challenger prepares with the probability that makes this research optimal. Its entry falls across the three strengths: it is 0.456294, 0.420777 and 0.392494 at $r = 1.55$, 1.60 and 1.65. Above r_U, the whole tuple is unique under C1, and under CM at every strength except the ceiling $r_C = 3.592659$. At r_C under CM, entry can take any value in $[\rho, 0.522757]$.

Proof of Remark A.1. Every trade choice is strict at the six strengths (proof of Proposition 4(b)), and research has no tie (Lemma A.9). The threshold equilibrium is (T) of Lemma A.12; its entry is 0.456294, 0.420777 and 0.392494 at the three strengths, with e^* equal to 0.865703, 0.758777 and 0.673748. For $r > r_U$, the bounds of Lemma A.13 force full orders and research at the cap in every candidate. Under rule C1, or under rule CM with $M_{\bar\lambda} \neq \tau(r)$, the whole tuple is then unique. At $r = r_C$, $M_{\bar\lambda} = \tau(r_C)$, the high-cost challenger is indifferent after a purchase, and rule CM admits every $\zeta(P(1)) \in [0, 1]$, with $\zeta(P(0)) = \zeta(P(-1)) = 0$. Each value has its own competitive purchase price, and entry $\rho + (1-\rho)\zeta(P(1))(\bar b + \bar\lambda/2)$ fills $[\rho, 0.522757]$. Research stays at the cap. □

Access to prices with research. At r_1, compare the economy above with one in which the challenger does not see the price. In that economy the challenger's belief stays at $1/2$, and $B_{r_1}(1/2) < c_H$ gives entry ρ after every order. The bounds of Lemma A.13 hold with $e \equiv \rho$, so (U) again forces full orders and research at the cap, uniquely. The order law, the research cost $(a/2)\bar\lambda^2$ and the trade cost $k\bar\lambda$ are therefore the same in both economies. Entry differs only after a purchase, where it is 1 with access and ρ without. The purchase has probability $\frac{1}{2}(\bar b + \bar\lambda)$ in class H and $\frac{1}{2}\bar b$ in class L, so access raises expected proceeds by $(1-\rho)(\bar b + \bar\lambda/2)[t_L - t_0 + \Delta_T M_{\bar\lambda}] = 0.257948$. For $\theta \geq p$, $\max\{\theta, R\} - R\mathbf{1}\{R \geq p\} = (\theta - \max\{p, R\})_+ + p\mathbf{1}\{R < p\}$, so a prepared challenger of class j adds $g_j + p\Pr(R < p) = g_j + p^2/r_1$ to allocation value. Net of the added preparation cost, access raises surplus by $(1-\rho)(\bar b + \bar\lambda/2)[B_{r_1}(M_{\bar\lambda}) - c_H + p^2/r_1] = 0.081960$. Both are positive, because $B_{r_1}(M_{\bar\lambda}) > c_H$.

Certificates. The exact program is `P3_research/P3_research.py` (run log `P3_research/P3_research.log`). It uses rational arithmetic for every rational quantity and rational intervals for every quantity that depends on e. It builds the tuple of every equilibrium of Lemmas A.12 and A.13 for each pair of tie rules and recomputes (E1) to (E7) at the tuple. It rounds every display outward from exact rational endpoints and writes the endpoints to `P3_research/P3_enclosures.json`. The independent program `XP3/recompute.py` (run log `XP3/recompute.log`) computes the research roots from the quadratic formula with rational bounds on every square root. It bounds e between

$\sum_{n\le 60} 1/n!$ and that sum plus $62/(61\cdot 61!)$, and it records exact rational endpoints for every printed value in `XP3/exact_enclosures.json`. These rational endpoints are the certificate of each display. The scan `P3_research/P3_float_scan.py` (run log `P3_research/P3_float_scan.log`) evaluates a grid of research intensities and order mixtures at each of the six strengths without the classification of Lemma A.12. It finds exactly the equilibria of the lemmas under rule C1. This scan enters no proof.

A.6 Robustness

This subsection states and proves Proposition A.3 and the results that Section 6 summarizes. Each proof uses the conventions of Appendix A.2.

The fixed-information control

Proposition A.3 (Fixed information)**.** *Fix the joint law of the class j and a signal S that the challenger observes. Fix the class value laws G_j and the cost law. In each economy i, require $\mathcal{L}_i(\theta, R \mid j, S) = G_j \otimes F_i$. The cost is independent of the class, both values and S. If the incumbent strengthens from F_0 to F_1, entry does not rise under rule C1. Its decrease is exactly*

$$\mathsf{E}_0 - \mathsf{E}_1 = \Pr\big\{B_{F_1}(\mu(S)) < C \le B_{F_0}(\mu(S))\big\}, \qquad \mu(S) = \Pr(j = H \mid S),$$

and the decrease is strict exactly when this probability is positive.

Proof of Proposition A.3. The cost is independent of the class, the values and the signal S, so $\mu(S) = \Pr(j = H \mid S)$ is the challenger's complete class belief at its decision. The condition $\mathcal{L}_i(\theta, R \mid j, S) = G_j \otimes F_i$ gives gross profit $B_{F_i}(\mu(S))$ in economy i. Under rule C1, the challenger prepares exactly when $C \le B_{F_i}(\mu(S))$. Proposition 1(a) gives $B_{F_1}(\mu) \le B_{F_0}(\mu)$ at every belief. Hence, pointwise,

$$\mathbf{1}\{C \le B_{F_0}(\mu(S))\} - \mathbf{1}\{C \le B_{F_1}(\mu(S))\} = \mathbf{1}\{B_{F_1}(\mu(S)) < C \le B_{F_0}(\mu(S))\} \ge 0.$$

The joint law of (S, C) is the same in both economies. Take expectations: $\mathsf{E}_0 - \mathsf{E}_1$ equals the probability in the proposition, which is nonnegative, and it is positive exactly when the decrease is strict. A tie probability q that is the same in both economies replaces each indicator by $\mathbf{1}\{C < B_{F_i}\} + q\,\mathbf{1}\{C = B_{F_i}\}$. With $B_{F_1} \le B_{F_0}$, the difference of the two expressions is 0 for $C < B_{F_1}$ and for $C > B_{F_0}$, $1 - q$ at $C = B_{F_1} < B_{F_0}$, 1 strictly between, q at $C = B_{F_0} > B_{F_1}$, and 0 when the two profits are equal; it is nonnegative, so entry does not rise with such a tie probability either. □

The control in the slot. The experiment fixes λ, β, π, the cost law and the full orders $(\sigma_H, \sigma_L) = (\delta_1, \delta_{-1})$. Lemma A.1(d) gives the posteriors (m, π, M). Let $e_r(\mu) = \rho + (1-\rho)\mathbf{1}\{B_r(\mu) \geq c_H\}$ and $P(x) = t_0 + e_r(\mu(x))\, w(\mu(x))$. This entry is nondecreasing in the posterior, so by the paragraph on constructed tuples in Appendix A.2 the three prices differ, and each price reveals its posterior. The challenger's signal is therefore the order, with a law that does not depend on r. Given the class and order, the value law is $G_j \otimes F_r$, as Proposition A.3 requires. At the benchmark, $\tau(r) = (48r-3)/[4(20r-r^2-1)]$ rises and $\tau(1) = 5/8 > 1/2 = \pi$, so only the top posterior M can recruit, and it recruits exactly when $r \leq r_C$. Entry is therefore $\rho + (1-\rho)(b + \lambda/2) = 0.522757$ on $(1, r_C]$ and $\rho = 0.250000$ above $r_C = 3.592659$. The probability of Proposition A.3 is 0 from r_0 to r_1, because the top posterior recruits at both strengths, and 0.272757 from r_1 to r_2. At r_0 the frozen profile is a control and not an equilibrium: its purchase and sale earn $\Delta_0(1-M) - k < 0$ and $\rho\Delta_0 m - k < 0$. At r_1, the bound $\rho\kappa\Delta_1 > k$ of Step 2 of the proof of Proposition 2 forces full orders in every candidate, so the frozen tuple is the unique equilibrium there.

When the challenger does not see the price, its belief stays at π, and $B_{r_i}(\pi) < c_H$ gives entry ρ after every order at both strengths. At r_0, $\Delta_0 < k$ forces zero orders; at r_1, the bound $\rho\kappa\Delta_1 > k$ holds with $e \equiv \rho$ and forces full orders. The prices are $t_0 + \rho\, w(\mu(x))$, and every condition holds by the computations of Steps 1 and 3 of the proof of Proposition 2 with $e \equiv \rho$. Entry is 0.250000 at both strengths. Table 3 lists all four rows. The values come from `P4_information/derive.py` (run log `P4_information/run.log`), which brackets e by $\sum_{n \leq 60} 1/n!$ and that sum plus $1/(60 \cdot 60!)$ and propagates exact rational endpoints. The independent derivation `XP4/xp4_check.py` (run log `XP4/run.log`) builds every tuple from the joint law of class, slot identity and order, and it finds the same values.

Preparation cost laws

The two-point cost law of Section 2.2 is not needed. Let the preparation cost C be independent of the class, the value, the incumbent's value and the slot, with the law

$$H_C = \rho H^0 + (1-\rho)H^1, \qquad \operatorname{supp} H^0 \subset [0, L], \qquad \operatorname{supp} H^1 \subset [c_-, c_+].$$

Either component may be atomless or have atoms. A challenger prepares when gross profit at its belief exceeds its cost, and it stays out when gross profit falls short. The low component plays the role of the zero cost, and the high component plays the role of c_H.

With a general cost law, the preparation rule assigns $\zeta(\bar{P}, C) \in [0, 1]$ to each price and cost, and (E3) reads: $\zeta(\bar{P}, C) = 1$ if $B(\bar{\mu}(\bar{P})) > C$, and $\zeta(\bar{P}, C) = 0$ if $B(\bar{\mu}(\bar{P})) < C$. Entry after the order x is $e(x) = \int \zeta(P(x), C)\, dH_C(C)$. Every step of Appendix A.2 that does not use the two-point law carries over; in particular, Lemma A.3 holds for any entry in $[\rho, 1]$.

Proposition A.4 (Preparation cost laws)**.** *Let the classes satisfy the class ranking* (1)*, and let* F_1 *strengthen* F_0*. Suppose*

$$0 < L < B_1(m), \qquad B_0(\pi) < c_- < c_+ < B_1(M), \qquad \Delta_0 < k < \rho\kappa\Delta_1.$$

(i) At F_0*, the unique orders are zero, and entry is* ρ*.*

(ii) At F_1*, the unique orders are full, and only a purchase recruits the high component. Entry is* $\rho + (1-\rho)(b + \pi\lambda)$*. High-class ownership rises if and only if* $[\rho + (1-\rho)(b+\lambda)]\, W_H(F_1) > \rho\, W_H(F_0)$*.*

Proof of Proposition A.4. Step 0: consequences of the hypotheses. The class ranking (1) gives $\Delta_i \geq 0$ and $w_L \geq 0$. Since $k > 0$, $k < \rho\kappa\Delta_1$ gives $\Delta_1 > 0$. The support gaps and Proposition 1(a) give $B_1(M) > c_+ > c_- > B_0(\pi) \geq B_1(\pi)$. Since $M > \pi$, $B_1(M) - B_1(\pi) = (M-\pi)(g_{H,1} - g_{L,1}) > 0$, so $g_{H,1} > g_{L,1}$. The proof needs no condition $\Delta_0 > 0$.

Step 1: the floor under every profile. Fix any candidate in either economy. By Lemma A.1, every price belief lies in $[m, M]$; this step uses $\beta \leq 1/3$ for a price that pools the zero order. Hence $B_{F_i}(\bar{\mu}) \geq B_{F_i}(m) \geq B_1(m) > L$, so every cost in the support of H^0 prepares strictly, at every price. Entry is therefore at least ρ after every order.

Step 2: orders. At F_0, the upper bound in (A.6) gives correct gross returns at most $\Delta_0 < k$, and wrong-signed orders earn at most $-k$; if $\Delta_0 = 0$, every nonzero order earns $-k$. So both classes abstain in every equilibrium. At F_1, Step 1 and (A.6) give correct gross returns at least $\rho\kappa\Delta_1 > k$, so full orders are forced in every equilibrium. These bounds hold under both investor rules.

Step 3: the tuples. At F_0: zero orders, $\mu \equiv \pi$, $\bar{\mu} \equiv \pi$, the single price $t_{0,0} + \rho\, w_0(\pi)$, and $\zeta(\bar{P}, C) = \mathbf{1}\{C \leq B_0(\pi)\}$. At F_1: full orders, posteriors (m, π, M), $\zeta(P(x), C) = \mathbf{1}\{C \leq B_1(\mu(x))\}$, entry $(\rho, \rho, 1)$, prices $t_{0,1} + e(x)\, w_1(\mu(x))$ and $\bar{\mu}(P(x)) = \mu(x)$; truthful bids in both. Verification: (E1) and (E2) hold as in Steps 1 and 3 of the proof of Proposition 2; at F_1, the stated entry is nondecreasing in the posterior and $\Delta_1 > 0$, so the prices differ. (E3): at F_0, $B_0(\pi) < c_-$ excludes the whole high component, and $B_0(\pi) > L$ includes the whole low component. At F_1, $B_1(M) > c_+$ includes the whole high component, and $B_1(m) < B_1(\pi) \leq B_0(\pi) < c_-$ excludes it at the other two posteriors. Every realized comparison is strict, so rules C1 and CM agree. (E4): (A.4). (E5): at F_0, the correct gross returns are $\rho\Delta_0(1-\pi)$ and $\rho\Delta_0\pi$, both below k; at F_1, the purchase earns $\Delta_1(1-M) - k > 0$ and the sale $\rho\Delta_1 m - k > 0$. (E6): truthful bids.

Step 4: outcomes. A purchase has probability $b+\lambda$ in class H and b in class L, and only a purchase admits the high component. So $\mathsf{E}_1 = \rho + (1-\rho)(b + \pi\lambda)$, class entry is $\bar{e}_H = \rho + (1-\rho)(b+\lambda)$, and ownership is $\pi\bar{e}_H W_H(F_1)$ against $\pi\rho W_H(F_0)$, as in Step 4 of the proof of Proposition 2. □

No step uses atomlessness or a density. The two-point law of the benchmark is the case $H^0 = \delta_0$,

$H^1 = \delta_{c_H}$.

The third economy and the benchmark band. For F_2, the conditions $L < B_2(m)$, $B_2(M) < c_-$ and $k < \rho\kappa\Delta_2$ give full orders by the same bound, include the low component everywhere, and exclude the high component at every posterior, so entry is ρ. Suppose instead that the low component prepares everywhere, that $B_r(\pi) < c_-$, and that the orders are full. The high component then prepares only after a purchase, with probability $H^1(B_r(M))$, so

$$\mathsf{E}(r) = \rho + (1-\rho)(b + \pi\lambda)\, H^1\big(B_r(M)\big).$$

This is the only place where atomlessness enters: for atomless H^1, H^1 is continuous, and $B_r(M)$ falls continuously in r, so entry is continuous and nonincreasing in strength. In the benchmark, take $L = 1/5$ and H^1 uniform on $[c_H - \varepsilon, c_H + \varepsilon]$ with $\varepsilon = 1/10$. The margins are $B_1(m) - L = 2.166178511$, $c_- - B_0(\pi) = 1.095833333$ and $B_1(M) - c_+ = 0.117154822$. Entry at r_1 is 0.522757 and ownership is 0.324401. At r_2, $c_- - B_2(M) = -0.097311098$: the top profit lies inside the band, at distances 0.097311098 and 0.102688902 from its ends, and the formula gives entry 0.382711 and ownership 0.222020 by direct evaluation of the uniform law. The return to ρ at r_2 holds for every half-width below $c_H - B_2(M) = 0.002688902$; at $\varepsilon = 1/1000$ its margin is 0.001688902.

Value classes and priors

Proposition A.5 (Value classes and priors)**.** *Fix classes that satisfy the class ranking* (1) *with* $\mathbb{E}_H[\theta] > \mathbb{E}_L[\theta]$*, a reserve* $p > 0$*, and any* $\pi, \rho \in (0,1)$ *and slot* (λ, β)*. For uniform incumbents, a nonempty open set of strengths and costs* (r_0, r_1, r_2, c_H, k) *satisfies (H1)–(H3) and the ownership condition of Proposition 2. On this set every conclusion of Proposition 2 holds.*

Proof of Proposition A.5. Let $\Gamma(x) = \int_p^x D(y)\, dy$, and write $\bar{\Gamma} = \mathbb{E}_H[\theta] - \mathbb{E}_L[\theta] > 0$ for the total mean gap. Under the class ranking (1), $D \geq 0$, so Γ is continuous and nondecreasing with limit $\bar{\Gamma}$. Let $v_* = \inf\{x \geq p : \Gamma(x) > 0\}$; finite means and $\bar{\Gamma} > 0$ give $p \leq v_* < \infty$.

Step 1: the uniform incumbent. For $R \sim U[0, r]$, (8) and integration by parts give

$$\Delta_T(r) = \int_p^r \Big(1 - \frac{y}{r}\Big) D(y)\, dy = \frac{1}{r}\int_p^r \Gamma(y)\, dy,$$

since $\Gamma(p) = 0$ and the boundary term at r vanishes. Since Γ is continuous,

$$\Delta_T'(r) = \frac{r\Gamma(r) - \int_p^r \Gamma(y)\, dy}{r^2} \geq \frac{p\,\Gamma(r)}{r^2} > 0 \qquad (r > v_*),$$

Table A.4. Value classes and priors outside the benchmark.

Panel A. Primitives								
Economy	Low class	High class	π	c_H	k	r_0	r_1	r_2
Moderate gap	1	2	1/2	89/100	1/500	21/20	3/2	
Narrow bands	$U[19/20, 21/20]$	$U[199/20, 201/20]$	1/2	6	1/50	6/5	3	18/5
Low prior	1	10	3/10	9/2	1/100	11/10	3	18/5
High prior	1	10	7/10	36/5	1/100	11/10	3	18/5
Wide classes	$U[3/5, 7/5]$	$U[1, 19]$	7/10	67/10	1/50	1	9/2	11/2
Same means, points	1	10	7/10	67/10	1/50	1	9/2	11/2

Panel B. Outcomes			
Economy	Entry E_0	Entry E_1	High-class ownership $\mathsf{O}_{H,1}$
Moderate gap	0.250000	0.522757	0.324401
Narrow bands	0.250000	0.522757	0.324401
Low prior	0.250000	0.472339	0.194641
High prior	0.250000	0.573175	0.454162
Wide classes	0.250000	0.573175	0.419819
Same means, points	0.250000	0.250000	0.175000

Notes. The slot of B, the reserve $1/2$ and $\rho = 1/4$ in every row. A single number is a point-mass class; $U[\cdot, \cdot]$ is a uniform class. The moderate-gap economy has two strengths. At the third strength every row with three strengths has entry ρ. The moderate-gap row satisfies (H1)–(H2) of Proposition 2; the other four reversal rows also satisfy (H3). The last row fails the recruitment condition at r_1.

because $\int_p^r \Gamma \le (r-p)\Gamma(r)$. Also $\Delta_T(r) \to 0$ as $r \downarrow v_*$, since $\Gamma = 0$ on $[p, v_*]$. The split identity of Section 3.1 gives $g_H(r) - g_L(r) = \bar{\Gamma} - \Delta_T(r) \ge \bar{\Gamma} - (r-p)\bar{\Gamma}/r = p\bar{\Gamma}/r > 0$, and at $r = v_*$, $g_H - g_L = \bar{\Gamma}$. Dominated convergence, with the integrable bound $1 - G_j$, gives continuity of $g_j(r)$ and $W_H(r)$ in r, and $g_j(r) \to 0$ as $r \to \infty$. Since $\theta \ge p$, $W_H(r) = \mathbb{E}_H[\min\{\theta/r, 1\}] \ge p/r > 0$.

Step 2: the construction. Let $e_H = \rho + (1-\rho)(b+\lambda) > \rho$. At v_*, $B_{v_*}(M) - B_{v_*}(\pi) = (M-\pi)\bar{\Gamma} > 0$. By continuity choose $r_1 > v_*$ close enough to v_* that $B_{r_1}(M) > B_{v_*}(\pi)$ and $e_H W_H(r_1) > \rho W_H(v_*)$. Then choose $r_0 \in (v_*, r_1)$ close enough to v_* that $\Delta_T(r_0) < \rho\kappa\Delta_T(r_1)$. Since $B_r(\pi)$ and $W_H(r)$ fall in r, $B_{r_0}(\pi) \le B_{v_*}(\pi) < B_{r_1}(M)$ and $e_H W_H(r_1) > \rho W_H(v_*) \ge \rho W_H(r_0)$. Choose c_H in $(B_{r_0}(\pi), B_{r_1}(M))$ and k in $(\Delta_T(r_0), \rho\kappa\Delta_T(r_1))$. Since $g_j(r) \to 0$, choose $r_2 > r_1$ with $B_{r_2}(M) < c_H$. These choices satisfy (H1) to (H3) and the ownership condition.

Step 3: openness and the conclusions. Each condition is a strict inequality between continuous functions of (r_0, r_1, r_2, c_H, k), so a neighborhood of the constructed point satisfies all of them. On this set, Proposition 2 supplies the three tuples, checks every condition against them, and proves uniqueness. In the point-mass case $D = \mathbf{1}\{\ell \le y < h\}$, so $v_* = \ell$. □

The economies of Table A.4. For a point value v and $R \sim U[0, r]$, integration over the pieces cut by p, v and r gives $t_v = v - (v^2 - p^2)/(2r)$, $g_v = (v^2 - p^2)/(2r)$ and $W_v = v/r$ for $p \le v \le r$, and $t_v = r/2 + p^2/(2r)$, $g_v = v - t_v$ and $W_v = 1$ for $v \ge r$. A uniform class averages these

piecewise polynomials over its support, with a cut at r; every piece has degree at most three. The script `P2_auction/derive.py` (run log `P2_auction/run.log`) computes these integrals in exact rational arithmetic, and every quantity that depends on e in outward interval arithmetic at 60 digits; a sign counts only when its enclosure excludes zero. The script certifies (H1)–(H2) for the moderate-gap row and (H1)–(H3) for the other four reversal rows. Table A.6 lists their margins. Proposition 2 then supplies the tuples and their uniqueness. In the wide classes the class order holds although the supports overlap: $G_L - G_H \geq 0$ everywhere, with equality only outside the common interior of the supports. The independent recomputation `XP2/xp2_check.py` (run log `XP2/run.log`) uses closed forms and Simpson's rule on each piece, derives the orders from the bounds on every profile, and agrees with every value.

The last row keeps the costs, the prior and the strengths of the wide classes and puts point masses at the same means. Its upper margin at r_1 is $B_1(M) - c_H = -0.018122 < 0$. The bounds of Steps 1 and 2 of the proof of Proposition 2 still force zero orders at r_0 and full orders at r_1 and r_2, since their trade margins are positive (Table A.6). Then $B_i(\mu) \leq B_i(M) < c_H$ at every posterior of the higher strengths, and the prior profit at r_0 is below c_H, so entry is ρ at all three strengths. The comparison of the two rows is a convexity statement. For each R, $v \mapsto (v - \max\{p, R\})_+$ is convex, so a mean-preserving spread within a class weakly raises g_j, and hence gross profit at every belief.

At the benchmark costs and strengths, the prior $3/10$ gives entry 0.250000 at all three strengths: the signed profit gap at r_1 is $B_{r_1}(M(3/10)) - c_H = -1.390821$. The prior $7/10$ gives entry 1.000000 at r_0, because the prior profit there exceeds the high cost by 0.600833, and entry 0.573175 at r_1. In both cases the bound $\Delta_T(r_0) < k$ forces zero orders at r_0, and the bound on every profile forces full orders at r_1 and r_2.

More than two classes

Let the classes $j = 1, \ldots, J$ have priors $\pi_j > 0$ and values above the reserve, and write $t_{j,i}$ and $g_{j,i}$ for the class objects at F_i. Every posterior vector ν that an order can produce lies in the set

$$\mathcal{M}_\pi = \left\{ \nu : \ \frac{\nu_j}{\pi_j} \leq (1+u)\frac{\nu_l}{\pi_l} \text{ for all } j, l \right\}.$$

The price reveals only the posterior mean of class proceeds, $\nu \cdot t$. Partition the classes into a low group $\mathcal{L}$ and a high group $\mathcal{H}$, with high-group prior mass $\bar{\pi}$. Write $\bar{g}_{\mathcal{H}}$ and $\bar{g}_{\mathcal{L}}$ for the prior-weighted group means of $g_{j,1}$.

Proposition A.6 (Many classes under separation)**.** *Let the incumbent strengthen from F_0 to F_1. Suppose*

$$\max_j t_{j,0} - \min_j t_{j,0} < k, \qquad \sum_j \pi_j g_{j,0} < c_H,$$

$$\min_{j\in\mathcal{H}} t_{j,1} - \max_{\nu\in\mathcal{M}_\pi} \nu\cdot t_1 > \frac{k}{\rho}, \qquad \min_{\nu\in\mathcal{M}_\pi} \nu\cdot t_1 - \max_{j\in\mathcal{L}} t_{j,1} > \frac{k}{\rho},$$

and $\bar{g}_{\mathcal{H}} > \bar{g}_{\mathcal{L}}$ with $\bar{g}_{\mathcal{L}} + M(\bar{\pi})(\bar{g}_{\mathcal{H}} - \bar{g}_{\mathcal{L}}) > c_H$. Then at F_0 the unique orders are zero, and entry is ρ. At F_1 the unique orders are a purchase in every high-group class and a sale in every low-group class, only a purchase recruits, and entry is $\rho + (1-\rho)(b + \bar{\pi}\lambda) > \rho$. These conclusions hold under either challenger rule.

Proof of Proposition A.6. The tuple now lists every σ_j, the posterior vector $\mu(q) = (\mu_j(q))_j$ after every order, the posterior vector at every price, the prices, ζ and the bids. Each class reaches each order through a liquidity order, so $f_j(q) \in [(1-\lambda)\eta(q),\ (1-\lambda)\eta(q) + \lambda]$, and with $\beta \le 1/3$ every ratio $f_j(q)/f_l(q)$ lies in $[1/(1+u), 1+u]$. Bayes' rule gives $\mu_j(q)/\pi_j = f_j(q)/\sum_l \pi_l f_l(q)$, so $\mu_j(q)/\pi_j \le (1+u)\,\mu_l(q)/\pi_l$ for all j, l, and every posterior vector lies in $\mathcal{M}_\pi$. A price belief is a mixture of order posteriors, and $\mathcal{M}_\pi$ is convex, so it also lies in $\mathcal{M}_\pi$.

Payoffs. Competitive prices give $P(q) = t_0 + e(q)\,[\mu(q)\cdot t - t_0]$, and $\mathbb{E}[V_T \mid j, q] = t_0 + e(q)(t_j - t_0)$ as in (A.4). Hence, against the fixed price and entry,

$$U_j(q) = q\,e(q)\,[t_j - \mu(q)\cdot t] - k|q|,$$

with $e(q) \in [\rho, 1]$ because the zero-cost challenger prepares after every price. This step needs no price sufficiency.

The weak economy. Every residual satisfies $|t_{j,0} - \mu\cdot t_0| \le \max_l t_{l,0} - \min_l t_{l,0} < k$, so every nonzero order loses, and zero is the unique order of every class. Then $f_j = f_l$ for all j, l, every posterior vector equals the prior vector π, and gross profit is $\sum_j \pi_j g_{j,0} < c_H$, so entry is ρ. The tuple has zero orders, the prior posterior at every order and price, the single price $t_{0,0} + \rho\,(\pi\cdot t_0 - t_{0,0})$, $\zeta \equiv 0$, and truthful bids. Each condition holds by these computations.

The strong economy, orders. Under every candidate, a high-group class j earns from a purchase at least $\rho\,[\min_{l\in\mathcal{H}} t_{l,1} - \max_{\nu\in\mathcal{M}_\pi} \nu\cdot t_1] - k > 0$, and from a sale at most $-k$ minus a nonnegative term, since $t_{j,1} - \mu\cdot t_1 > 0$. A low-group class earns from a sale at least $\rho\,[\min_{\nu\in\mathcal{M}_\pi} \nu\cdot t_1 - \max_{l\in\mathcal{L}} t_{l,1}] - k > 0$, and from a purchase at most $-k$. Zero pays zero. So every equilibrium has purchases in $\mathcal{H}$ and sales in $\mathcal{L}$.

The strong economy, the tuple. With these orders, the group posterior z of $\mathcal{H}$ follows the two-class formulas with prior $\bar{\pi}$: $z = (m(\bar{\pi}), \bar{\pi}, M(\bar{\pi}))$ after $(-1, 0, 1)$. Within each group every class has the same order law, so the class proportions stay at the prior proportions: $\mu_j(z) = z\pi_j/\bar{\pi}$ for $j \in \mathcal{H}$ and $\mu_j(z) = (1-z)\pi_j/(1-\bar{\pi})$ for $j \in \mathcal{L}$. Gross profit at z is $\bar{g}_{\mathcal{L}} + z(\bar{g}_{\mathcal{H}} - \bar{g}_{\mathcal{L}})$, increasing in z. At $z = \bar{\pi}$ it is $\sum_j \pi_j g_{j,1} \le \sum_j \pi_j g_{j,0} < c_H$ by Proposition 1(a); at $z = M(\bar{\pi})$ it exceeds c_H by hypothesis. So only a purchase recruits, and entry is $(\rho, \rho, 1)$. The prices are $P = t_{0,1} + e(z)[\bar{t}_{\mathcal{L}} - t_{0,1} + (\bar{t}_{\mathcal{H}} - \bar{t}_{\mathcal{L}})z]$ with group means $\bar{t}$. Separation gives $\bar{t}_{\mathcal{H}} > \max_\nu \nu\cdot t_1 \ge \min_\nu \nu\cdot t_1 > \bar{t}_{\mathcal{L}}$, so with e nondecreasing

in z and $e \geq \rho$, the three prices differ, each price reveals z and the whole vector, and $\bar{\mu}(P) = \mu(z)$. Each condition then holds by direct computation: (E1) and (E2) by the formulas above, (E3) by the strict profit comparisons, (E4) by the price formula, (E5) by the strict payoff bounds, (E6) by truthful bids. A purchase has probability $b + \bar{\pi}\lambda$, so entry is $\rho + (1-\rho)(b + \bar{\pi}\lambda) > \rho$. Every comparison is strict, so both challenger rules give this outcome. □

The four-class economy. The economy has the benchmark slot, costs, reserve and first two strengths, and four point-mass classes with prior $1/4$ each. The low group has the values 1 and $1001/1000$, and the high group has the values $9999/1000$ and 10 (`P2_auction/derive.py`). For posterior vectors of the form $\nu_j = \pi_j w_j / \sum_l \pi_l w_l$ with $1 \leq w_j \leq 1 + u$, the mean $\nu \cdot t$ is linear-fractional in each weight, so its extremes over $\mathcal{M}_\pi$ occur at the 16 vertices of the weight box. The script evaluates all 16 and certifies the separation distance $\gamma = 0.178871$, the trade margin $\rho\gamma - k = 0.024718$, the weak range margin 0.003333, the weak cost margin 1.195875, the group profit gap 8.332667 and the upper profit margin 0.216834.

A price collision at three classes. Without separation, the price can pool orders with different posterior vectors. Take the slot of calibration B, $R \sim U[0,8]$, $p = 1$, $\rho = 1/4$, $c_H = 10$, $k = 1/8$, priors $(1/4, 1/2, 1/4)$, and the classes $G_1 = \delta_1$, $G_2 = (1-w)\delta_1 + w\delta_{10}$, $G_3 = \delta_{10}$ with $w = (1 + 2b_0 + \lambda)/2 = 0.940709$. Since $\beta > 1/4$, $1 - w = (1-\lambda)(4\beta - 1)/2 > 0$, so the classes are strictly ordered. The auction gives $t_0 = 7/8$, $(t_1, t_2, t_3) = (1,\ 1 + 49w/16,\ 65/16)$ and $(g_1, g_2, g_3) = (0,\ 95w/16,\ 95/16)$. The tuple is $\sigma_j = \delta_{j-2}$ (class 1 sells, class 2 abstains, class 3 buys), Bayes posteriors at each order, entry ρ at every price, prices $P(q) = t_0 + \rho(\mu(q) \cdot t - t_0)$, the pooled belief at the common price of the orders 0 and 1, and truthful bids. Every class profit is at most $95/16 < c_H$, and class 3 has positive probability at every price, so the high cost never prepares and the zero cost prepares strictly. Normalize proceeds to $(0, w, 1)$. The conditional means at the orders 0 and 1 are

$$z_0 = \frac{b_0(w/2 + 1/4) + \lambda w/2}{b_0 + \lambda/2}, \qquad z_1 = \frac{b(w/2 + 1/4) + \lambda/4}{b + \lambda/4},$$

and $z_0 = z_1$ exactly when $w = (2b + 3b_0 + 2\lambda)/[2(2b + b_0 + \lambda)]$, which equals the stated w because $2b + b_0 + \lambda = 1$. The posterior vectors differ: $(0.154664, 0.690671, 0.154664)$ after the order 0 and $(0.174878, 0.349755, 0.475367)$ after the order 1, with the common price 1.522107. The sale has the distinct price 1.292045. Against the fixed tuple, $U_j(q) = q\rho[t_j - \mu(q) \cdot t] - k|q|$, and each class's chosen order beats its best alternative by 0.260795, 0.020627 and 0.024768. The collision is a knife-edge in w: a change of $1/1000$ in either direction separates the two prices and keeps every choice strict.

Complementary private signals

A buyer may know its integration plans, while investors who follow the target know its customers or its product market. Two conditionally independent binary signals represent the two sources. The investor observes $T \in \{+, -\}$ with accuracy a_T, and the challenger observes $Y \in \{+, -\}$ with accuracy d:

$$\Pr(T = + \mid H) = \Pr(T = - \mid L) = a_T, \qquad \Pr(Y = + \mid H) = \Pr(Y = - \mid L) = d,$$

with $1/2 < a_T, d < 1$. The investor orders on its signal, and the challenger sees the price and its signal before it decides. Write $m = m(1/2)$, $M = M(1/2)$, $\mu_- = (1 - a_T) + (2a_T - 1)m$ and $\mu_+ = (1-a_T)+(2a_T-1)M$. A challenger with signal + holds the belief $\phi_+(\mu) = d\mu/[d\mu+(1-d)(1-\mu)]$. Write $D^+(r) = \Delta_T(r) + (1 - \rho)(2d - 1)[t_H(r) - t_0(r)]$ for the largest price response to the public posterior.

Throughout, the values are the point masses h and ℓ, the prior is $1/2$, and T and Y are independent given the class and independent of the slot, the cost and R. The investor's order distributions are σ_+ and σ_- after its two signals. Write $f_t(x) = (1 - \lambda)\eta(x) + \lambda\sigma_t(x)$ for the probability of the order x given $T = t$, and $\nu(x) = f_+(x)/[f_+(x) + f_-(x)]$ for the posterior of $T = +$; the two signals have prior $1/2$.

Beliefs, entry and prices. The order depends on the class only through T, so $\mu(x) = a_T\nu(x) + (1 - a_T)(1 - \nu(x)) = (1 - a_T) + (2a_T - 1)\nu(x)$. Lemma A.1(b)–(c), applied to T in place of the class, gives $m \le \nu(x) \le M$, so $\mu_- \le \mu(x) \le \mu_+$; the same holds for every price belief. The signal Y is independent of the order given the class, so a challenger with $Y = y$ at the price $\bar{P}$ holds the belief $\phi_y(\bar{\mu}(\bar{P}))$, with $\phi_-(\mu) = (1 - d)\mu/[(1 - d)\mu + d(1 - \mu)] < \phi_+(\mu)$. Let $I_y(\mu) \in [0, 1]$ be the high-cost preparation probability after $Y = y$ at the public belief μ. Since B rises in the belief, $I_-(\mu) > 0$ implies $B(\phi_+(\mu)) > B(\phi_-(\mu)) \ge c_H$, so $I_+ \ge I_-$ under both challenger rules. Class entry at μ is

$$e_H(\mu) = \rho + (1 - \rho)[dI_+ + (1 - d)I_-], \qquad e_L(\mu) = \rho + (1 - \rho)[(1 - d)I_+ + dI_-],$$

so $e_H \ge e_L \ge \rho$. Competitive prices give $P(x) = t_0 + e_L w_L + \mu(x)\,\Delta_P$, with $\Delta_P = e_H w_H - e_L w_L = e_L\Delta_T + (e_H - e_L)w_H$. Hence

$$\rho\Delta_T \le \Delta_P \le \Delta_T + (1 - \rho)(2d - 1)w_H = D^+. \tag{A.9}$$

If $P(x) = P(x')$, class entry at the two orders is the same, because entry depends on the price, the signal and the cost; then $\Delta_P \ge \rho\Delta_T > 0$ gives $\mu(x) = \mu(x')$, and $\bar{\mu}(P(x)) = \mu(x)$. The price reveals

the public posterior under every profile.

Order payoffs. After $T = t$, the class probabilities are a_T and $1 - a_T$ or the reverse, and entry given the class and the order does not depend on T. Subtract the price:

$$U_+(1) = (2a_T - 1)[1 - \nu(1)]\,\Delta_P(\mu(1)) - k, \qquad U_-(-1) = (2a_T - 1)\,\nu(-1)\,\Delta_P(\mu(-1)) - k,$$

and the wrong-signed orders earn $-(2a_T - 1)[1 - \nu(-1)]\Delta_P - k$ and $-(2a_T - 1)\nu(1)\Delta_P - k$, both at most $-k$. Zero pays zero. With wrong-signed orders excluded, $\nu(1)$ depends only on the purchase rate after $T = +$, and $\nu(-1)$ only on the sale rate after $T = -$; the zero order can depend on both, but it pays zero. By (A.9) and the bounds on ν, every correct gross return lies in $[(2a_T - 1)m\rho\Delta_T,\ (2a_T - 1)D^+]$ under every profile.

Proposition A.7 (Complementary private signals)**.** *Let $0 < p < \ell < r_0 < r_1 < h$, $0 < \rho < 1$, and $1/2 < a_T, d < 1$, with any slot (λ, β). Suppose*

$$B_{r_0}(d) < c_H < B_{r_1}\big(\phi_+(\mu_+)\big), \qquad (2a_T - 1)\,D^+(r_0) < k < (2a_T - 1)\,\rho\,m\,\Delta_T(r_1).$$

At r_0, the unique orders are zero after both signals, and entry is ρ. At r_1, the unique orders are a purchase after $T = +$ and a sale after $T = -$, and entry strictly exceeds ρ. These order conclusions hold under every pair of tie rules. The hypotheses allow $d > a_T$.

Proof of Proposition A.7. The weak strength. Under every candidate at r_0, every correct gross return is at most $(2a_T - 1)D^+(r_0) < k$, so zero is the unique order after both signals, under both investor rules. Then $\nu \equiv \mu \equiv 1/2$, the private beliefs are d and $1 - d$, and $B_{r_0}(d) < c_H$ excludes the high cost after either signal, so entry is ρ. The tuple has zero orders, these beliefs, the price $t_0 + \rho(w_H + w_L)/2$ at every order, $\zeta(\bar{P}, y, c_H) = 0$, and truthful bids. Against it, each correct order earns $(2a_T - 1)\rho\Delta_T(r_0)/2 - k \le (2a_T - 1)D^+(r_0) - k < 0$.

The strong strength. Under every candidate at r_1, every correct gross return is at least $(2a_T - 1)m\rho\Delta_T(r_1) > k$, so the purchase after $T = +$ and the sale after $T = -$ are the unique best orders. Then $f_+ = (b, b_0, b + \lambda)$ and $f_- = (b + \lambda, b_0, b)$, so $\nu = (m, 1/2, M)$ and $\mu = (\mu_-, 1/2, \mu_+)$. Class entry follows from $I_y(\mu(x))$ at each order, and the prices from the formula above. The probabilities $I_y(\mu)$ are nondecreasing in μ, because ϕ_y and B rise, so e_H and e_L are nondecreasing in μ. For $\mu_2 > \mu_1$,

$$P(\mu_2) - P(\mu_1) = (\mu_2 - \mu_1)\Delta_P(\mu_1) + \mu_2[e_H(\mu_2) - e_H(\mu_1)]w_H + (1 - \mu_2)[e_L(\mu_2) - e_L(\mu_1)]w_L > 0,$$

since the first term is positive by (A.9) and both brackets are nonnegative. The three prices therefore differ, each price has one order in its preimage, and $\bar{\mu}(P(x)) = \mu(x)$. At the purchase price,

$B_{r_1}(\phi_+(\mu_+)) > c_H$, so a challenger with $Y = +$ prepares. The event $(X, Y) = (1, +)$ has positive probability in each class, so entry strictly exceeds ρ. Against the tuple, $U_+(1)$ and $U_-(-1)$ are at least $(2a_T - 1)m\rho\Delta_T(r_1) - k > 0$. These order conclusions use only bounds on every profile, so they hold under every pair of tie rules.

Keep the values, costs, strengths and accuracies of $S(B)$. Set $(\lambda, \beta) = (1/3, 3/10)$. Then $(a_T, d) = (7/10, 3/4)$, $M = 8/11$ and $m = 3/11$. The five margins are

$$\begin{aligned} c_H - B_{r_0}(d) &= \frac{21}{400} > 0, \\ B_{r_1}(\phi_+(\mu_+)) - c_H &= \frac{1363}{36800} > 0, \\ k - (2a_T - 1)D^+(r_0) &= \frac{2}{1375} > 0, \\ (2a_T - 1)\rho m\Delta_T(r_1) - k &= \frac{603}{31625} > 0, \\ c_H - B_{r_1}(\phi_-(\mu_+)) &= \frac{76753}{18400} > 0. \end{aligned}$$

Thus the hypotheses allow $d > a_T$. □

One pair that recruits. If also $B_{r_1}(\phi_-(\mu_+)) < c_H$, the challenger with $Y = -$ never prepares at a high cost at r_1. With $Y = +$, the zero and sale orders have public posteriors at most $1/2$, and $B_{r_1}(\phi_+(1/2)) = B_{r_1}(d) \le B_{r_0}(d) < c_H$. Only $(X, Y) = (1, +)$ recruits, every entry comparison is strict, and

$$e_H = \rho + (1 - \rho)\, d\, (b + a_T\lambda), \qquad e_L = \rho + (1 - \rho)(1 - d)\big(b + (1 - a_T)\lambda\big),$$

because a purchase has probability $b + a_T\lambda$ in class H and $b + (1 - a_T)\lambda$ in class L. Entry is $(e_H + e_L)/2$, and ownership is $e_H/2$, since $h > r_1$.

The vector $S(B)$. At $S(B)$ with $(a_T, d) = (0.70, 0.75)$, the four conditions of the proposition and the one-pair condition have the margins 0.052500000, 0.045265153, 0.001454545, 0.018594291 and 4.159475417. The formulas above give entry 0.850000 and 0.879797, and ownership 0.425000 and 0.449238. Without the challenger's signal ($d = 1/2$), the top public profit stays below c_H at both strengths, with margins 1.469972860 and 1.862796856, so entry is ρ; the bounds 0.013181818 and 0.018594291 force zero and full orders, and entry is 0.850000 at both strengths. When the investor observes the class and the challenger has no signal, the margins 0.010454545, 0.068985729, 2.301363636 and 0.666013879 give zero orders, full orders and entry 0.850000 at both strengths. On the grid of accuracies, six pairs satisfy the conditions, all with $d > a_T$; the smallest margin

among them is 0.000100000.

The benchmark B. At B the conditions cannot hold with $d \geq a_T$. Write $z = 2a_T - 1$. The strong bound needs $z > z_* = k/(m\rho\Delta_T(r_1)) = 0.446193819$. If $d \geq a_T$, then $2d - 1 \geq z$, and the weak gross bound is $zD^+(r_0) \geq z_*[\Delta_T(r_0) + (1-\rho)z_* w_H(r_0)]$, which exceeds k by 0.049029700. With an accurate investor signal and a weak challenger signal, $(a_T, d) = (0.94, 0.505)$, the five margins are 1.150916667, 0.020731442, 0.002610833, 0.019444742 and 0.048821802. Entry rises from 0.250000 to 0.386933. Without the challenger's signal, the top public profit at r_1 stays below c_H, with margin 0.013903757, the bounds 0.005333333 and 0.019444742 fix the orders, and entry is 0.250000 at both strengths. Both sources are therefore needed for the rise. Every comparison is strict, so all four pairs of tie rules give the same outcomes. At a higher investor accuracy, the public posterior alone can recruit, and a pair can satisfy the theorem without this complementarity; the margin $c_H - B_{r_1}(\mu_+)$ is the condition that separates the two.

The benchmark B with the accuracies of $S(B)$. At B with $(a_T, d) = (0.70, 0.75)$, three of the four conditions fail, and each equilibrium has a mixed order: both signal types mix at r_0, and the type with $T = -$ mixes at r_1. After wrong-signed orders are excluded, write $s = \lambda\sigma_+(1)$ and $t = \lambda\sigma_-(-1)$; the purchase return depends only on s, and the sale return only on t, with $1 - \nu(1) = b/(2b + s)$ and $\nu(-1) = b/(2b + t)$.

- *At r_0.* Under every profile $B_{r_0}(\phi_+(\mu_-)) > c_H$ and $B_{r_0}(\phi_-(\mu_+)) < c_H$, with margins 0.363883929 and 2.755550596. So $I_+ = 1$ and $I_- = 0$ at every price, $e_H = 13/16$, $e_L = 7/16$, and $\Delta_P = 311/1920$ at every price. Each correct gross return is $\frac{2}{5} \cdot \frac{b}{2b+s} \cdot \frac{311}{1920}$, strictly decreasing in its rate, above k at rate zero (margin 0.012395833) and below k at rate λ (margin 0.002574837). The unique zero-profit rates are $s = t = 119b/96 < \lambda$ (margin $u - 119/96 = 0.478698495$). Entry is 0.625000 and ownership 0.406250.

- *At r_1.* A challenger with $Y = -$ never recruits (margin 3.155195358). A purchase has public posterior at least $1/2$, and $B_{r_1}(d) > c_H$ (margin 0.375000000), so $Y = +$ recruits after a purchase, $\Delta_P = 137/192$ there, and the purchase earns at least $\frac{2}{5}m\frac{137}{192} - k = 0.056760364 > 0$: $s = \lambda$. After a sale, $Y = +$ recruits exactly when $\mu(-1) \geq 47/106$, that is $t \leq 15b/19$. On that range the sale earns at least the floor return $19/795$, which exceeds k by 0.003899371; so no equilibrium has $t \leq 15b/19$, and no tie mixture exists at the threshold. Above it, $e_H = e_L = \rho$ at the sale price. The net sale payoff $U_-(-1) = b/[15(2b + t)] - 1/50$ strictly decreases with t. Its unique zero-profit rate is $t = 4b/3 < \lambda$ (margin 0.384948495), with $\mu(-1) = 21/50$, below $47/106$ by 0.023396226. After the zero order $Y = +$ recruits (margin 0.219556089). Only a sale blocks recruitment after $Y = +$; a sale has probability $b + (1 - a_T)t$

in class H and $b + a_T t$ in class L. Hence entry is $5/8 - 23b/40 = 0.512521$ and ownership $13/32 - 63b/160 = 0.329227$.

At r_0, the investor mixes between trade and abstention after either signal, and it trades with probability 0.721409. At r_1, it buys with probability one after a favorable signal, and after an unfavorable signal it sells with probability 0.775969. Each mixed tuple is unique under rule IM with either challenger rule, by the strict monotonicity of each return and the endpoint signs. Under rule I0 neither strength has an equilibrium, because the unique admissible rates are interior. Without the challenger's signal, entry is ρ under every profile at both strengths, zero orders are forced at r_0, and at r_1 the unique rates are $s = t = 4b/3$; this control also needs rule IM, since the gross return at zero, 0.033333333, exceeds k and the return at full orders, 0.017929428, falls short of it. The rational constants above are exact; the script `P4_information/derive.py` prints them with every tuple, and `XP4/xp4_check.py` enumerates every rate on each order side and finds no other equilibrium.

A.7 Welfare and sale terms

This subsection states and proves Propositions A.8, A.9 and A.10, and it proves the statements of Section 7 that rest on them.

Access to prices

Hold the incumbent at F_1. Compare the equilibrium of Proposition 2 with the equilibrium of an economy in which the challenger does not see the price. All other primitives stay fixed, and market makers set competitive prices in both economies. Write $\mathcal{R}$ for expected target proceeds and $\mathcal{W}$ for expected acquisition surplus net of preparation costs. The superscript 1 marks price access, and 0 marks the hidden price. Acquisition surplus excludes transfers between bidders, shareholders and traders.

Proposition A.8 (Access to prices)**.** *Let the hypotheses of Proposition 2 hold. At F_1, both economies have unique full orders. Price access raises expected target proceeds and acquisition surplus net of preparation costs:*

$$\mathcal{R}^1 - \mathcal{R}^0 = (1-\rho)\big\{\pi(b+\lambda)(t_H - t_0) + (1-\pi)\, b\,(t_L - t_0)\big\} > 0,$$
$$\mathcal{W}^1 - \mathcal{W}^0 = (1-\rho)(b+\pi\lambda)\big[B_1(M) - c_H + pF_1(p)\big] > 0.$$

Proof of Proposition A.8. The two tuples. The access tuple is $\mathcal{E}_1$ of the proof of Proposition 2. The hidden tuple has full orders, the posteriors (m, π, M) at the orders, the challenger belief π at every price because the challenger sees neither the price nor the order, $\zeta \equiv 0$, entry ρ after every order, the prices $P^0(x) = t_0 + \rho\, w(\mu(x))$, and truthful bids. Here and below every object is at F_1.

Verification of the hidden tuple. (E1): the full-order likelihoods. (E2): the challenger's belief is its prior. (E3): $B_1(\pi) \le B_0(\pi) < c_H$, strictly. (E4): (A.4) with $e \equiv \rho$. (E5): with $e \equiv \rho$, the purchase earns $\rho\Delta_1(1-M) - k \ge \rho\kappa\Delta_1 - k > 0$ and the sale $\rho\Delta_1 m - k > 0$, and the wrong-signed orders earn at most $-k$. (E6): truthful bids. These bounds hold for every candidate of the hidden economy, so its orders are unique, and the tuple is unique. The access economy is unique by Proposition 2.

The proceeds gain. Couple the two economies on the class, the value, the cost, the slot identity and the liquidity order. The orders coincide at every realization, so the trading costs coincide. Access adds high-cost preparation exactly after the order 1, which has probability $b + \lambda$ in class H and b in class L. With entry, expected proceeds rise from t_0 to t_j. Hence

$$\mathcal{R}^1 - \mathcal{R}^0 = (1-\rho)\{\pi(b+\lambda)w_H + (1-\pi)\, b\, w_L\}.$$

Each term is nonnegative, and $w_H = w_L + \Delta_1 \ge \Delta_1 > 0$, so the gain is positive.

The surplus gain. Let $\mathcal{V}_0(R) = R\mathbf{1}\{R \ge p\}$ be the allocation value without the challenger. With truthful bids, the target goes to the highest admissible value, so with an admissible challenger of value θ the allocation value is $\max\{\mathcal{V}_0(R), \theta\}$, and

$$\max\{\mathcal{V}_0(R), \theta\} - \mathcal{V}_0(R) = (\theta - \max\{p, R\})_+ + p\mathbf{1}\{R < p\}. \tag{A.10}$$

For $R < p$, both sides equal θ, since $\theta \ge p$; for $R \ge p$, both sides equal $(\theta - R)_+$. Average over R and over the class at the posterior M of the order 1: one added preparation adds $B_1(M) + pF_1(p)$ to allocation value and costs c_H. Added preparations have probability $(1-\rho)\Pr(X = 1) = (1-\rho)(b+\pi\lambda)$. Hence $\mathcal{W}^1 - \mathcal{W}^0 = (1-\rho)(b+\pi\lambda)[B_1(M) - c_H + pF_1(p)] > 0$, because $B_1(M) > c_H$. Transfers between bidders, shareholders and traders cancel from this sum. □

The benchmark, the transfer and the trade scale. At r_1, proceeds are 0.872531 with access and 0.614583 without, a gain of 0.257948; net surplus is 2.634044 and 2.552083, a gain of 0.081960. If the execution cost is a resource cost, both economies lose λk, and the difference is unchanged. In the access economy the investor arrives with probability λ; class H buys with entry one after a purchase, and class L sells with entry ρ after a sale. By the payoffs of Lemma A.3 before the cost k, its expected gross profit is

$$G = \lambda\Delta_1\{\pi(1-M) + (1-\pi)\rho m\} = 0.037665.$$

Competitive prices give $\mathbb{E}[V_T - P(X) \mid X] = 0$, so $\mathbb{E}[X(V_T - P(X))] = 0$. The informed and the liquidity orders partition this expectation, so liquidity traders lose G in expected gross terms. Let

one order unit be z shares per share of float, with the same signal experiment. The economy without an informed investor has the posterior π after every order and entry ρ, so its proceeds equal $\mathcal{R}^0$. If initial shareholders bear the whole transfer, their total gain over that economy is $(\mathcal{R}^1 - \mathcal{R}^0) - zG$, positive exactly for $z < (\mathcal{R}^1 - \mathcal{R}^0)/G = 6.848394$. A deterministic external dividend raises V_T and every price by its amount. It leaves $V_T - P$, every order payoff, the challenger's information and entry unchanged, and it enters neither proceeds of the sale nor allocation surplus. The values come from `P2_auction/derive.py` (run log `P2_auction/run.log`); with research at the cap, Appendix A.5 gives the same gains.

Access outside the recruitment region. Suppose instead $B_1(m) < c_H < B_1(\pi)$ and $k < \rho\kappa\Delta_1$. Without the price, the challenger's belief is π, so every challenger prepares, and entry is one after every order. With the price, the posteriors are (m, π, M), and entry is $(\rho, 1, 1)$. In both economies the bound on every profile forces full orders, so the orders, the posteriors and the trading costs coincide, and each tuple is verified as above. Access removes high-cost preparation exactly after a sale, which has probability b in class H and $b + \lambda$ in class L. Proceeds change by $-(1-\rho)\{\pi b\, w_H + (1-\pi)(b+\lambda)\, w_L\} < 0$, and surplus by $-(1-\rho)[b+(1-\pi)\lambda]\ [B_1(m) - c_H + pF_1(p)]$. Keep the strong benchmark and set $c_H = 12/5$. Then $B_1(m) - c_H = -0.033821$ and $B_1(\pi) - c_H = 1.891667$, and access changes proceeds by -0.173917 and surplus by -0.013505.

The payment rule

Consider a verifiable-value institution with zero reserve. After diligence, values are verifiable, and a sale to the next buyer at its value is an enforceable fallback. The seller and the winner divide the surplus above the fallback by Nash bargaining, with seller weight $\omega \in [0, 1)$. Without a challenger, the seller receives ωR.

Proposition A.9 (The payment rule)**.** *Let the challenger's values be the point masses ℓ and h, and let the incumbent distribution have support $[0, \bar{r}]$ with $0 < \ell < \bar{r} < h$. The transfer, the challenger's expected profit and the spread are*

$$\begin{aligned} T_\omega(R, \theta) &= (1 - \omega)\min\{R, \theta\} + \omega \max\{R, \theta\}, \\ g_{\theta,\omega} &= (1 - \omega)\,\mathbb{E}[(\theta - R)_+], \\ \Delta_\omega &= \omega(h - \ell) + (1 - 2\omega)\,\mathbb{E}[(R - \ell)_+]. \end{aligned}$$

A stronger incumbent weakly lowers the challenger's profit. Its effect on the spread weakly has the sign of $1 - 2\omega$. The effect is strict when $\mathbb{E}[(R - \ell)_+]$ changes strictly and $\omega \neq 1/2$. At $\omega = 1/2$, the spread equals $(h - \ell)/2$ at every incumbent distribution.

Proof of Proposition A.9. The continuation. After preparation, the buyer with the largest value acquires the target, and the next buyer's value z is the enforceable fallback; without a challenger, the fallback is zero. Let the winner's value be V. For $V > z$ and $0 < \omega < 1$, Nash bargaining maximizes $(P - z)^{\omega}(V - P)^{1-\omega}$ over $z \le P \le V$. The logarithm has derivative $\omega/(P - z) - (1 - \omega)/(V - P)$ and a strictly negative second derivative, so the unique optimum solves $\omega(V - P) = (1 - \omega)(P - z)$, that is $P = (1 - \omega)z + \omega V$. At $V = z$, feasibility fixes $P = z$, and at $\omega = 0$ the next buyer accepts the fallback, so $P = z$. Hence the transfer is $T_{\omega}(R, \theta) = (1 - \omega)\min\{R, \theta\} + \omega \max\{R, \theta\}$.

Profit and spread. A challenger wins when $\theta > R$ and keeps $\theta - T_{\omega} = (1 - \omega)(\theta - R)$, so $g_{\theta,\omega} = (1 - \omega)\mathbb{E}[(\theta - R)_+]$. For $R \le \ell$, both classes win, and $T_{\omega}(R, h) - T_{\omega}(R, \ell) = \omega(h - \ell)$. For $\ell < R < h$, the high class wins and the low class loses, and $T_{\omega}(R, h) - T_{\omega}(R, \ell) = (1 - \omega)R + \omega h - (1 - \omega)\ell - \omega R = \omega(h - \ell) + (1 - 2\omega)(R - \ell)$. The support lies below h, so the average over the two cases gives $\Delta_{\omega} = \omega(h - \ell) + (1 - 2\omega)\mathbb{E}[(R - \ell)_+]$.

Signs. The function $(\theta - R)_+$ is nonincreasing in R, and $(R - \ell)_+$ is nondecreasing. A first-order stochastic strengthening of the incumbent therefore weakly lowers $g_{\theta,\omega}$ and weakly moves $\mathbb{E}[(R-\ell)_+]$ up, so the spread moves with the sign of $1 - 2\omega$, strictly when $\mathbb{E}[(R - \ell)_+]$ rises strictly and $\omega \neq 1/2$. At $\omega = 1/2$, the coefficient $1 - 2\omega$ is zero, and $\Delta_{\omega} = (h - \ell)/2$. □

For $R \sim U[0, r]$, $\mathbb{E}[(R - \ell)_+] = (r - \ell)^2/(2r) = \Delta_T(r)$, so $\Delta_{\omega}(r) = \omega(h - \ell) + (1 - 2\omega)\Delta_T(r)$. In the benchmark, $\Delta_{\omega}(r_1) - \Delta_{\omega}(r_0) = (1 - 2\omega)[\Delta_T(r_1) - \Delta_T(r_0)] = 13/20\,(1 - 2\omega)$, which is zero exactly at $\omega = 1/2$. This result specifies the acquisition stage after every history; the trading and preparation equilibrium under this rule is outside the proposition.

Reserves

At a general reserve, a class value below p gives no admissible bid. For $R \sim U[0, r]$ and a point value v, integration over $R < p$, $p \le R < v$ and $R \ge v$ gives the kernel

$$(t_v, g_v, W_v) = \begin{cases} (t_0, 0, 0), & v < p, \\ (p,\ v - p,\ 1), & r \le p \le v, \\ \big(v - (v^2 - p^2)/(2r),\ (v^2 - p^2)/(2r),\ v/r\big), & p \le v \le r, \\ \big(r/2 + p^2/(2r),\ v - r/2 - p^2/(2r),\ 1\big), & p \le r \le v, \end{cases} \tag{A.11}$$

with $t_0 = p(1 - p/r)_+$. The pieces agree at $v = r$ and at $p = r$. A class law averages the kernel. Both t_v and g_v are nondecreasing in v, and $t_v \ge t_0$; under the class order, $\Delta_T \ge 0$, $w_L \ge 0$ and $g_H \ge g_L \ge 0$. For $\ell < p < r < h$ with point values, $t_L = t_0 = p - p^2/r$, $g_L = 0$, $t_H = r/2 + p^2/(2r)$, $g_H = h - t_H$ and $\Delta_T = r/2 - p + 3p^2/(2r)$. For $r < p < h$, $t_0 = t_L = g_L = 0$, $t_H = p$, $g_H = h - p$ and $\Delta_T = p$.

Proposition A.10 (Reserves)**.** *Fix a reserve, and allow class values below it. Keep the class order, the costs, the slot and a strengthening $F_1 \le F_0$, and compute each class object with the reserve. Suppose $k < \rho\kappa \min\{\Delta_0, \Delta_1\}$.*

(a) If $B_0(\pi) < c_H < B_1(M)$, entry is the same at the two strengths: $\mathsf{E}_0 = \mathsf{E}_1 = \rho + (1-\rho)(b + \pi\lambda)$.

(b) If $B_0(\pi) < c_H < B_0(M)$ and $B_1(M) < c_H$, entry falls strictly: $\mathsf{E}_0 = \rho + (1-\rho)(b + \pi\lambda) > \rho = \mathsf{E}_1$.

Each economy has a unique outcome under every pair of tie rules.

Proof of Proposition A.10. Orders. The zero-cost challenger prepares after every price, so entry is at least ρ. Lemma A.3 uses only (E1), (E4), (E6), $e \in [\rho, 1]$ and $\Delta_T \ge 0$, so it holds with the reserve kernel. Every correct order earns at least $\rho\kappa\Delta_i - k > 0$ under every candidate, and every wrong-signed order at most $-k$. Full orders are forced at both strengths, and the posteriors are (m, π, M).

Entry. Gross profit falls with incumbent strength at every belief, because $(\theta - \max\{p, R\})_+$ is nonincreasing in R. In part (a), $B_1(\pi) \le B_0(\pi) < c_H < B_1(M) \le B_0(M)$, so at both strengths only the order 1 recruits, and $B_i(m) \le B_i(\pi)$ keeps the sale below the cost. In part (b), the weak strength has the same pattern, and at the strong strength $B_1(\mu) \le B_1(M) < c_H$ at every posterior. Every realized comparison is strict.

The tuples. At each strength the tuple has full orders, posteriors (m, π, M), entry $(\rho, \rho, 1)$ or (ρ, ρ, ρ) as above, $\zeta(P(x)) = [e(x) - \rho]/(1-\rho)$, prices $P_i(x) = t_{0,i} + e_i(x)\, w_i(\mu(x))$, $\bar{\mu}(P_i(x)) = \mu(x)$, and truthful bids. Since $\Delta_i > 0$ and entry is nondecreasing in the posterior, the three prices differ, so (E2) holds. (E1), (E3) and (E4) hold by the computations above, (E5) by the strict bounds, and (E6) by truthful bids. The bounds hold for every candidate, so each tuple is unique, under every pair of tie rules. A purchase has probability $b + \pi\lambda$, which gives both entry formulas. □

The benchmark reserves. The node margins below come from `P2_auction/derive.py` in outward interval arithmetic; at each node, $T = \rho\kappa\Delta_T - k$, $C = c_H - B(1/2)$ and $J = B(M) - c_H$.

- *Exclusion.* At the reserve $101/100$ with point values, the low class lies below the reserve and the high class above both incumbent supports, so the formulas above apply. The margins are $(T, C, J) = (0.038167, 1.512521, 0.561220)$ at r_0 and $(0.047239, 1.835008, 0.089706)$ at r_1. Part (a) applies, and entry is 0.522757 at both strengths. Proceeds exceed their levels at the reserve $1/2$ by 0.047856 and 0.121853. For the narrow bands at $11/10$, the margins are $(0.048076, 1.552083, 0.503375)$ and $(0.047572, 1.850833, 0.066568)$, with proceeds gains 0.027458 and 0.150186.

- *Deterrence.* At the reserve $31/20$, the reserve exceeds r_0, so the weak incumbent never meets it, and at r_0 the objects are $t_0 = t_L = g_L = 0$, $t_H = p$ and $g_H = h - p$. The margins are $(0.084215, 1.775000, 0.177445)$ at r_0 and $(0.057405, 1.950208, -0.078730)$ at r_1. Part (b) applies: entry is 0.522757 at r_0 and 0.250000 at r_1. Proceeds exceed their levels at the reserve $1/2$ by 0.110114 and 0.020542. The point values and the narrow bands give the same objects here, because the reserve lies above the low band and below the high band. The strong upper profit equals c_H at $p_H = \sqrt{2r_1(h - c_H/M) - r_1^2} \approx 1.325270$, and the weak upper profit equals c_H at $h - c_H/M \approx 1.792723$; the reserve lies inside this interval, at distances 0.224730 and 0.242723 from its ends, and $p_H - r_0 = 0.125270$.

- *Why deterrence excludes the weak incumbent.* For point values and $1 < p \le r_0$, the formulas above give $\Delta_T(r_0) - 17/20 = (p-1)(5p+1)/4 \ge 0$ and $\Delta_T(r_1) - 1 = (p-1)^2/2 \ge 0$, so the trade margins stay at least 0.037150 and 0.047235. The strong upper profit is $M(h - r_1/2 - p^2/(2r_1))$, which at $p = r_0$ exceeds c_H by 0.038544 and is larger at every lower reserve. The weak upper profit is larger still, and both prior profits lie below c_H. So both strengths recruit after a purchase only, and entry is flat. For the narrow bands with $21/20 < p \le r_0$, the low band lies below p and the high band above r_1, so every band object equals the point-value object, and the same conclusion holds. Strict deterrence after full exclusion of the low class therefore needs $p > r_0$.

- *Boundary reserves.* At $p = p_H$, the purchase leaves the high-cost challenger indifferent at r_1: rule C1 gives entry $(\rho, \rho, 1)$, and rule CM allows any last component in $[\rho, 1]$; the trade margin there is 0.050792. At $p = h$ with point values, every acquisition profit is zero, the definition fixes zero-cost preparation, $\Delta_T = h$, the bound $h\rho\kappa > k$ forces full orders, and entry is 0.250000 with proceeds 1.250000. These are boundary events outside the open sets of the proposition.

The independent recomputation `XP2/xp2_check.py` (run log `XP2/run.log`) derives the orders and entry patterns at every node from the bounds on every profile and agrees with every value. A selection of the equilibrium at every reserve, and the seller's optimal reserve, are outside these results.

A.8 Empirical implications

This subsection states and proves the results that Section 8 summarizes. The auction is the benchmark, the prior is $1/2$, and the slot lies in the band of Proposition A.1(a). The price increment of entry $A_r(\mu) = t_L - t_0 + \Delta_T(r)\mu$ is the function $w(\mu)$ of Appendix A.2 at the strength r. It is positive, rises in μ with slope Δ_T, and rises in r, because $t_L - t_0 = \frac{1}{2} - 1/(8r)$ and Δ_T rise in

r at the benchmark. The indicator $\chi_r = \mathbf{1}\{B_r(M) \geq c_H\}$ records whether a purchase recruits, and the volume $v = \Pr(X \neq 0)$ is the frequency of nonzero orders per order opportunity. Write $\bar{f} = b + \lambda/2$, $\vartheta = u/(2+u) = 2M - 1 = M - m$, and $e_+ = \rho + (1-\rho)\chi_r$. At the benchmark, $B_r(1/2) = 5 - r/4 + 1/(8r) < c_H$ for $1 < r < h$. The top profit $B_r(M)$ falls strictly in r for $r > \ell$, equals $9M + 3/8$ at $r = \ell$, and lies below c_H at $r = h$; so for $M > 5/8$ the ceiling r_C is the unique strength at which $B_r(M) = c_H$, and $\chi_r = 1$ exactly when $r \leq r_C$. In the band, $M - 5/8 = 0.106059$ at the benchmark slot.

Full two-sided trade on (r_S, h). The strength r_S solves $\rho\Delta_T(r)m = k$ exactly: $r_S = \mathfrak{r}(D_S)$ with $D_S = k/(\rho m) = 0.297463$. For $r_S < r < h$, every candidate satisfies

$$K_H \geq \rho\Delta_T(r)(1-M) = \rho\Delta_T(r)m > k, \qquad K_L \geq \rho\Delta_T(r)m > k$$

by (A.6), since $1 - M = m$ at the prior $1/2$ and Δ_T rises. So full orders are the unique equilibrium orders, under both investor rules. The tuple $\mathcal{F}$ has the posteriors $(m, 1/2, M)$, entry (ρ, ρ, e_+), the prices $P(-1) = t_0 + \rho A_r(m) < P(0) = t_0 + \rho A_r(1/2) < P(1) = t_0 + e_+ A_r(M)$, $\bar{\mu}(P(x)) = \mu(x)$, $\zeta(P(1)) = \chi_r$, and truthful bids. Each condition holds by direct computation. (E1): the likelihoods $f_H = (b, b_0, b+\lambda)$ and $f_L = (b+\lambda, b_0, b)$ give $(m, 1/2, M)$. (E2): entry is nondecreasing in the posterior, so the three prices differ, and each price reveals its order. (E3): $B_r(m) < B_r(1/2) < c_H$, and the purchase recruits exactly when $\chi_r = 1$. (E4): (A.4). (E5): the strict bounds above. (E6): truthful bids. At r_C rule C1 gives $\chi_{r_C} = 1$. The interval contains r_1, r_C and r_2, because $r_S < r_1 < r_C < r_2 < h$ by Lemma A.5; at the benchmark $r_C - r_S = 1.468510$. The bound is exact: $r_S \approx 2.124148$, and the six-place display lies below r_S. At $r = 2.1241481$, inside the rounded interval, $\rho\Delta_T m - k = -8.59 \times 10^{-10}$, so the low class strictly prefers abstention to the full sale there; 2.124149 is a sufficient rounded endpoint. With a hidden price, entry is ρ after every order under every profile, because the belief stays at $1/2$ and $B_r(1/2) < c_H$; the same bounds with $e \equiv \rho$ force full orders, and the tuple is $\mathcal{F}$ with χ_r replaced by 0.

Under full orders, class H places the order 1 with probability $b + \lambda$ and class L with probability b, so $\Pr(X = 1) = \Pr(X = -1) = \bar{f}$, $\Pr(X = 0) = 1 - 2\bar{f}$, $v = 2b + \lambda$ and $\mathbb{E}[X] = 0$. The informed share of nonzero orders is $\lambda/(2b+\lambda) = u/(2+u) = \vartheta$. Only the order 1 can recruit, so entry is $\rho + (1-\rho)\chi_r\bar{f}$; from r_0, where the orders are zero and entry is ρ, to r_1 it rises by $(1-\rho)\bar{f} = 0.272757$.

The spread $S = P(1) - P(-1)$ is the difference between the conditional execution prices of a unit purchase and a unit sale. The purchase impact is $I_+ = P(1) - P(0)$, and the sale impact is $I_- = P(0) - P(-1)$. The model has no separate quote stage: each price is the competitive value conditional on the order.

Proposition A.11 (The spread and its entry component)**.** *Let $r_S < r < h$.*

(i) $S = \rho\Delta_T\vartheta + (1-\rho)\chi_r A_r(M)$*: a class component and an entry component.*

(ii) $I_- = \rho\Delta_T\vartheta/2$ *and* $I_+ = I_- + (1-\rho)\chi_r A_r(M)$*. Hence* $I_+ > I_-$ *if and only if a purchase recruits.*

(iii) If the challenger does not see the price, full two-sided trade is still the unique outcome, and S equals the class component $\rho\Delta_T\vartheta$*.*

(iv) On $(r_S, r_C]$*, both components, both impacts and S rise strictly with r. On* (r_C, h)*, the entry component is zero, and the class component, both impacts and S rise strictly.*

(v) At r_C*, S falls by* $(1-\rho)A_{r_C}(M) > 0$*. The prices* $P(-1)$ *and* $P(0)$ *and the sale impact are continuous there;* $P(1)$ *and the purchase impact fall by this amount.*

Proof of Proposition A.11. Let $r_S < r < h$, so the outcome is $\mathcal{F}$.

(i) Subtract the prices: $S = \rho[A_r(M) - A_r(m)] + (1-\rho)\chi_r A_r(M)$. Since A_r has slope Δ_T, $A_r(M) - A_r(m) = \Delta_T(M-m) = \Delta_T\vartheta$.

(ii) Also $A_r(1/2) - A_r(m) = \Delta_T(1/2 - m) = \Delta_T\vartheta/2$ and $A_r(M) - A_r(1/2) = \Delta_T\vartheta/2$, so $I_- = \rho\Delta_T\vartheta/2$ and $I_+ = \rho\Delta_T\vartheta/2 + (1-\rho)\chi_r A_r(M)$. Since $A_r(M) > 0$, $I_+ > I_-$ exactly when $\chi_r = 1$.

(iii) With a hidden price the outcome is $\mathcal{F}$ with entry ρ at every order, so part (i) holds with χ_r replaced by 0.

(iv) On $(r_S, r_C]$, $\chi_r = 1$; Δ_T and $A_r(M)$ rise strictly in r, and ρ, ϑ and $1-\rho$ do not depend on r. So both components, both impacts and S rise strictly. On (r_C, h), $\chi_r = 0$: the entry component is identically zero, and the class component $\rho\Delta_T\vartheta$, the impacts $I_- = I_+ = \rho\Delta_T\vartheta/2$ and $S = \rho\Delta_T\vartheta$ rise strictly.

(v) Since $r_S < r_C$, both sides of r_C lie in $\mathcal{F}$. Rule C1 gives $\chi_{r_C} = 1$, and $\chi_r = 0$ for $r > r_C$. Every other term is continuous in r. So $P(-1)$, $P(0)$ and I_- are continuous at r_C, and $P(1)$, I_+ and S fall by $(1-\rho)A_{r_C}(M) > 0$. □

In the benchmark at r_1 the class component is 0.077020 and the entry component 0.709279, so $S = 0.786299$; at r_C the spread falls from 0.969913 to 0.108078. Under rule CM, entry after a purchase at r_C can take any value in $[\rho, 1]$, and $S(r_C)$ can take any value between its two one-sided limits. The spread is the difference between conditional execution prices; the model has no separate quote stage.

Proposition A.12 (Identification of the informed share)**.** *Let $r_S < r < h$, and write $v = 2b + \lambda$.*

(a) Counts. *The order law depends on (λ, β) only through v, and entry $\rho + (1-\rho)\chi_r v/2$ depends on them only through v and χ_r. For a ratio $u' > 0$, a slot (λ', β') with this ratio and the volume v exists if and only if $3v \le 2 + u'(1-v)$. Equivalently, its informed share $\vartheta' = u'/(2+u')$ satisfies $\vartheta' \ge 3 - 2/v$. This slot is then unique.*

(b) Incentives. *A slot with informed share ϑ' produces these counts through full two-sided trade at r only if $\vartheta' \le \bar{\vartheta}(r) = 1 - 2k/(\rho\Delta_T(r))$.*

(c) Band. *The slots with strict recruitment at r_1 and strict exclusion at r_2 have informed shares exactly in $(2\tau(r_1) - 1,\ 2\tau(r_2) - 1)$. At the benchmark volume, each share in this band belongs to a slot with the same counts and the same entry at r_1 and at r_2. Full two-sided trade is the unique outcome of this slot at both strengths.*

(d) Prices. *Given the entry floor $\rho = \Pr(\text{preparation} \mid X \neq 1)$ and the spread $\Delta_T(r)$, the sale impact and the volume identify the slot: $\vartheta = 2I_-/(\rho\Delta_T(r))$, $\lambda = v\vartheta$ and $\beta = v(1-\vartheta)/[2(1-v\vartheta)]$.*

Proof of Proposition A.12. Let $r_S < r < h$.

(a) The order law of $\mathcal{F}$ is $(v/2, 1-v, v/2)$ with $v = 2b + \lambda$, and entry is $\rho + (1-\rho)\chi_r v/2$. Given v and $u' > 0$, the equations $v = \lambda' + 2b'$ and $u' = \lambda'/b'$ give $b' = v/(u'+2)$, $\lambda' = vu'/(u'+2)$ and $\beta' = b'/(1-\lambda') = v/[2 + u'(1-v)]$, a unique pair. It has $0 < \lambda' < v < 1$ and $\beta' > 0$, so it is a slot of the model exactly when $\beta' \le 1/3$, that is $3v \le 2 + u'(1-v)$. With $u' = 2\vartheta'/(1-\vartheta')$, the right side is $(2 - 2v\vartheta')/(1-\vartheta')$, and the condition reads $3v(1-\vartheta') \le 2 - 2v\vartheta'$, that is $v\vartheta' \ge 3v - 2$.

(b) A pair with share ϑ' has $m' = (1-\vartheta')/2$. Under full orders at r, the low-class sale earns $\rho\Delta_T(r)m' - k$, because a sale never recruits. This is nonnegative only if $\vartheta' \le 1 - 2k/(\rho\Delta_T(r)) = \bar{\vartheta}(r)$. At the benchmark, $\bar{\vartheta}(r_1) = 0.760000$ and $\bar{\vartheta}(r_2) = 0.829586$. For example, the share 0.9 has $\beta' = 0.105296$ at the benchmark volume, but its sale at r_1 earns -0.011667.

(c) Strict recruitment at r_1 and strict exclusion at r_2 are $\tau(r_1) < M' < \tau(r_2)$, that is $2\tau(r_1) - 1 < \vartheta' < 2\tau(r_2) - 1$, because $\vartheta' = 2M' - 1$. At the benchmark volume $v = 0.727352$, $v \le 200/259$, so every u' in the band $(82/59, 2686/1559)$ satisfies $3v \le 2 + u'(1-v)$, and part (a) gives a slot with $\beta' \le 1/3$, the same counts, and the same entry at r_1 and at r_2. For every such slot, the weakest correct-order bound at r_1 is $\rho\Delta_T(r_1)[1 - \tau(r_2)] - k = 0.024768 > 0$, and $\Delta_T(r_2) > \Delta_T(r_1)$; so full two-sided trade is the unique outcome at both strengths.

(d) In $\mathcal{F}$, entry after the orders 0 and -1 is ρ, so $\rho = \Pr(\text{preparation} \mid X \neq 1)$. Part (ii) of Proposition A.11 gives $I_- = \rho\Delta_T\vartheta/2$, so $\vartheta = 2I_-/(\rho\Delta_T(r))$. Then $v = \lambda + 2b$ and $\vartheta = \lambda/v$ give $\lambda = v\vartheta$ and $b = v(1-\vartheta)/2$, and $\beta = b/(1-\lambda) = v(1-\vartheta)/[2(1-v\vartheta)]$. □

The slot with $u' = 8/5$ has share 0.444444, lies in the band, and has the benchmark volume; its spread at r_1 is 0.778935. The smallest share with the benchmark counts and $\beta' \le 1/3$ is $3 - 2/v \approx 0.250299$. For example, the ratio $u' = 1/10$ and the benchmark volume give $\beta' > 1/3$, so no slot of the model has this ratio and these counts.

At the share $\bar{\vartheta}(r_1) = 0.760000$, the low-class sale earns exactly zero under full orders. Rule IM admits the full sale, and full two-sided trade is then the unique outcome at r_1. A sale rate below λ' raises $\mu(-1)$ above m' and makes the sale strictly profitable. A purchase rate below λ' lowers $\mu(1)$ below M' and makes the purchase strictly profitable. Rule I0 excludes the full sale at this share, so this slot has no equilibrium at r_1 under rule I0. Under rule IM, the shares that the benchmark counts and the incentives at r_1 allow therefore form the closed interval $[3 - 2/v, \bar{\vartheta}(r_1)]$. The values come from `P5_observables/p5_observables.py` (run log `P5_observables/run_log.txt`).

Corollary A.3 (Counts under purchases only)**.** *Take the benchmark auction at the prior* $1/2$ *and a slot in the band. In an equilibrium with* $\sigma_H = \delta_1$ *and* $\sigma_L = \delta_0$*, the counts identify the slot:* $\lambda = 2\mathbb{E}[X]$ *and* $b = (v - \mathbb{E}[X])/2$*, where* $v = 2b + \lambda/2$*. Proposition 3 gives such an equilibrium exactly on* $[r_F, r_N]$ *(form F). On branch R, class H buys with probability* $\sigma_H(1) < 1$ *and class L abstains. There the counts identify the purchase rate* $s_+ = \lambda\sigma_H(1) = 2\mathbb{E}[X]$ *and b, but not* λ*.*

Proof of Corollary A.3. At the prior $1/2$, the order x has probability $[f_H(x) + f_L(x)]/2$. With $\sigma_L = \delta_0$ and $s_+ = \lambda\sigma_H(1)$, $f_H(1) = b + s_+$, $f_L(1) = b$ and $f_H(-1) = f_L(-1) = b$. So $\Pr(X = 1) = b + s_+/2$, $\Pr(X = -1) = b$, $\mathbb{E}[X] = s_+/2$ and $v = 2b + s_+/2$, which give $s_+ = 2\mathbb{E}[X]$ and $b = (v - \mathbb{E}[X])/2$. In form F, $s_+ = \lambda$, which gives the first claim. By Proposition 3, the high class buys with probability one and the low class abstains exactly in form F on $[r_F, r_N]$.

On branch R, $s_+ = b\,[\Delta_T(r)/k - 2]$ depends on the slot only through b, and r_A does not depend on the slot. Take a slot with the same b and a presence probability slightly below λ. Its liquidity probability $b/(1 - \lambda)$ falls, so it stays at most $1/3$; its ratio stays in the band; and its end r_F stays above r. So it has an R equilibrium at r with the same counts, and the counts do not identify λ on branch R. □

Let A be the event that the challenger prepares and W the event that it acquires the target, and write $w_x = \Pr(W \mid X = x, A)$.

Proposition A.13 (Orders predict the winner)**.** *In the point-mass benchmark:*

(i) $w_x = \mu(x) + [1 - \mu(x)]\,\ell/r$*, and* $\Pr(W \mid X = x) = e(x)\,w_x$*.*

(ii) Under full two-sided trade, $w_1 - w_{-1} = \vartheta(1 - \ell/r)$*, and this gap rises strictly with r.*

(iii) Under full two-sided trade, $\mathbb{E}[X] = 0$ *and* $\mathrm{Cov}(X, \mathbf{1}_A) = (1 - \rho)\chi_r(b + \lambda/2)$*. For* $r > r_C$*,* $\mathrm{Cov}(X, \mathbf{1}_A) = 0$ *and* $\mathrm{Cov}(X, \mathbf{1}_W) = (b + \lambda/2)\,\rho\,\vartheta\,(1 - \ell/r) > 0$*.*

Proof of Proposition A.13. (i) The cost is independent of the class and of R, so preparation leaves the class posterior unchanged: $\Pr(H \mid X = x, A) = \mu(x)$. A prepared high-class challenger wins, because $h > r$. A prepared low-class challenger wins exactly when $R < \ell$, with probability ℓ/r. Hence $w_x = \mu(x) + [1 - \mu(x)]\ell/r$, and $\Pr(W \mid X = x) = e(x)w_x$.

(ii) In $\mathcal{F}$, $w_1 - w_{-1} = (M - m)(1 - \ell/r) = \vartheta(1 - \ell/r)$, and its derivative in r is $\vartheta\ell/r^2 > 0$. The function $\mu \mapsto \mu + (1 - \mu)\ell/r$ rises strictly, and the prices of $\mathcal{F}$ rise strictly with the posterior, so the win probability of a prepared challenger rises with the price.

(iii) In $\mathcal{F}$, $\mathbb{E}[X] = 0$, so for any indicator Y with conditional means $y_x = \mathbb{E}[Y \mid X = x]$, $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] = \bar{f}(y_1 - y_{-1})$. For $Y = \mathbf{1}_A$, $y_1 - y_{-1} = e_+ - \rho = (1 - \rho)\chi_r$, so $\mathrm{Cov}(X, \mathbf{1}_A) = (1 - \rho)\chi_r\bar{f}$. For $Y = \mathbf{1}_W$, $y_x = e(x)w_x$. For $r > r_C$, $e \equiv \rho$, so $\mathrm{Cov}(X, \mathbf{1}_A) = 0$ and $\mathrm{Cov}(X, \mathbf{1}_W) = \bar{f}\rho(w_1 - w_{-1}) = \bar{f}\rho\vartheta(1 - \ell/r) > 0$. □

Add a public event after the price that reveals only whether the challenger prepared, and nothing else. Market makers then price again.

Proposition A.14 (Announcement returns)**.** *Take any equilibrium at a strength r.*

(i) After the order x, the price changes by $\Delta_x^1 = [1 - e(x)]\, A_r(\mu(x))$ when the event reports preparation, and by $\Delta_x^0 = -e(x)\, A_r(\mu(x))$ when it reports no preparation. The second report has positive probability exactly when $e(x) < 1$.

(ii) After an order x with $e(x) < 1$, the mean change over both reports, with their model probabilities, is zero.

(iii) Pool all orders, and take the sample of preparation reports. Its mean price change $\sum_x \Pr(X = x \mid A)\, \Delta_x^1$ is positive. After an order with $e(x) = 1$, the report of preparation changes the price by zero.

(iv) Under full two-sided trade, the price after no order lies $(1 - \rho)\chi_r(b + \lambda/2)A_r(M)$ below the expected price before the order opportunity, although the posterior after no order is $1/2$.

Proof of Proposition A.14. After the order x, the event reveals whether the challenger prepared.

(i) The cost is independent of the class, so $\Pr(H \mid x, \text{prepared}) = \mu(x)$, and with preparation $\mathbb{E}[V_T] = t_j$ in class j. So the price after a report of preparation is $t_L + \Delta_T\mu(x) = t_0 + A_r(\mu(x))$. Without preparation $\mathbb{E}[V_T] = t_0$ in both classes. Subtract $P(x) = t_0 + e(x)A_r(\mu(x))$: $\Delta_x^1 = [1 - e(x)]A_r(\mu(x))$ and $\Delta_x^0 = -e(x)A_r(\mu(x))$. The report of no preparation has probability $1 - e(x)$, positive exactly when $e(x) < 1$.

(ii) When $e(x) < 1$, $e(x)\Delta_x^1 + [1 - e(x)]\Delta_x^0 = e(x)[1 - e(x)]A_r - [1 - e(x)]e(x)A_r = 0$.

(iii) Pooled over orders, a sample of preparation reports has mean change $\sum_x \Pr(X = x \mid A)\,[1 - e(x)]A_r(\mu(x))$, with $\Pr(X = x \mid A) = \Pr(X = x)e(x)/\Pr(A)$. Every term is nonnegative, because $A_r > 0$ and $e(x) \le 1$; an order with $e(x) = 1$ contributes zero, since $\Delta_x^1 = 0$. To find a positive term, note that (E1) gives $\sum_x \Pr(X = x)\mu(x) = 1/2$, so some order x has $\mu(x) \le 1/2$. At the benchmark $B_r(1/2) < c_H$ for $1 < r < h$, so $\tau(r) > 1/2 \ge \mu(x)$, and Lemma A.2(d) gives $e(x) = \rho$ under both challenger rules. This order has $\Pr(X = x) \ge (1 - \lambda)\eta(x) > 0$, and its term $\Pr(X = x)\rho(1 - \rho)A_r(\mu(x))/\Pr(A)$ is positive. So the mean is positive in every equilibrium; in $\mathcal{F}$, the order -1 is such an order.

(iv) In $\mathcal{F}$, the expected price before the order opportunity is $\bar{P} = \sum_x \Pr(X = x)P(x) = t_0 + \rho[\bar{f}A_r(m) + (1 - 2\bar{f})A_r(1/2) + \bar{f}A_r(M)] + (1 - \rho)\chi_r\bar{f}A_r(M)$. Since A_r is affine and $m + M = 1$, $\bar{f}A_r(m) + \bar{f}A_r(M) = 2\bar{f}A_r(1/2)$, so $\bar{P} = t_0 + \rho A_r(1/2) + (1-\rho)\chi_r\bar{f}A_r(M)$. The price after no order is $t_0 + \rho A_r(1/2)$, so $P(0) - \bar{P} = -(1-\rho)\chi_r\bar{f}A_r(M)$, while $\mu(0) = 1/2$. □

In the benchmark at r_1, the expected price is 0.872531 and the price after no order is 0.614583. A sample of preparation reports has mean change 0.160592 at r_1 and 0.701042 at r_2. The zero mean of part (ii) needs both reports with their model probabilities.

Two clusters at intermediate strengths. Let $r_F < r < r_N$, so $r < r_C$. Proposition 3 gives exactly two equilibria under rules IM and C1: N, and F with $\sigma_L = \delta_0$ (purchases only). In the second, $\Pr(X = 1) = b + \lambda/2$ and $\Pr(X = -1) = b$, so $\mathbb{E}[X] = \lambda/2 = 0.168061$. The posterior after a sale is $1/2$, and after no order it is $\mu_0 = (1 - 2b - \lambda)/(2 - 4b - \lambda)$, with $1/2 - \mu_0 = \lambda/[4(1 - v_U)] > 0$, $v_U = 2b + \lambda/2$. Both orders have entry ρ, so $P(-1) - P(0) = \rho\Delta_T\lambda/[4(1 - v_U)] > 0$; at $r = 1.60$ it is 0.005363. After a purchase every challenger prepares. In N the three prices coincide and entry after a purchase is ρ. Under rule CM, the threshold pair T of Proposition 3 adds a third tuple: at $r = 1.60$ class H buys with probability 0.492845, entry after a purchase is 0.506106, the mean order is 0.082828, the low-class sale earns −0.005938, and entry is 0.321311. The exhaustive two-cluster statement therefore uses rule C1.

Certificates. The values come from the script `P5_observables/p5_observables.py`. It computes every value in interval arithmetic at 60 digits from the exact input $\beta = 294655/10^6$ and an enclosure of e; a sign counts only when the enclosure excludes zero. The independent program `XP5/recompute.py` (run log `XP5/run_log.txt`) rebuilds every value from the auction cash flows and the slot probabilities, with a rational Taylor bound on e through the term $1/120!$, and it records one comparison row per value in `XP5/registry_recompute.csv`.

A.9 Parameter declarations and the margins table

This subsection declares the inputs of every computation and records the margin of every strict hypothesis at the benchmark. Table A.5 gives calibration B, following Section 2: an *input* is a declaration, and a derived value follows from its definition. The derived values keep their unrounded definitions in every computation; the decimals are displays.

Table A.5. Declarations at calibration B.

Object	Symbol	Value	Kind
Class values	$h,\ \ell$	10, 1	input
Reserve	p	1/2	input
Zero-cost probability	ρ	1/4	input
Preparation costs	$c_L,\ c_H$	0, 6	input
Trading cost	k	1/50	input
Prior	π	1/2	input
Incumbent strengths	$r_0,\ r_1,\ r_2$	6/5, 3, 18/5	input
Slot mass ratio	$u = \mathrm{e} - 1$	1.718282	input
Liquidity order probability	β	0.294655, an exact decimal	input
Presence probability	$\lambda = u\beta/(1+u\beta)$	0.336122	derived
Liquidity masses	$b,\ b_0$	0.195615, 0.272648	derived
Posterior bounds	$m(\pi),\ M(\pi)$	0.268941, 0.731059	derived
Research cost and cap	$a,\ \bar{\lambda}$	1/40, 0.336122	input
Cost law	$L;\ \varepsilon$	1/5; 1/10, 1/1000	input
Signal accuracies at B	$a_T,\ d$	0.94, 0.505	input
Four-class values	$\theta_1, \ldots, \theta_4$	1, 1001/1000, 9999/1000, 10	input
Four-class prior	π_j	1/4 each	input
High cost, access outside recruitment	c_H	12/5	input
Reserves	p	101/100, 11/10, 31/20	input

Notes. The registry keys are `prim_*`, `u_B`, `beta_B`, `lambda_B`, `b_B`, `b0_B`, `m_half_B`, `M_half_B`, `res_a`, `res_cap`, `atom_*`, `sig_comp_*`, `four_*`, `access_out_cH` and `rsv_*`. The upright e is Euler's number. The research cap equals the benchmark presence probability, so research at the cap reproduces the benchmark slot. The rows below the rule declare the inputs of single results. The economies of Table A.4 carry their own declarations in that table.

The vector $S(B)$. The signal example of Section 6.3 uses its own vector. It keeps the slot (u, β, λ), the values h and ℓ, the reserve p and the prior π of calibration B, and it declares

$$\rho = 0.85, \qquad c_H = 7.14, \qquad k = 0.015, \qquad r_0 = 1.1, \qquad r_1 = 2.3,$$

and the accuracies $(a_T, d) = (0.70, 0.75)$, each an input. Its primitive margins are positive: ρ, $1-\rho$, $r_0 - \ell$, $r_1 - r_0$, $h - r_1$, c_H and k are all positive. The values of $S(B)$ describe a comparison of strengths within $S(B)$; they do not describe calibration B.

The margins table. Table A.6 lists every strict hypothesis of the paper with its margin at the benchmark, in the direction in which a positive number means that the hypothesis holds. A negative entry records a condition that fails by design, and the table says which. Each margin comes from the script in its last column; the codes are

[P1] `P1_core/p1_numbers.py` (run log `p1_numbers.log`) and `p1_certificates.py` (run log `p1_certificates.log`) in the same folder;

[P2] `P2_auction/derive.py` (run log `run.log`);

[P3] `P3_research/P3_research.py` (run log `P3_research.log`);

[P4] `P4_information/derive.py` (run log `run.log`);

[P5] `P5_observables/p5_observables.py` (run log `run_log.txt`).

Every margin that depends on e is computed in outward interval or exact rational interval arithmetic, and its sign is certified only when the whole enclosure lies on one side of zero. The independent recomputations in the folders `XP1/`, `XP2/`, `XP3/`, `XP4/` and `XP5/` reproduce every display; `XP1/xp1_registry_recompute.csv` and `XP3/exact_enclosures.json` record exact rational endpoints. The structural bound $\beta \le 1/3$ is a domain restriction; the table records its distance.

The margin of (H3) at the upper end of the band is one distance in three units: $c_H - B_{r_2}(M) = 0.002689$ in cost units, $\tau(r_2) - M = 0.000334$ in posterior units, and $r_2 - r_C = 0.007341$ in strength units. Given the auction primitives it depends only on the top posterior and on the place of r_2, and in cost units it is the smallest margin of Proposition 2.

Open parameter sets. The open-set statement concerns the sufficient conditions of each intended result. Take each row of Table A.6 whose hypothesis holds at the benchmark, with the sign that the row requires. It is a strict inequality between continuous functions of the primitives. Together these rows define an open set of primitives that contains the benchmark. With the structural bound $\beta \le 1/3$, the set is relatively open in the model domain. The statement excludes these entries:

- the third economy of Proposition A.4 at the half-width $1/10$, which fails by design;
- the control economy with the same means in the last panel, whose negative C_1 is its purpose;
- the row that shows that $d \ge a_T$ is impossible at B;
- exact ties, such as $r = r_C$, $u = u_{\mathrm{lo}}$ or the informed share $\bar{\vartheta}(r_1)$. These boundary equilibria are valid under the stated tie rules, and the strict-margin convention does not remove them.

The sections through the benchmark are: the slot band and the prior range of Proposition A.1 (Table 2); the cap strip and the cost window of Proposition A.2; the support conditions of Proposition A.4,

Table A.6. Strict hypotheses and their margins at the benchmark.

Result	Hypothesis	Margin	Value at B	Source
Proposition 2 and Proposition A.1 (calibration B, prior $\pi = 1/2$)				
(H1) left	$B_0(\pi) < c_H$	$c_H - B_{r_0}(\pi)$	1.195833	[P1]
(H1) right	$c_H < B_1(M)$	$B_{r_1}(M) - c_H$	0.217155	[P1]
		$M - \tau(r_1)$	0.026059	[P1]
(H2) left	$\Delta_0 < k$	$k - \Delta_T(r_0)$	0.003333	[P1]
(H2) right	$k < \rho\kappa\Delta_1$	$\rho\kappa\Delta_T(r_1) - k$	0.024824	[P1]
(H3)	$B_2(M) < c_H$	$c_H - B_{r_2}(M)$	0.002689	[P1]
		$\tau(r_2) - M$	0.000334	[P1]
Implied	$B_1(\pi) < c_H$	$c_H - B_{r_1}(\pi)$	1.708333	[P1]
	$k < \rho\kappa\Delta_2$	$\rho\kappa\Delta_T(r_2) - k$	0.043127	[P1]
	$B_2(\pi) < c_H$	$c_H - B_{r_2}(\pi)$	1.865278	[P1]
Slot band	$u > u_{\text{lo}}$	$u - 82/59$	0.328451	[P1]
	$u < u_{\text{hi}}$	$2686/1559 - u$	0.004617	[P1]
	λ in its band	distance to lower, upper end	0.045583, 0.000599	[P1]
Prior range	π in its range	distance to lower, upper end	0.032150, 0.000424	[P1]
Domain	$\beta \le 1/3$ (structural)	$1/3 - \beta$	0.038678	[P1]
Ownership	$e_H W_H(F_1) > \rho W_H(F_0)$	$e_H - \rho\,(W_H = 1)$	0.398802607	[P4]
	$b + \lambda > 1/2$ (sufficient form)	$b + \lambda - 1/2$	0.031737	[P2]
Proposition 3 and Corollary A.2 (benchmark path, Lemmas A.4 and A.5)				
Path	$\Delta_T > 0$, $g_H > g_L$ on the path	$\min \Delta_T$; $\min(g_H - g_L)$	1/60; 1451/180	[P1]
	$\tau(r_0) > 1/2$	$\tau(r_0) - 1/2$	0.133117	[P1]
	$J(r_0) < k$	$k - J(r_0)$	1283/92400	[P1]
	$\rho J(r_1) > k$; $\rho J(r_2) > k$	$\rho J - k$	7/240; 0.043048	[P1]
	$\rho J(2) > k$ (so $r_J < 2$)	$\rho J(2) - k$	11/11200	[P1]
	$\tau(r_2) < 3/4$	$3/4 - \tau(r_2)$	27/1451	[P1]
	$\tau(r_N) > 1/2$	$\tau(r_N) - 1/2$	0.154463	[P1]
	$Q_J(r_0) > 0$	$Q_J(r_0)$	4634399/15625	[P1]
	$u < 19/3$	$19/3 - u$	4.615052	[P1]
Order	$r_0 < r_A$	$r_A - r_0$	0.193958	[P1]
	$r_A < r_F$	$r_F - r_A$	0.073169	[P1]
	$r_F < r_N$	$r_N - r_F$	0.280750	[P1]
	$r_N < r_J$	$r_J - r_N$	0.218220	[P1]
	$r_J < r_S$	$r_S - r_J$	0.158051	[P1]
	$r_S < r_1$	$r_1 - r_S$	0.875852	[P1]
	$r_1 < r_C$	$r_C - r_1$	0.592659	[P1]
	$r_C < r_2$	$r_2 - r_C$	0.007341	[P1]
Priors	$\pi > \pi_A^{\text{lo}}$	$\pi - \pi_A^{\text{lo}}$	0.103825	[P1]
	$\pi < \pi_A^{\text{hi}}$	$\pi_A^{\text{hi}} - \pi$	0.140739	[P1]
	$\tau_A < 3/4$	$3/4 - \tau_A$	0.109261	[P1]
	$\tau_A < \pi_{\max}$	$\pi_{\max} - \tau_A$	0.009320	[P1]
	rule CM at $\pi_{\max}$: purchase at r_*	$\Delta_T(r_*)[1 - M(\pi_{\max})] - k$	0.000343	[P1]
Propositions 4 and A.2 ($a = 1/40$, $\bar{\lambda} = 0.336122$)				
Weak	$B_{r_0}(1/2) < c_H$	$c_H - B_{r_0}(1/2)$	1.195833	[P3]
	$\Delta_T(r_0) < k$	$k - \Delta_T(r_0)$	0.003333	[P3]
Intermediate	$\rho\Delta_T(1.65) < 2k$	$2k - \rho\Delta_T$	0.007992	[P3]
	$\varphi_a(1.55) > 0$	$\varphi_a(1.55)$	0.004629	[P3]
	$a < a_{\text{up}}$	$a_{\text{up}} - a$	0.011763	[P3]
	$\lambda_R(1.65) < \bar{\lambda}$	$\bar{\lambda} - \lambda_R$	0.019439	[P3]
	$a > a_{\text{lo}}(\bar{\lambda})$	$a - a_{\text{lo}}$	0.003531	[P3]
	$a < a_{\text{mono}}$	$a_{\text{mono}} - a$	0.136713	[P3]
Strong	$\lambda_E(r_1) < \bar{\lambda}$	$\bar{\lambda} - \lambda_E(r_1)$	0.045583	[P3]
	$\tau(r_1) < M_{\bar{\lambda}}$	$M_{\bar{\lambda}} - \tau(r_1)$	0.026059	[P3]
	$\bar{\lambda} < \lambda_E(r_2)$	$\lambda_E(r_2) - \bar{\lambda}$	0.000599	[P3]
	$M_{\bar{\lambda}} < \tau(r_2)$	$\tau(r_2) - M_{\bar{\lambda}}$	0.000334	[P3]
	(U) at r_1, r_2	$\rho\Delta_T m_{\bar{\lambda}} - k - a\bar{\lambda}$	0.016421, 0.034723	[P3]
Branch	$r_R < 1.55$	$1.55 - r_R$	0.048977	[P3]
	$1.65 < \bar{r}_R$	$\bar{r}_R - 1.65$	0.027633	[P3]
	$\bar{r}_R < r_N$	$r_N - \bar{r}_R$	0.070245	[P3]

Table A.6. Strict hypotheses and their margins at the benchmark (continued).

Result	Hypothesis	Margin	Value at B	Source
Proposition A.3 (full orders held fixed)				
Weak row	$c_H < B_0(M)$	$B_{r_0}(M) - c_H$	0.879842898	[P4]
Proposition A.4 ($L = 1/5$, half-width $\varepsilon = 1/10$)				
Supports	$L < B_1(m)$	$B_{r_1}(m) - L$	2.166178511	[P4]
	$B_0(\pi) < c_-$	$c_- - B_{r_0}(\pi)$	1.095833333	[P4]
	$c_+ < B_1(M)$	$B_{r_1}(M) - c_+$	0.117154822	[P4]
Third economy	$L < B_2(m)$	$B_{r_2}(m) - L$	2.072133347	[P4]
	$B_2(M) < c_-$ (fails by design)	$c_- - B_{r_2}(M)$	−0.097311098	[P4]
	same at $\varepsilon = 1/1000$	$c_H - \varepsilon - B_{r_2}(M)$	0.001688902	[P4]
Proposition A.6 (four classes) and the three-class collision				
Separation	$\rho\gamma > k$	$\rho\gamma - k$	0.024718	[P2]
	weak range below k	k minus the range	0.003333	[P2]
	weak prior profit below c_H	c_H minus the profit	1.195875	[P2]
	$\bar{g}_{\mathcal{H}} > \bar{g}_{\mathcal{L}}$	$\bar{g}_{\mathcal{H}} - \bar{g}_{\mathcal{L}}$	8.332667	[P2]
	top group profit above c_H	profit minus c_H	0.216834	[P2]
Collision	$\beta > 1/4$	$\beta - 1/4$	0.044655	[P2]
	$0 < w < 1$	w; $1 - w$	0.940709; 0.059291	[P2]
	strict class choices	choice gaps	0.260795; 0.020627; 0.024768	[P2]
Proposition A.7: the vector $S(B)$ and calibration B				
Private exclusion	$B_{r_0}(d) < c_H$	$c_H - B_{r_0}(d)$	$S(B)$: 0.052500000; B: 1.150916667	[P4]
Joint entry	$c_H < B_{r_1}(\phi_+(\mu_+))$	$B_{r_1}(\phi_+(\mu_+)) - c_H$	$S(B)$: 0.045265153; B: 0.020731442	[P4]
Weak order	$(2a_T - 1)D^+(r_0) < k$	$k - (2a_T - 1)D^+(r_0)$	$S(B)$: 0.001454545; B: 0.002610833	[P4]
Strong order	$k < (2a_T - 1)\rho m\Delta_T(r_1)$	$(2a_T - 1)\rho m\Delta_T(r_1) - k$	$S(B)$: 0.018594291; B: 0.019444742	[P4]
One pair	$B_{r_1}(\phi_-(\mu_+)) < c_H$	$c_H - B_{r_1}(\phi_-(\mu_+))$	$S(B)$: 4.159475417; B: 0.048821802	[P4]
Control at B	no-signal top excluded	$c_H - B_{r_1}(\mu_+)$	0.013903757	[P4]
	$d \geq a_T$ impossible	weak bound minus k	0.049029700	[P4]
Proposition A.10 and the benchmark reserves: $T = \rho\kappa\Delta_T - k$, $C = c_H - B(1/2)$, $J = B(M) - c_H$				
Exclusion, points	T, C, J at r_0		0.038167, 1.512521, 0.561220	[P2]
	T, C, J at r_1		0.047239, 1.835008, 0.089706	[P2]
Exclusion, bands	T, C, J at r_0		0.048076, 1.552083, 0.503375	[P2]
	T, C, J at r_1		0.047572, 1.850833, 0.066568	[P2]
Deterrence	T, C, J at r_0		0.084215, 1.775000, 0.177445	[P2]
	T, C, J at r_1 ($J < 0$ required)		0.057405, 1.950208, −0.078730	[P2]
	$p_H < p < h - c_H/M$	distance to each end	0.224730, 0.242723	[P2]
	$p_H > r_0$	$p_H - r_0$	0.125270	[P2]
Section 8 (full two-sided trade and the two clusters)				
Full trade	$r_S < r_C$; $r_S < h$	$r_C - r_S$; $h - r_S$	1.468510; 7.875852	[P5]
	$M > 5/8$	$M - 5/8$	0.106059	[P5]
Band	full trade at r_1 for every share in the band	weakest bound	0.024768	[P5]
Clusters	$r_F < 1.55$; $1.65 < r_N$	distances	0.082873; 0.097878	[P5]
	purchase at 1.55; low class abstains at 1.65	$\Delta_T m - k$; $k - \rho\Delta_T/2$	0.006243; 0.003996	[P5]
	purchase recruits at 1.65	$B(M) - c_H$	0.713202	[P5]

Table A.6. Strict hypotheses and their margins at the benchmark (continued): the economies of Table A.4.

Economy	C_0	C_1	T_0	T_H	T_L	C_2	O	Source
Benchmark	1.195833	0.217155	0.003333	0.024824	0.024824	0.002689	0.398803	[P2]
Moderate gap	0.033452	0.030137	0.000810	0.003603	0.003603	—	0.398803	[P2]
Narrow bands	1.195660	0.217192	0.002986	0.024833	0.024833	0.002658	0.398803	[P2]
Low prior	1.460455	0.109179	0.005455	0.066983	0.012698	0.058137	0.398803	[P2]
High prior	0.562273	0.123413	0.005455	0.012698	0.066983	0.132569	0.398803	[P2]
Wide classes	0.021000	0.058891	0.006667	0.023541	0.127671	0.314551	0.349742	[P2]
Same means (control)	0.025000	−0.018122	0.020000	0.026343	0.137174	0.447727	0.398803	[P2]

Notes. $C_0 = c_H - B_0(\pi), C_1 = B_1(M) - c_H, T_0 = k - \Delta_0, T_H = \rho(1-M)\Delta_1 - k, T_L = \rho m \Delta_1 - k, C_2 = c_H - B_2(M)$ and $O = [\rho + (1-\rho)(b+\lambda)]W_H(F_1) - \rho W_H(F_0)$. The moderate-gap economy has two strengths. The last row is a control: its negative C_1 is the reason entry stays at ρ, and its O uses the recruitment coefficient that the economy does not reach.

which allow every half-width below 0.002688902 for all three parts; the reserve intervals of Appendix A.7; and, for Section 8, the set $0 < \beta < 1/3$ with the slot band and $r_S(M) < r < h$. On the last set the uniqueness margin $\rho\Delta_T(r)m - k$ tends to zero as $r \downarrow r_S$, so no single positive margin covers the whole interval.